

HOL-2770-05-ANS-L

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VMware vDefend Firewall Technology Deep Dive (HOL-2770-05-ANS-L)

Lab Guidance

VMware vDefend Firewall provides stateful Layer 2-7 security to protect virtualized workloads, containers, and bare-metal servers against lateral threat movement in private cloud environments.

The solution operates through two distinct firewall tiers:

- **Distributed Firewall:** Secures east-west workload traffic natively across individual ESXi hosts.
- **Gateway Firewall:** Protects network boundaries and north-south traffic, running as a VM appliance or as a bare-metal ISO installation.

In this hands-on lab, you will explore core vDefend Firewall capabilities. You will gain practical experience with the Security Services Platform, Distributed and Gateway Firewalls, and the new Rule Analysis tool.

Module	Title	Length	Level
1	vDefend Firewall & Security Intelligence Introduction	10 min	Beginner
2	Provide Enhanced Application Security with vDefend DFW	20 min	Intermediate
3	Provide Enhanced Application Security With Layer 7 Firewall Rules	20 min	Intermediate
4	vDefend Gateway Firewall	15 min	Intermediate
5	vDefend DFW Rule Analysis	10 min	Intermediate
6	AI Assistant for vDefend Firewall	20 min	Intermediate

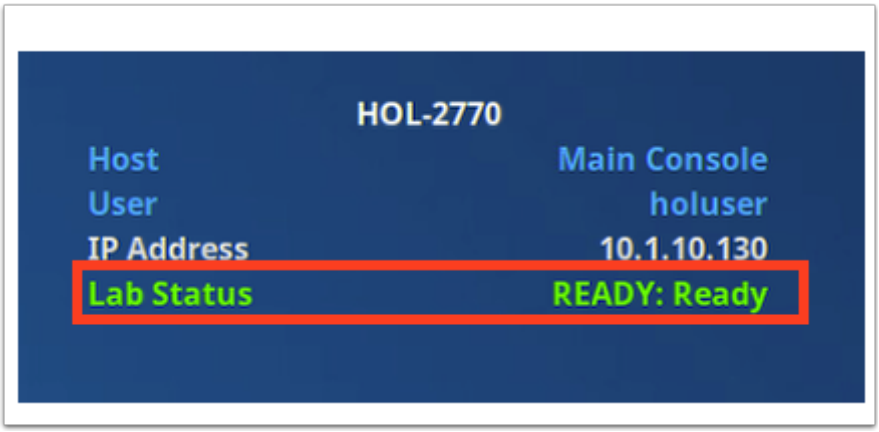
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First time using Hands-on Labs?

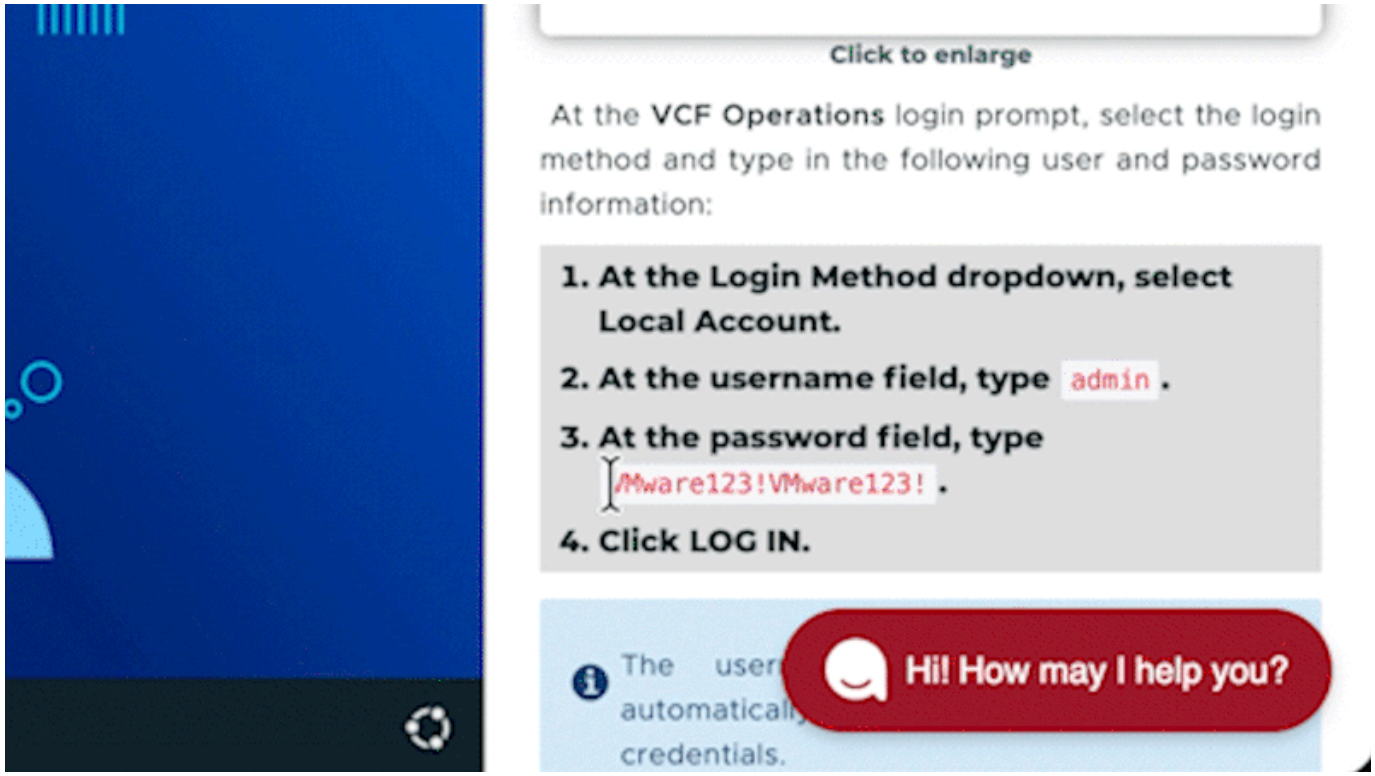
If this is your first time taking a lab you can review the VMware Lab Platform interface before proceeding.

This includes help with copy/paste from the manual into the console and other user-interface questions.



The lab console will indicate when your lab has finished all the startup routines and is ready for you to start. If you see anything other than "Ready", please wait for the status to update. If after 5 minutes your lab has not changed to "Ready", please ask for assistance.

Drag and Drop Text



Save time by dragging and dropping required text from the manual into the lab console.

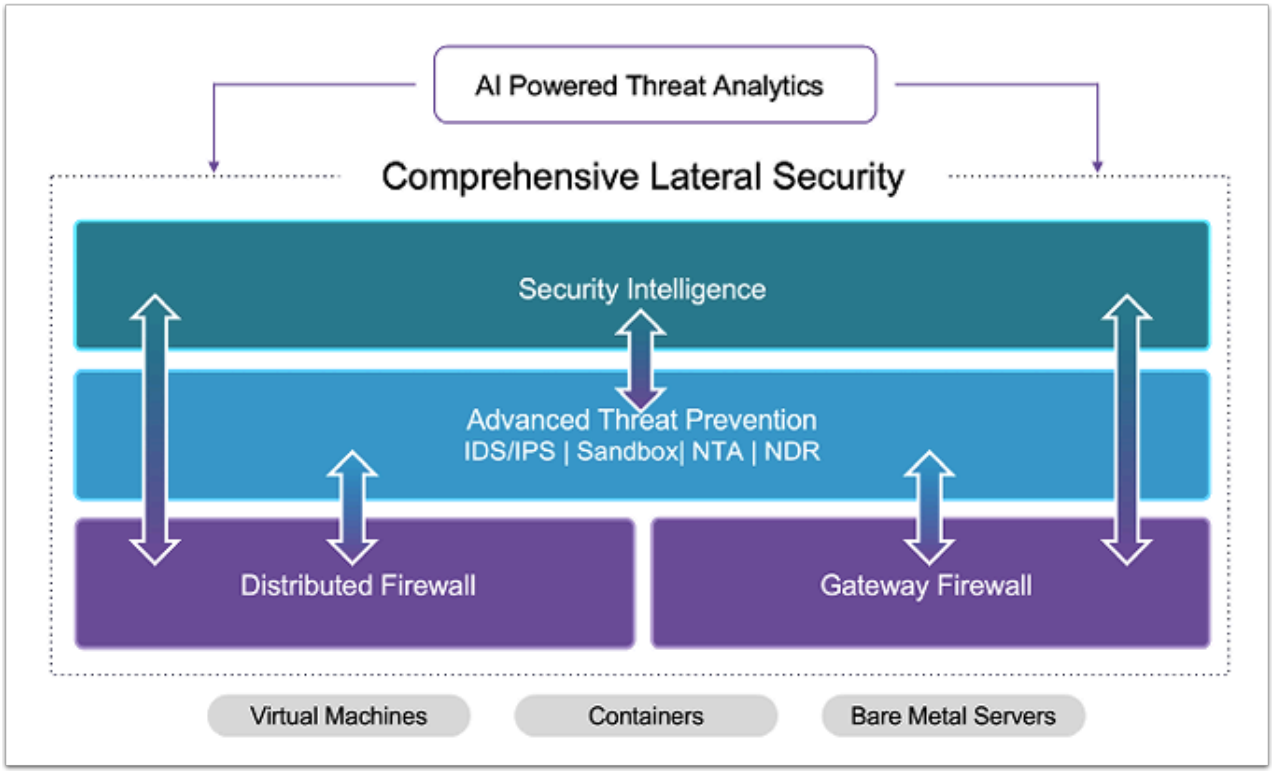
Module 1 – vDefend Firewall & Security Intelligence Introduction (10 Min)

vDefend Firewall Overview

VMware vDefend is a software-defined Layer 2-7 security suite integrated natively into VMware Cloud Foundation (VCF). It protects workloads against lateral threat movement across private cloud and bare-metal environments by combining stateful micro-segmentation with Advanced Threat Prevention (ATP) capabilities.

The VMware vDefend Advanced Threat Prevention (ATP) capabilities combines advanced detection technologies such as IDS/IPS, MPS, and NTA with aggregation and correlation engines from Network Detection and Response (NDR).

VMware vDefend Security suite



VMware vDefend Firewall

VMware vDefend Firewall delivers stateful, granular network security across virtual machines, containers, and physical servers. Its primary components include:

- **Distributed Firewall (DFW):** Integrated directly into the hypervisor on each host to provide kernel-level, scale-out micro-segmentation at the individual workload interface.

- **Gateway Firewall (GFW):** Delivers perimeter security and zone-based isolation for traffic entering or exiting VCF workload domains. GFW deploys as a Virtual Machine (VM) appliance on a vSphere host or as a standalone installation on bare-metal hardware via an ISO image.
- **Security Intelligence:** A distributed analytics engine that delivers full network traffic visibility across all hosts and workloads while automatically generating rule recommendations for micro-segmentation.

VMware vDefend Advanced Threat Prevention (ATP)

VMware vDefend ATP combines multi-layered detection technologies with correlation engines from Network Detection and Response (NDR) to stop advanced threats:

- **Distributed Intrusion Detection and Prevention System (IDPS):** Inspects network traffic for vulnerabilities and malicious activity to simplify security posture management and compliance audits.
- **Network Sandboxing:** Performs deep behavioral analysis on suspicious files and URLs by safely executing and deconstructing malware artifacts.
- **Network Traffic Analysis (NTA):** Identifies threat activity by detecting protocol, traffic volume, and workload behavior anomalies.
- **Network Detection and Response (NDR):** Aggregates and correlates events from IDPS, Sandboxing, and NTA into condensed, visual intrusion campaigns to simplify incident analysis.

vDefend Security Services Platform Overview

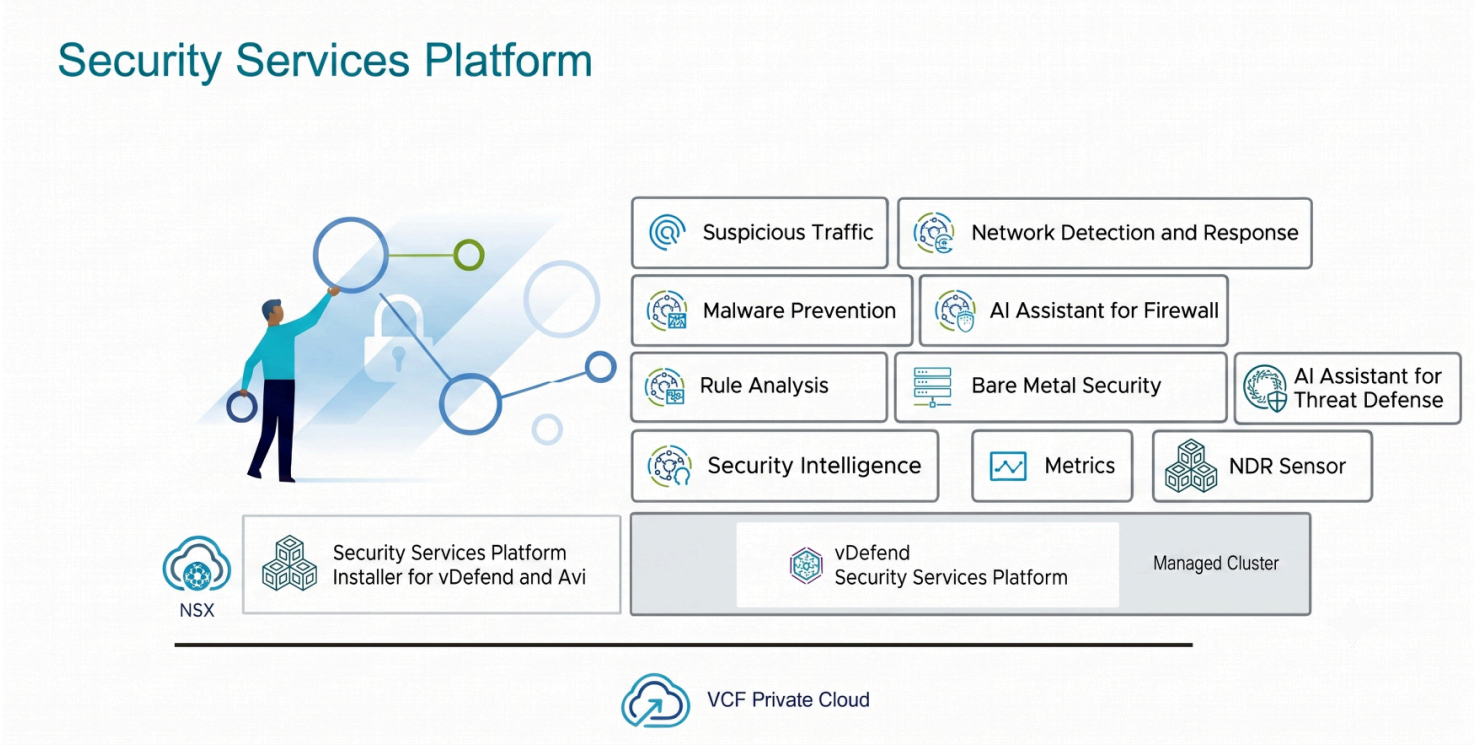
The Security Services Platform (SSP) for VMware® vDefend™ accelerates and simplifies the journey to Zero Trust security within the VCF environment. It is a high-performance and scalable platform that runs vDefend security services, including Security Intelligence, Network Detection and Response, Malware Prevention, and more.

Key Features

The Security Services Platform addresses key challenges that are posed by evolving cyber threats. The platform hosts innovations in VMware® vDefend™ not only react to threats but fundamentally change how you build, operate, and secure your enterprise private cloud.

From SSP, you can manage the following security features:

- **Security Intelligence:** Automatically visualizes network traffic flows and generates recommendations for micro-segmentation policies.
- **Rule Analysis(RA):** Identifies overlapping, redundant, or unused firewall rules to optimize policy enforcement and performance.
- **Bare Metal Security(BMS):** Extends vDefend security policies and visibility to non-virtualized physical workloads.
- **Network Traffic Analysis (NTA):** Monitors and analyzes network flow data to detect suspicious lateral movement.
- **Malware Prevention Service (MPS):** Uses sandboxing and behavioral analysis to inspect files and block malicious payloads.
- **Network Detection and Response (NDR):** Correlates multi-stage network threat signals to provide unified detection, investigation, and response.
- **AI Assistant For VMware vDefend:** Provides generative AI-driven troubleshooting, natural language policy queries, and threat analysis.
- **Metrics:** Displays real-time operational data and system health analytics across all active vDefend security services.


Security Services Platform
Suspicious Traffic
Network Detection and Response
Malware Prevention
AI Assistant for Firewall
Rule Analysis
Bare Metal Security
AI Assistant for Threat Defense
Security Intelligence
Metrics
NDR Sensor
NSX
Security Services Platform Installer for vDefend and Avi
vDefend Security Services Platform
Managed Cluster
VCF Private Cloud

vDefend Security Services Platform Architecture

The **vDefend Security Services Platform (SSP)** architecture establishes the foundation for deploying and managing advanced VMware network security tools across your infrastructure:

vDefend Security Services Platform Concepts

SSP Installer - SSP Installer is a VM which deploys and manages theSecurity Services PlatformInstance andAvi OperationsInstance. TheSSP Installerhosts packages needed for theSecurity Services Platform instance lifecycle management, which includes deployment and upgrade. The installer features a browser-based user interface for managing the environment.

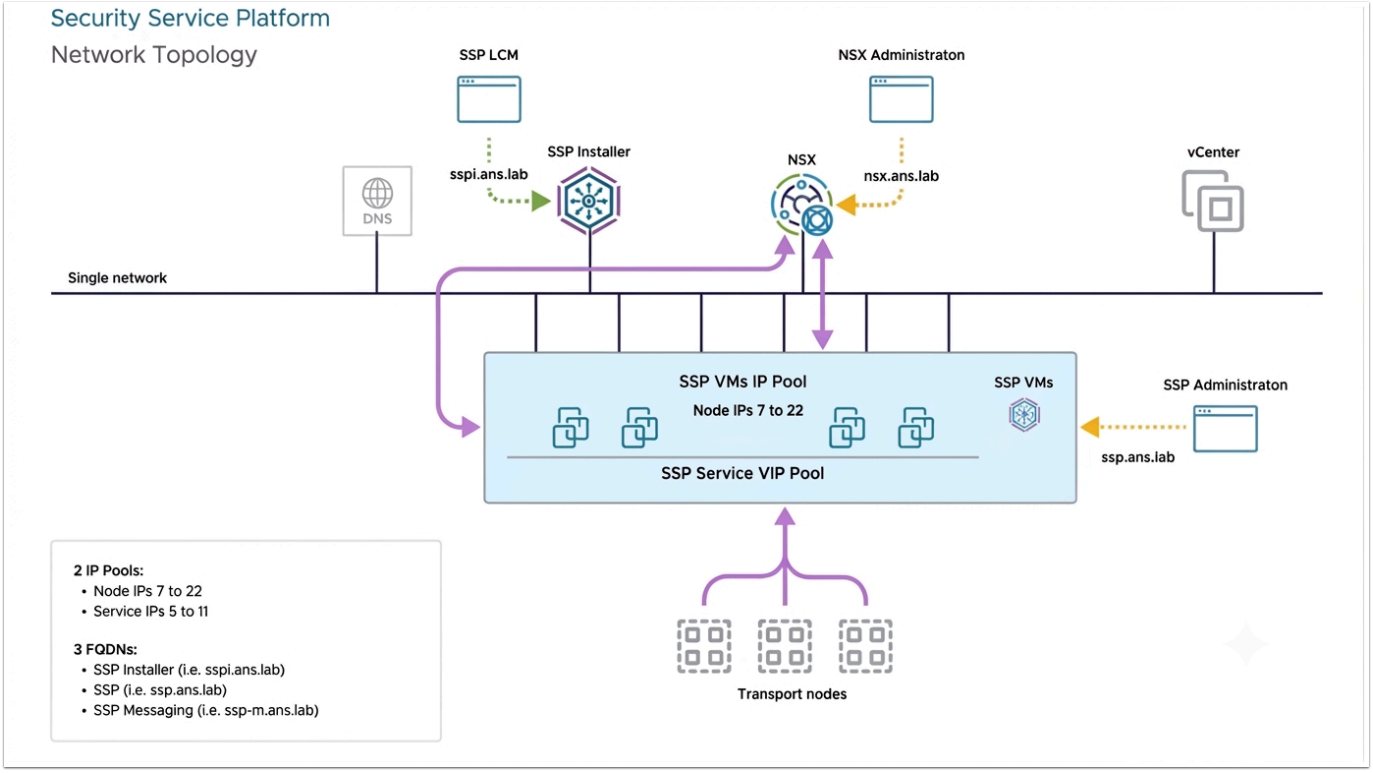
SSP Instance - Security Services Platform is a managed cluster that hosts various core security features likeSecurity Intelligence,Network Detection and Response,Malware Prevention Service, and so on. Multiple instances of the platform can be deployed and managed by the SSP Installer.

Security Feature - A security feature is a network security tool or technology, such as Security Intelligence, Network Detection and Response, Network Traffic Analysis, and so on.

Avi Operations - Avi Operations instance integrates with the VMware Avi Load BalancerController and enables AI Assistant capabilities for Avi, facilitating enhanced operational capabilities using generative AI.

Network Topologies

Security Services Platform can work in a single-network environment (for example, a single Virtual Private Cloud (VPC), Local Area Network (LAN), dedicated VLAN, or air-gapped environment) or in a routed-network environment.

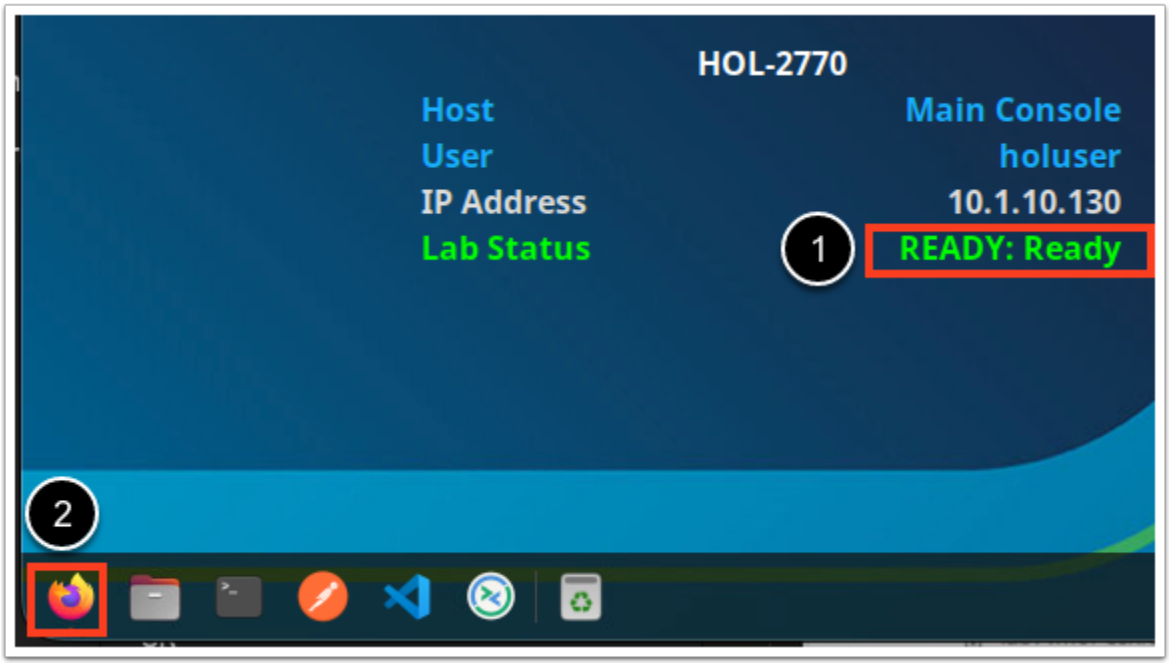


- In a single-network environment, all components reside on the same network. There is direct connectivity across all components, such as vCenter,NSX Manager, transport nodes,SSP Installer, and the SSP instance. The IP addresses of all the components are on the same network.
- Security Services Platform can also be deployed where SSP Installer is on a management network and the SSP instance is on a different network (for example, the network where the transport nodes are on). Both networks are routable.

vDefend Security Services Platform Installer Hands On

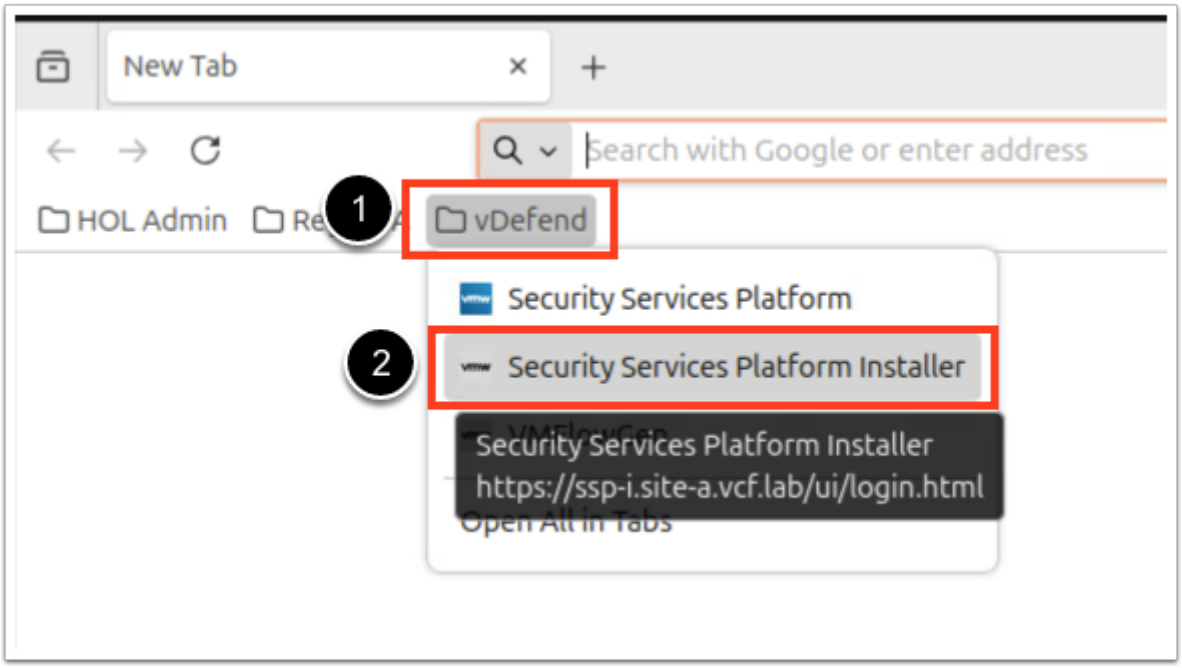
Lets get some hands on exposure and experience with the vDefend Security Services Platform Installer.

Accessing FireFox



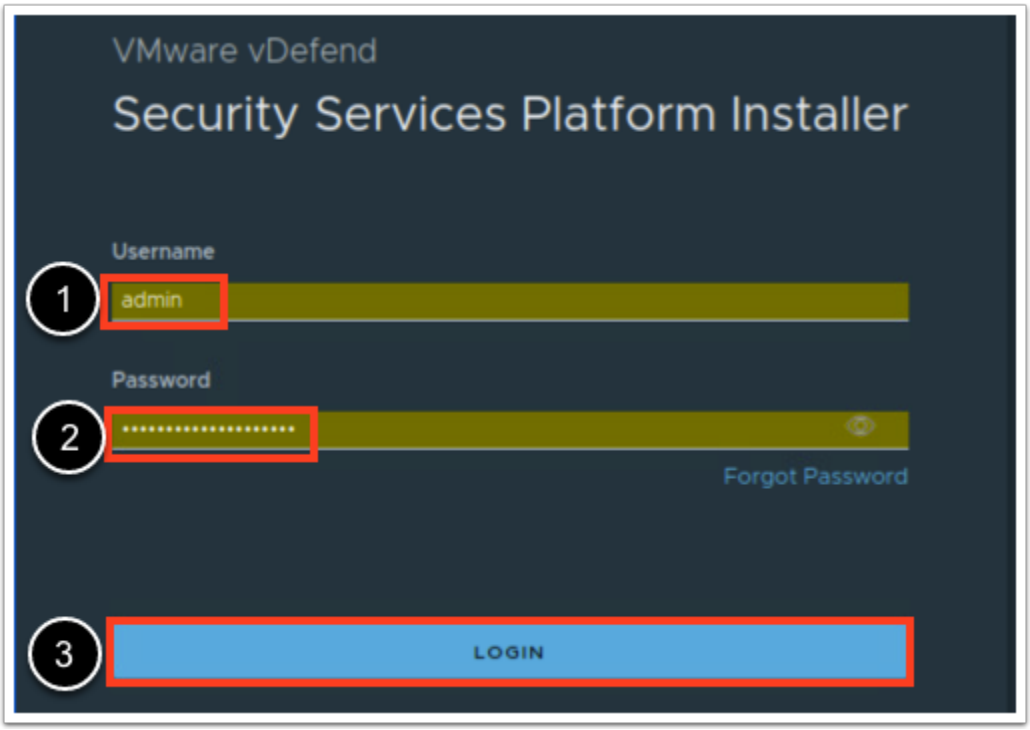
1. Confirm the Lab Status shows **Ready**
2. Click on the FireFox icon to launch the browser

Accessing Security Services Platform Installer



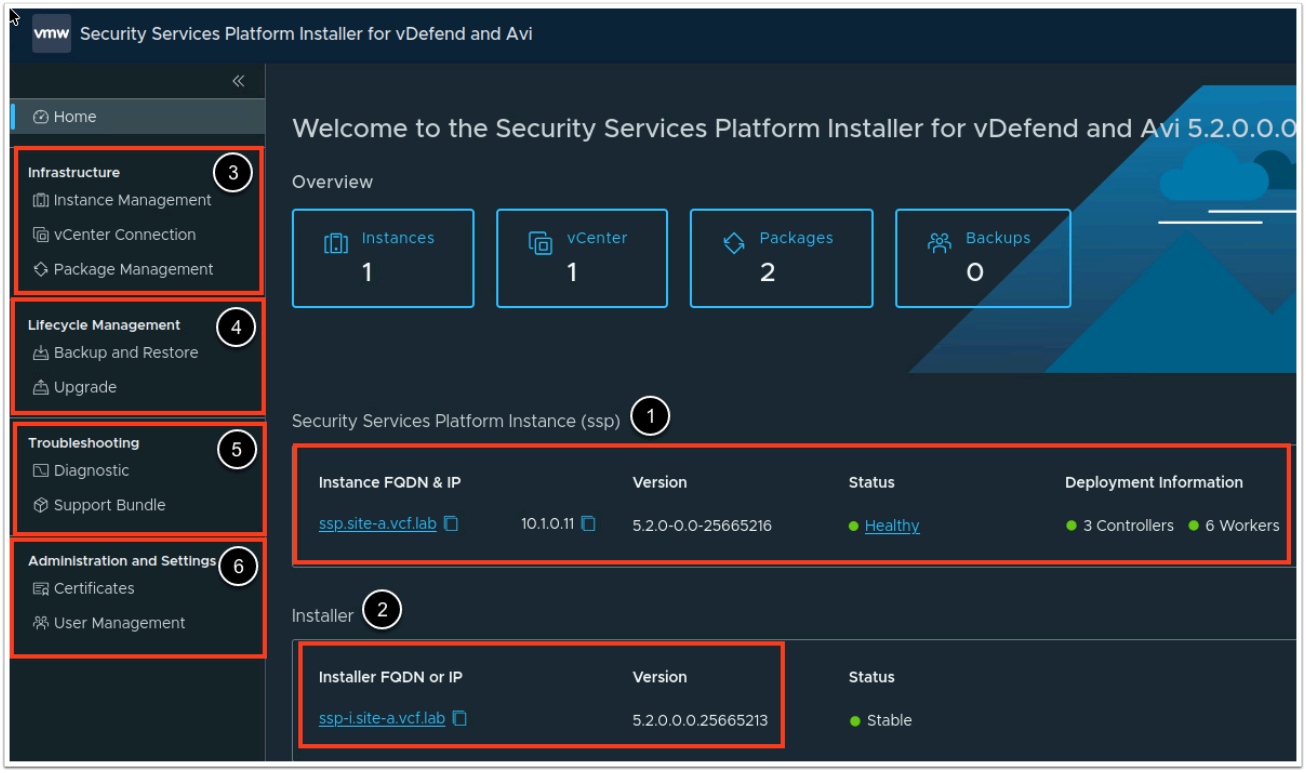
1. Click the **vDefend** Folder on the browser
2. Click on the **Security Services Platform Installer** bookmark.

Accessing Security Services Platform, Log In



1. Username **admin** should be pre-populated
2. Password **VMware123!VMware123!** should be pre-populated
3. Click **LOGIN**

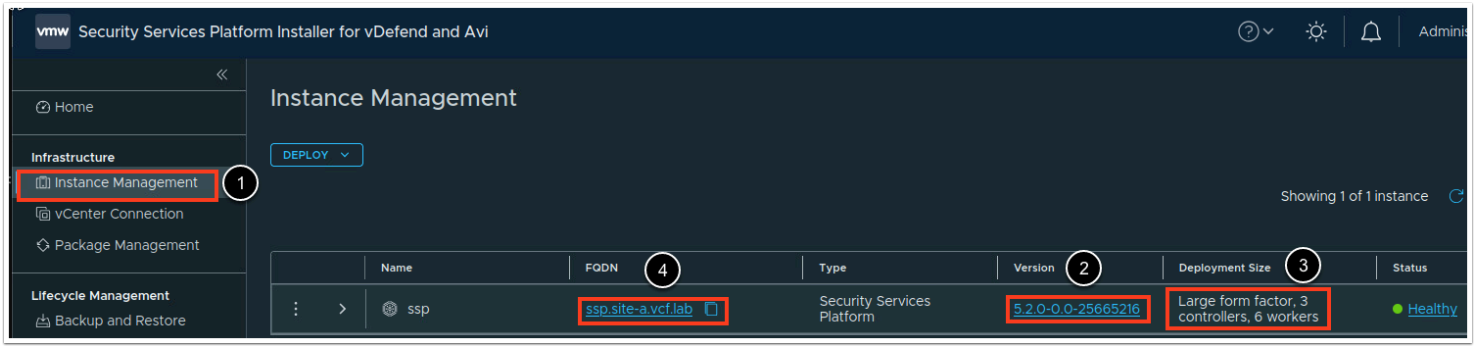
vDefend Security Services Platform Installer UI Overview



In this lab we have pre-deployed the Security Services Platform (SSP) Installer and created a SSP instance.

1. **Security Services Platform Instance (SSP):** Displays the primary operational status, network details, version, and Deployment Information (controllers & workers) of the platform instance.
2. **Installer:** Shows the network address, running software version, and current status of the platform installer utility.
3. **Infrastructure:** Configures core integrations and connections, including cluster instances, vCenter environments, and software packages.
4. **Lifecycle Management:** Handles routine platform maintenance tasks, including system backups, restores, and version upgrades.
5. **Troubleshooting:** Provides system diagnostics (Health chart & Components Grid) and support bundle generation capability.
6. **Administration and Settings:** Manages platform security parameters, including SSL/TLS certificates and user access controls.

vDefend Security Services Platform - Instance Management

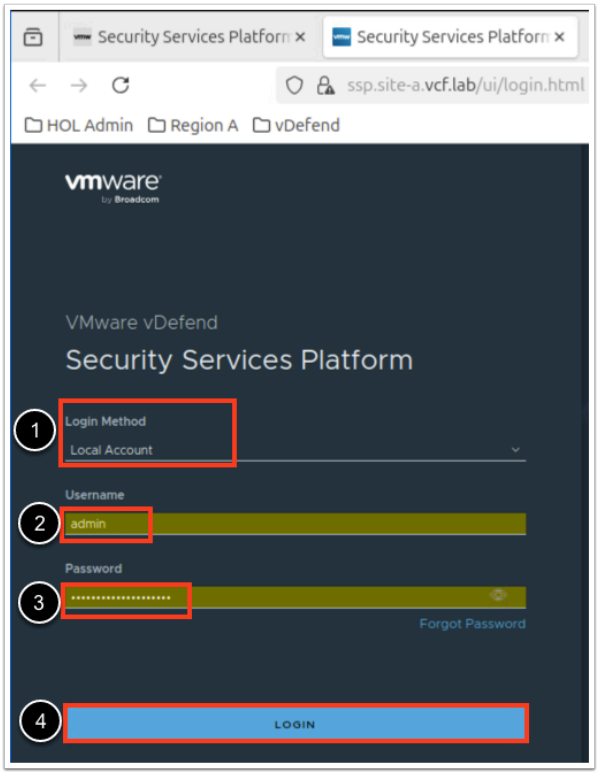


1. Click **Instance Management**
2. **Version:** This indicates the version of the SSP instance deployed.
3. **Deployment Size:** Details the form factor and Deployment Information (number of controllers & workers
4. **FQDN:** Click on the FQDN **ssp.site-a.vcf.lab** to access the pre-deployed SSP instance.

vDefend Security Services Platform Instance Hands On

Lets get some hands on exposure and experience with vDefend Security Services Platform features.

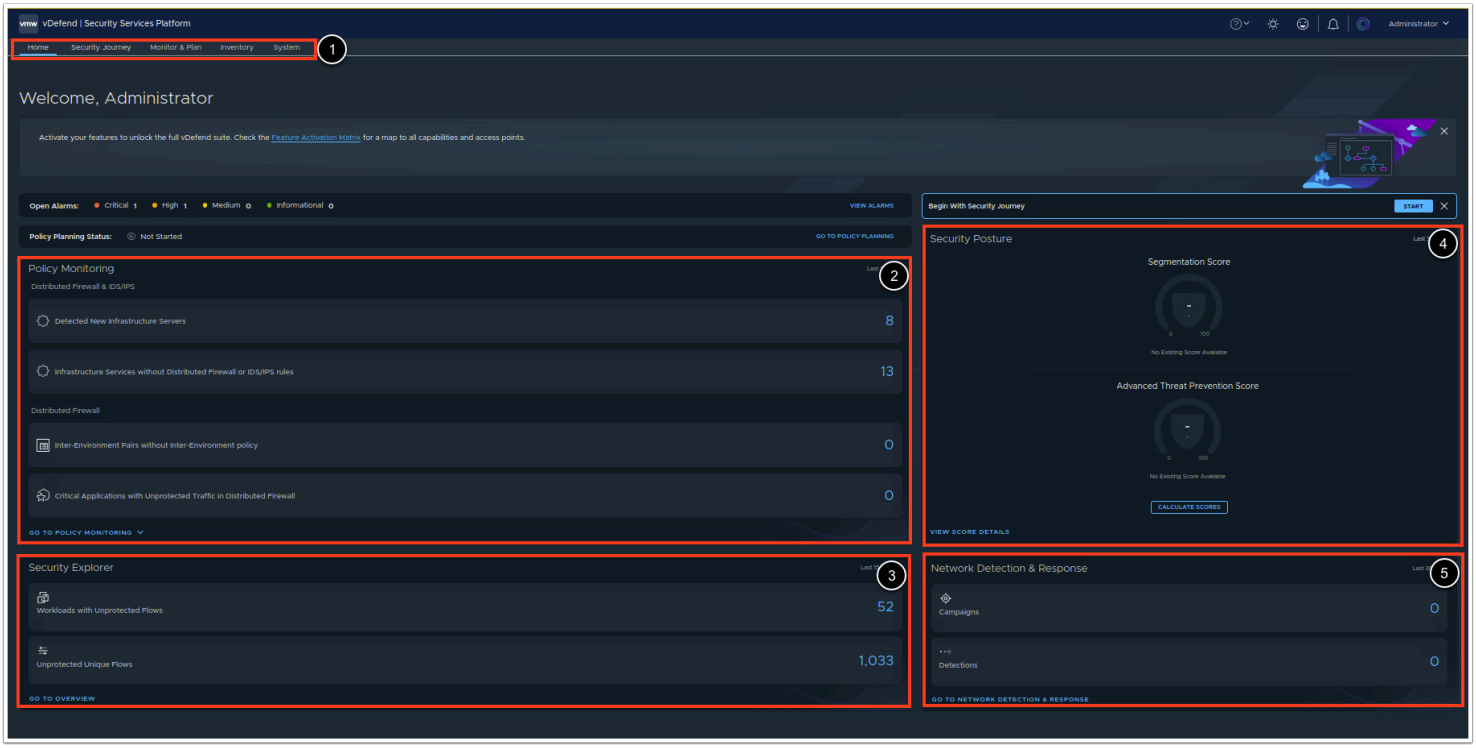
Accessing the vDefend Security Platform Instance



If you skipped the SSP Installer overview, you will need to open the Instance view from the vDefend folder. Else you should be take directly to the SSP instance without having to login and proceed to the next step.

1. **Login Method:** This should already be set to **Local Account**
2. **Username:** should be pre-populated with **admin**
3. **Password:** should be pre-populated with **VMware123!VMware123!**
4. **Click LOGIN**

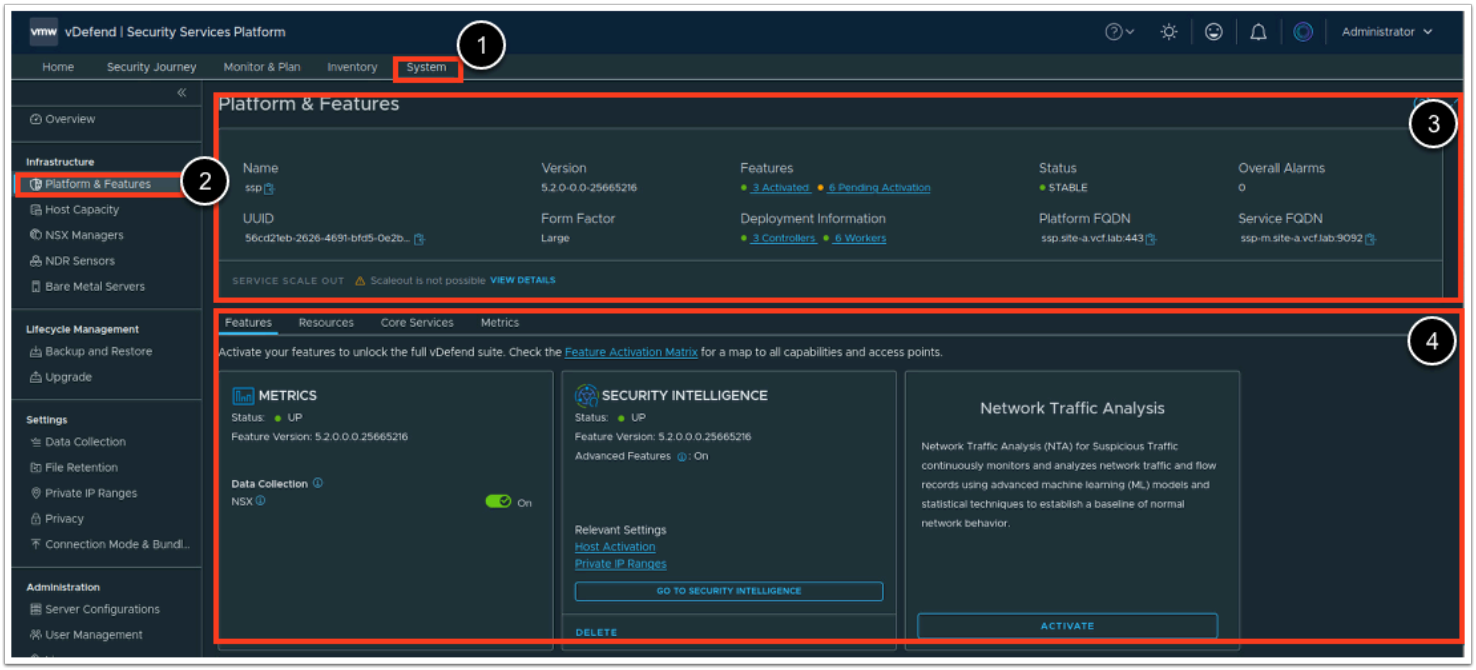
vDefend Security Services Platform Instance - Home Page



The Security Services Platform Instance landing page provides an overview of various features

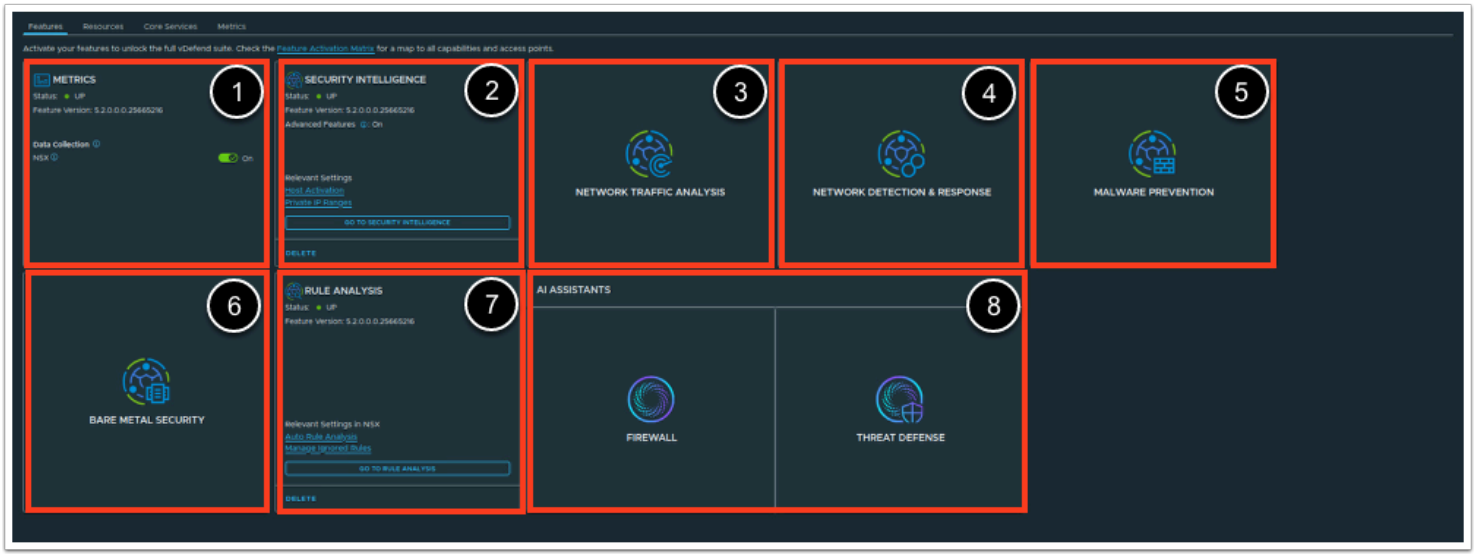
1. The **Menu bar** allows you to switch between various platform functionalities.
2. **Policy Monitoring** highlights critical applications with unprotected flows in Distributed Firewall and IDS/IPS.
3. **Security Explore** provides a summary of the detected Flows.
4. **Security Posture** allows to calculate Segmentation Score & Advanced Threat Prevention Score.
5. **Network Detection & Response** highlights campaigns & Detections.

Security Services Platform Instance - Platform and Features



1. Click **System**
2. Click **Platform & Features**
3. **Platform & Features:** Here we find the Platform instance details and summary - similar to what we saw in the Installer.
4. **Features:** Here is where we activate the features which will run on the Security Services Platform Instance. You will notice that Metrics and Security Intelligence have already been activated.

Security Services Platform Instance - Features



1. **Metrics** will be enabled by default when deploying the SSP instance.
2. **Security Intelligence** was pre-activated and from here you can access the feature dashboard or configure settings.
3. **Network Traffic Analysis** has not been activated yet, this will be performed later in the lab.
4. **Network Detection & Response** has not been activated yet, this will be performed later in the lab.
5. **Malware Prevention** has not been activated yet, this will be performed later in the lab.
6. **Bare Metal Security** has not been activated yet, this will be performed later in the lab.
7. **Rule Analysis** was pre-activated and from here you can access the feature dashboard or configure settings.
8. **AI Assistants** has not been activated yet.

Security Services Platform Instance - NSX Manager

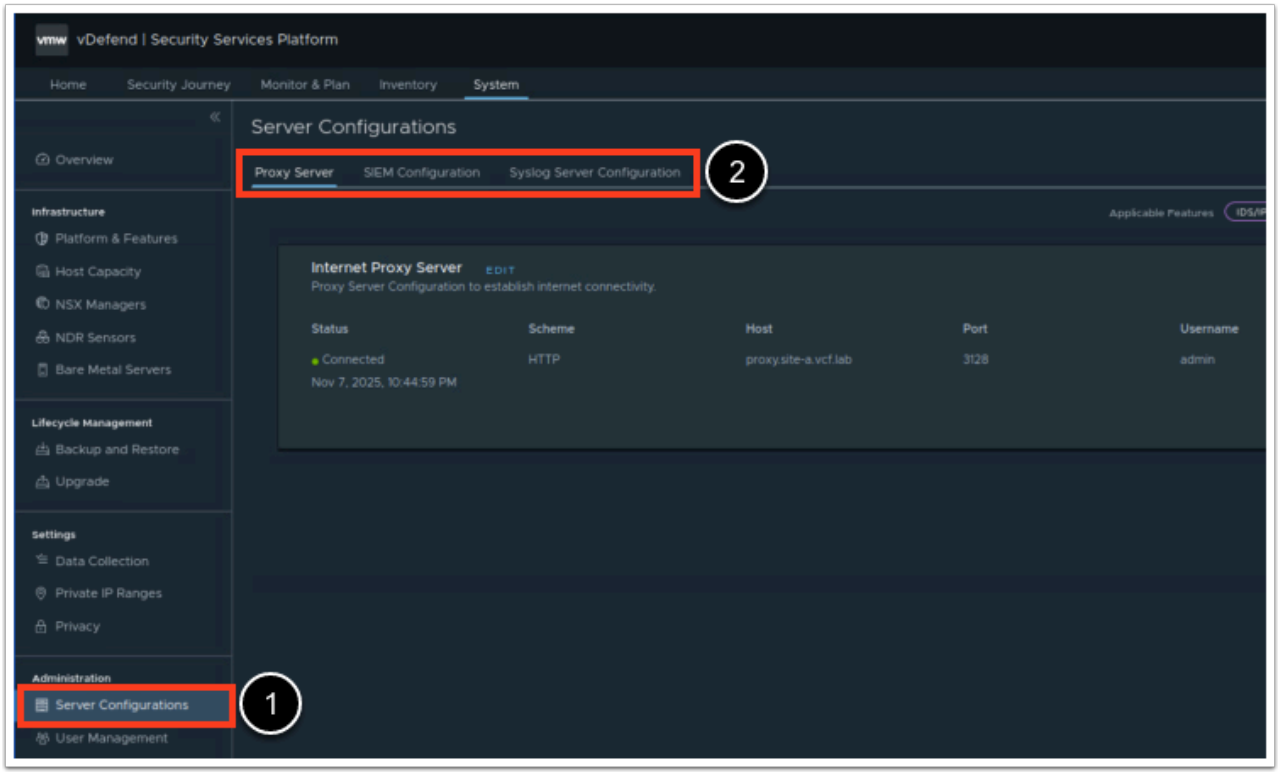


In the current version of SSP 5.2 , **Each SSP Instance** maintains a strict **1:1 relationship** with a single NSX Manager cluster.

In the current version of SSP 5.2 , **Single SSP Installer appliance** can manage **maximum 5 SSP instances**. You do not need to deploy a new installer VM every time you add another environment.

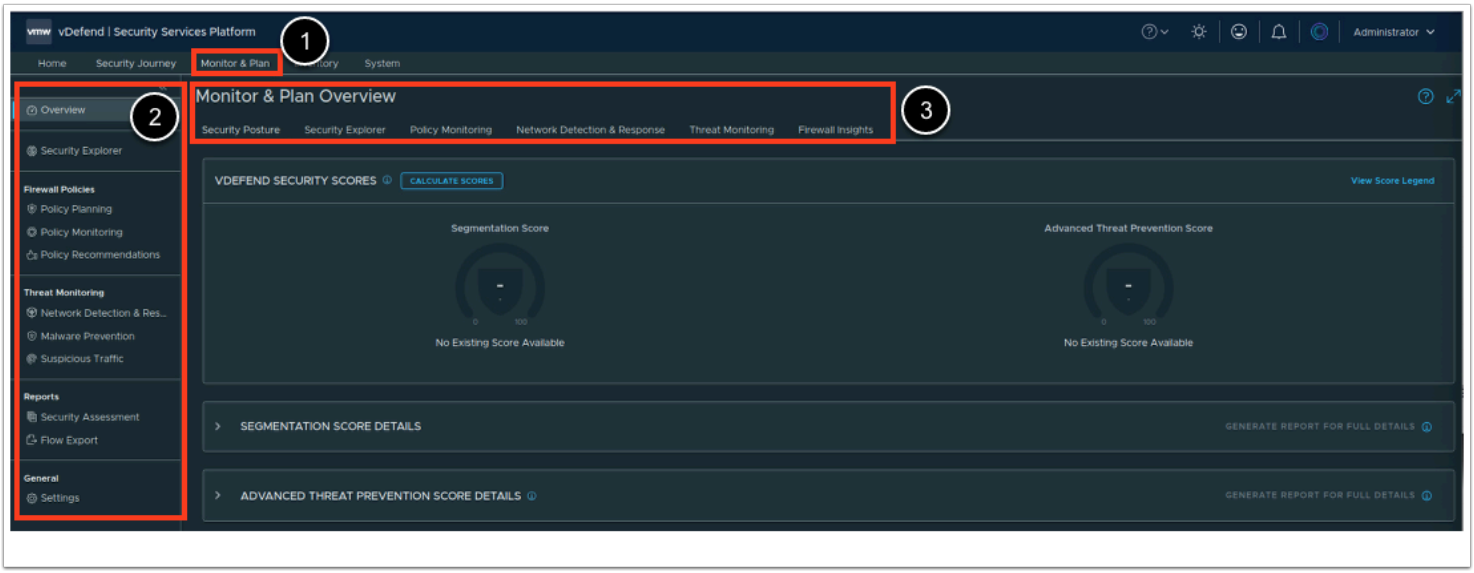
1. Click **NSX Managers**.
2. Here we see the NSX Manager that has been onboarded to this SSP instance.
3. Here we can validate the connectivity and status with the NSX Manager.

Security Services Platform Instance - Server Configurations



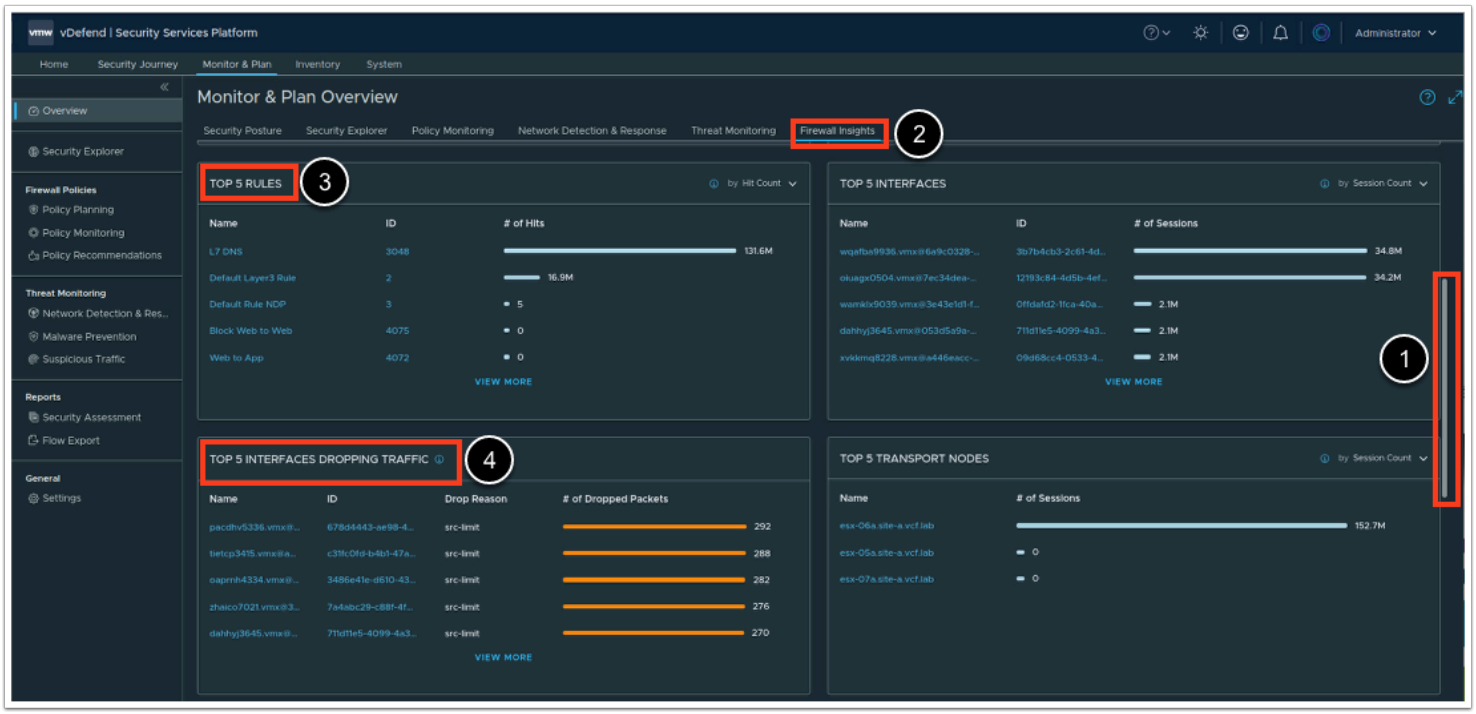
1. Click **Server Configurations.**
2. The SSP Platform supports configuring a Proxy Server, SIEM Server and Syslog Server Configurations.

Security Services Platform Instance - Monitor & Plan



1. Click **Monitor & Plan.**
2. Here are options to review features and capabilities related to Security Intelligence.
3. Here are monitoring tabs for various features.

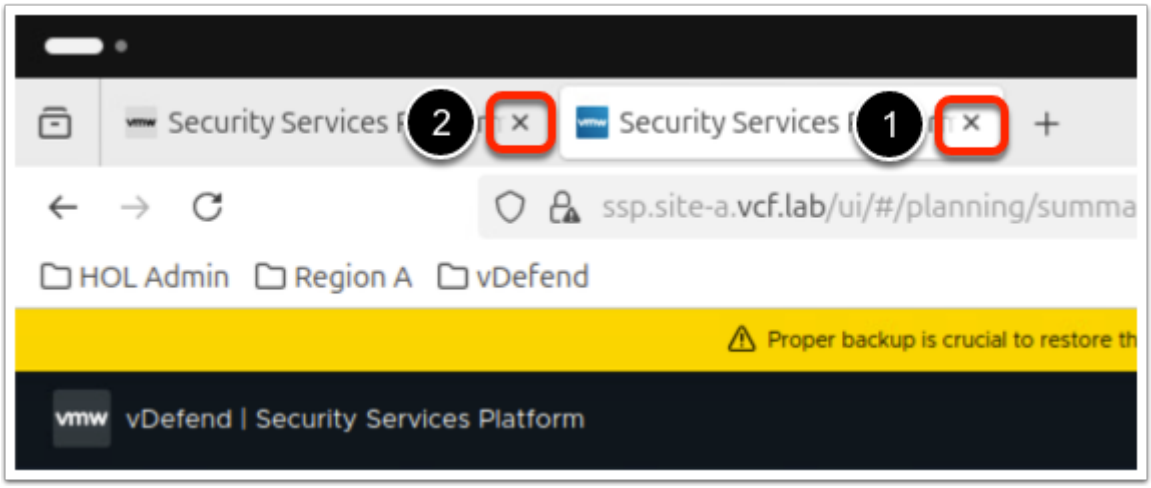
Security Services Platform Instance - Monitor & Plan, Firewall Insights



1. Click **Firewall Insights**
2. **Click and hold** the scroll bar on the right and page down to see all the feature tiles.
3. **Top 5 Rules**; Here you see the Top 5 Firewall rules by hit count.
4. **Top 5 Interfaces**; Here you see the Top 5 VM interfaces by session count.

This module provides a brief overview of these Security Services Platfrom instance dashboard, you may want click around and get familiar with other capabilities or features or continue with the next steps

Closing the browser tabs



1. Click the **x** to close the SSP Instance page.
2. Click the **x** to close the SSP Installer page.

vDefend Security Services Platform Summary

In this module, you explored the foundational concepts of the **vDefend Security Services Platform (SSP)** and gained hands-on experience navigating its deployment

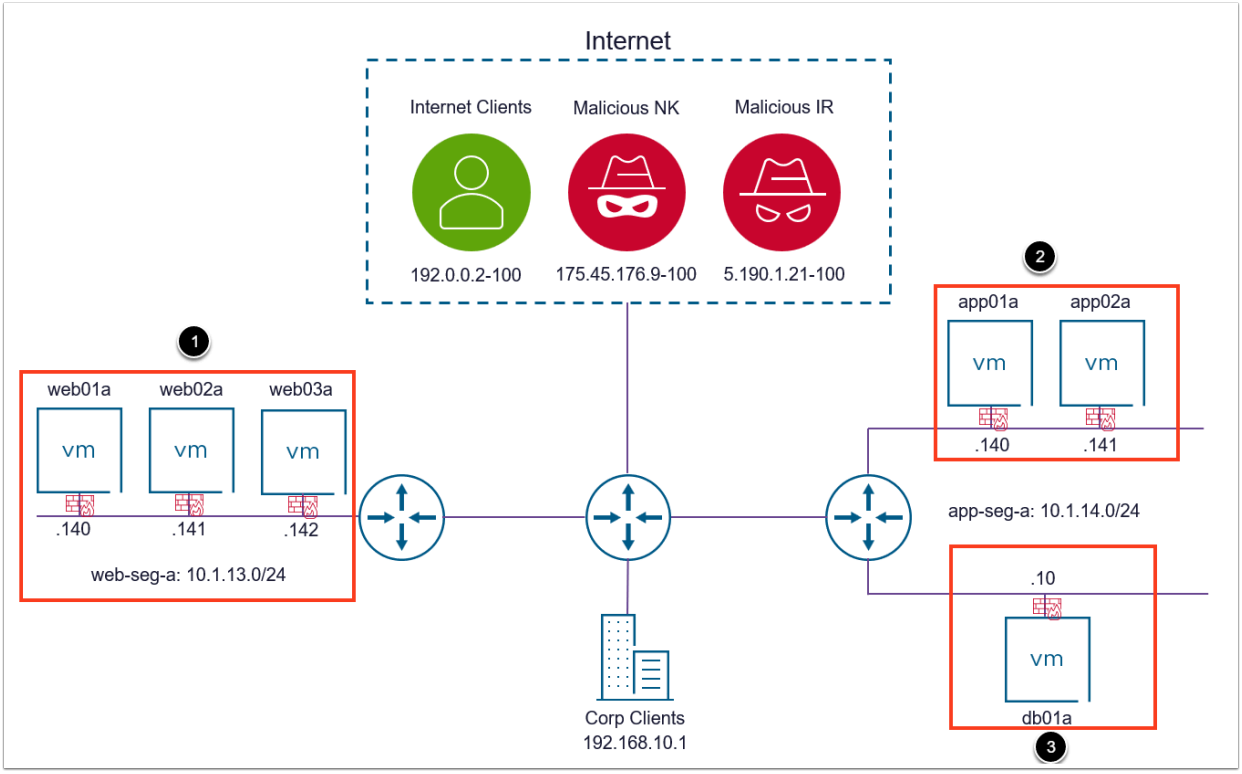
Proceed to **Module 2** to learn about Enhancing Application Security with vDefend DFW

Module 2 – Provide Enhanced Application Security with vDefend DFW (20 Min)

Introduction

This module guides you through implementing micro-segmentation across a simulated 3-tier application topology. Using VMware vDefend DFW and vDefend Security Intelligence, you will analyze network traffic flows and apply firewall policies to restrict communication exclusively to required application ports.

Logical Diagram



This lab secures a traditional 3-tier application architecture consisting of Web, Application, and Database tiers alongside internal and external clients.

- **3 Web VMs:** Receive inbound traffic from corporate and external clients on TCP 443 and communicate with Application VMs on TCP 8080.
- **2 Application VMs:** Run Tomcat, receiving traffic from Web VMs on TCP 8080 and sending queries to the Database VM on TCP 3306.
- **1 Database VM:** Hosts the MySQL database, accepting traffic exclusively from Application VMs on TCP 3306.

This lab will guide you through the process of creating and understanding VMware vDefend Distributed Firewall security policies, groups, and tags to provide a comprehensive security perimeter around and a fine-grained microsegmentation policy within the application.

> The application topology is deployed via a traffic simulation tool, which does not include a front-end web UI. You will monitor application activity using the Security Intelligence interface.

Tools Used in this lab

The following VMware tools provide operational visibility and traffic generation throughout this lab:

1. VCF Operations for Logs
2. vDefend Security Intelligence
3. VMFlowGen

VCF Operations for Logs

VCF Operations for Logs acts as a central repository for logs from various sources within the VMware Cloud Foundation environment, including vSphere, NSX, and other components. It supports content packs, which are plug-ins that provide pre-built dashboards, extracted fields, and alerts tailored for specific products or log types.

vDefend Security Intelligence

vDefend Security Intelligence is a feature hosted on the Security Services Platform, which provides visualization, analysis, and monitoring of traffic flows for complex modern applications, enabling micro-segmentation at scale.

Security Intelligence is available for ESXi-based hosts, physical server hosts, and ESX clusters that are enabled by VMware vSphere® Lifecycle Manager.

Security Intelligence provides the following functionalities:

- A graphical visualization of NSX components, such as groups, VMs, physical servers, IPs, and network traffic flows. The data used is based on network traffic flows aggregated during the specified time period. **We will use this functionality in this lab.**
- Recommendations for security policies, policy security groups, and services for applications. The recommendations assist you with the implementation of micro-segmentation at the

application level. When you implement these recommendations, you can enforce a more dynamic security policy by correlating communication patterns occurring between VMs, physical servers, and IPs in your NSX environment.

- Detection of suspicious or anomalous network behaviors in your data center network using the Suspicious Traffic feature. To filter out activities that are interesting from a security perspective, the Suspicious Traffic feature applies threat-centric detectors to the collected traffic flow data. The detection events generated by these detectors may be associated with specific techniques or tactics in the MITRE ATT&CK® Framework. If you have activated the Network Detection and Response feature, detection events are sent to the VMware vDefend® Advanced Threat Prevention cloud service for further analysis. If the cloud service determines that detection events are related, those events are correlated into a campaign organized along a timeline on the Network Detection and Response UI. Each campaign can then be further investigated by your network security team.

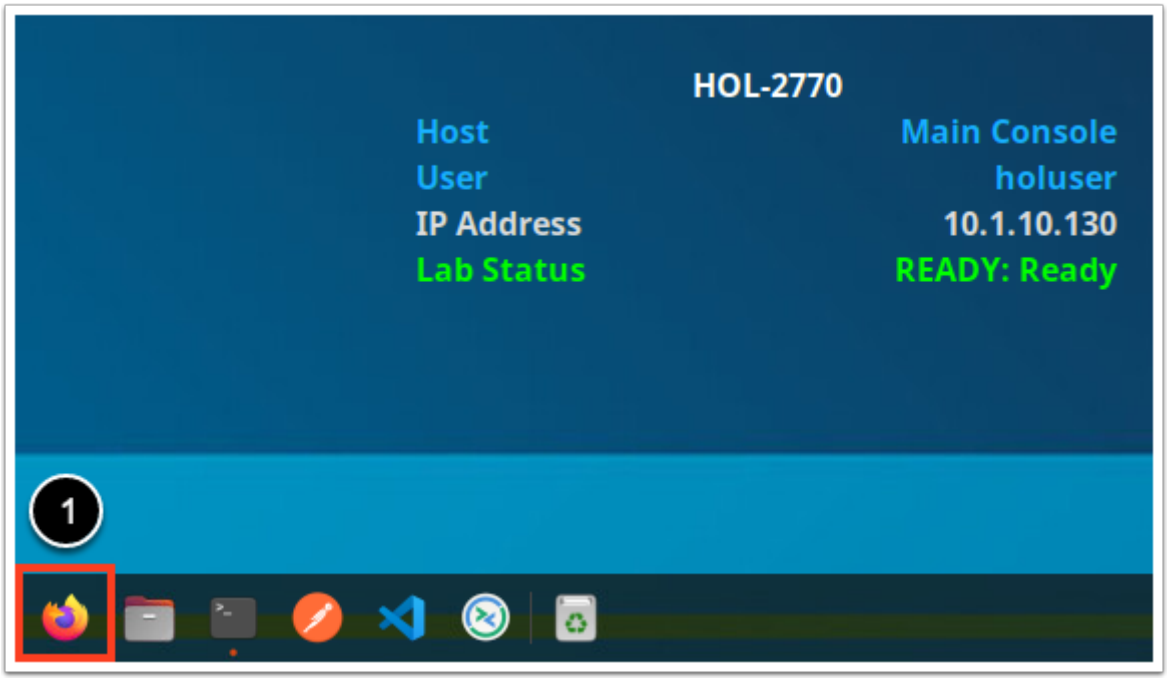
VMFlowGen

VMFlowGen is an internal tool designed to demonstrate Security Intelligence. It deploys lightweight virtual machines and generates active network flows to simulate realistic enterprise application traffic.

3-Tier Application Deployment

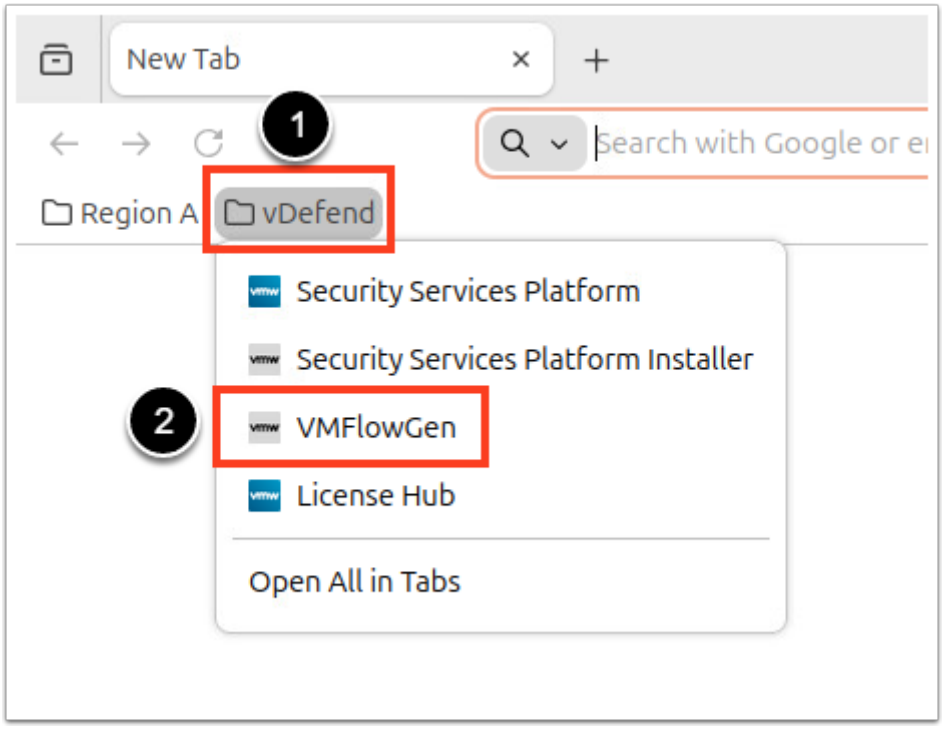
In this task you will review the current state of the lab in preperation for the application of security policies to secure the 3-Tier application.

Launch Firefox



1. Launch Firefox by clicking on the Firefox icon in the lower left-hand corner of the console.

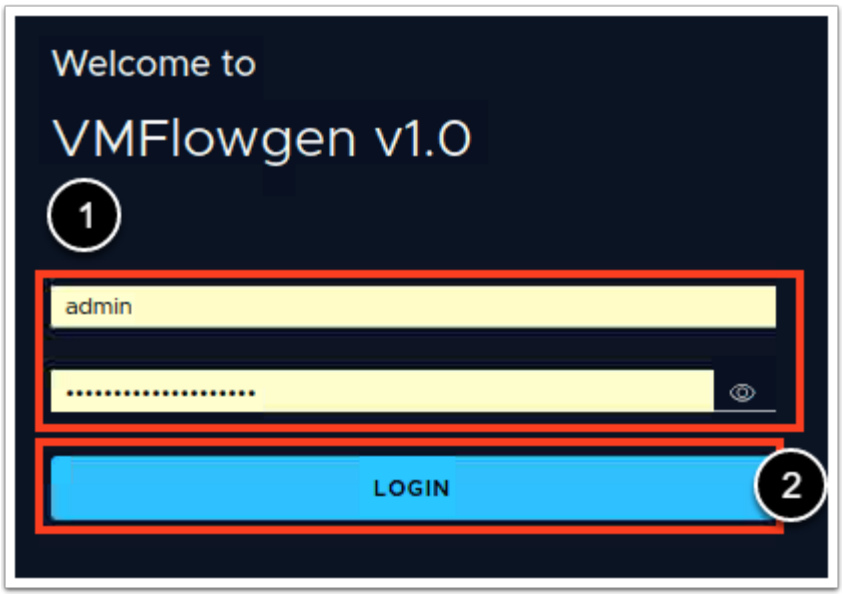
Open VMFlowgen



VMFlowGen is a tool developed by the Application Networking and Security Business Unit at Broadcom to create large-scale application and data center network traffic simulations. This tool is used strictly for testing and validation purposes and is not intended to be used outside of a lab environment.

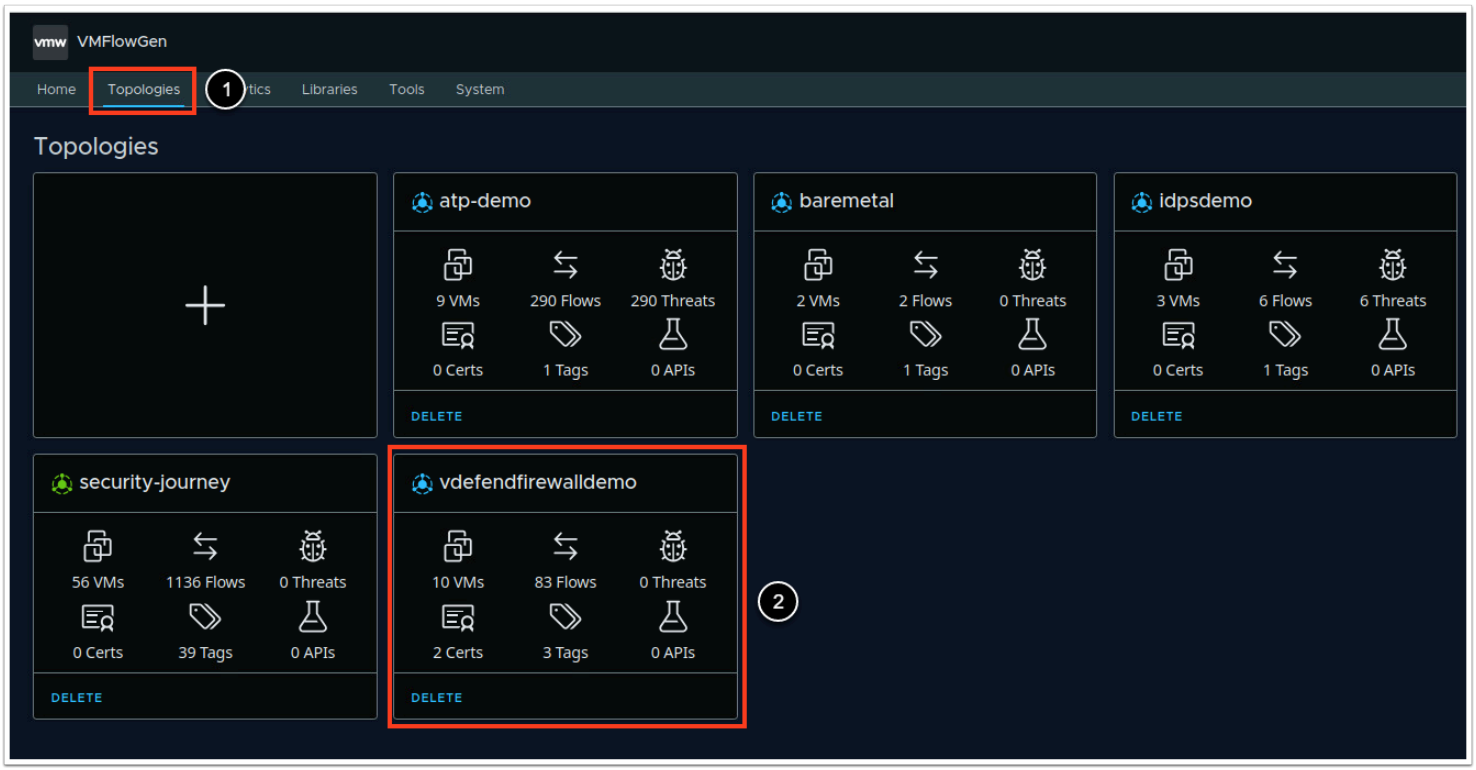
1. Select **vDefend** from the Bookmark Bar.
2. Select **VMFlowGen** from the drop-down menu.

Login to VMFlowgen



1. The username and password should already be filled in. If not use the username **admin** and the password **VMware123!VMware123!**
2. Select **LOGIN.**

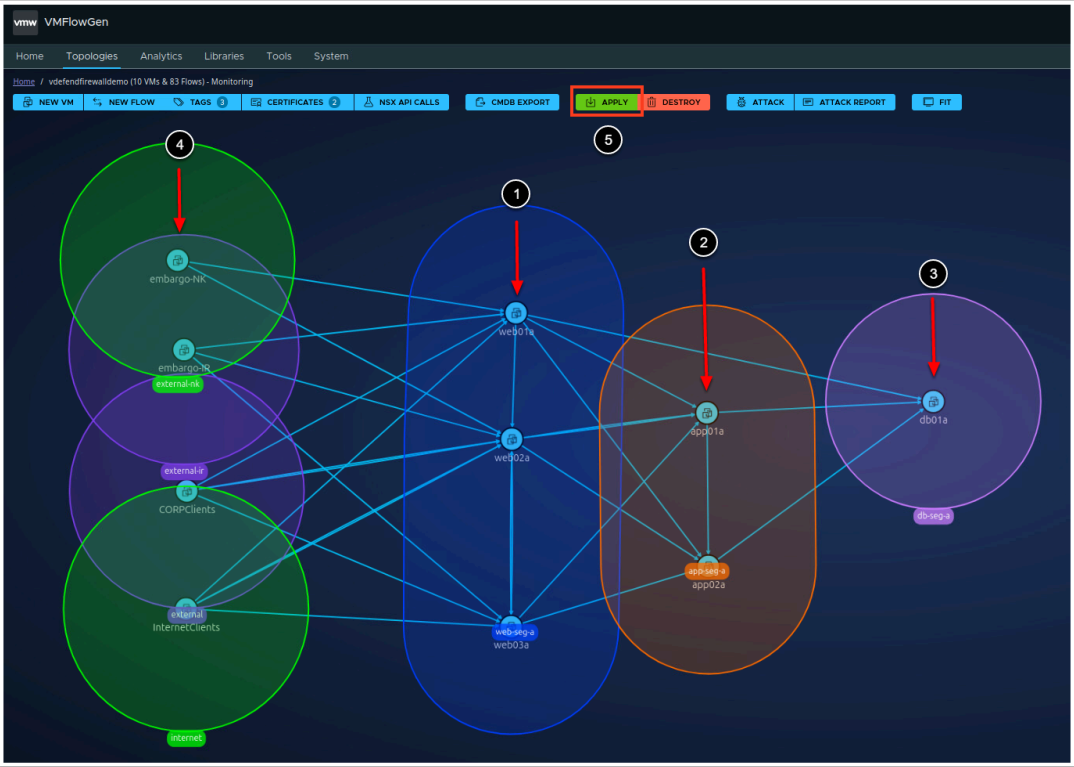
Navigate to the 3-Tier Application Topology



1. Click **Topologies.**
2. Click the **vdefendfirewalldemo** topology.

> The application is currently not deployed as indicated by the color of the symbol in the top left of the application card

Review and Deploy the Application



This is the application view of VMFlowGen, which shows the virtual machines that make up the application as indicated by the VM icons with names associated with them. Each VM has several links coming from it connecting it to another VM. These are communication links that represent a dedicated traffic flow that will be generated. As indicated by the blue color of the VM icons and links, none of these VMs have been deployed. The application is constructed as follows:

1. In the second column are the web servers (labeled **1** in the diagram). There are flows coming in from multiple sources on the left side (described in detail above).
   - Between the web servers, there is TCP 445 (SMB), TCP 80 (HTTP), and TCP 3389 (RDP) traffic simulated.
   - Outbound from the web servers, there is TCP 8080 (HTTP) and TCP 3389 (RDP) to the app servers, as well as TCP 3389 (RDP) to the db server.

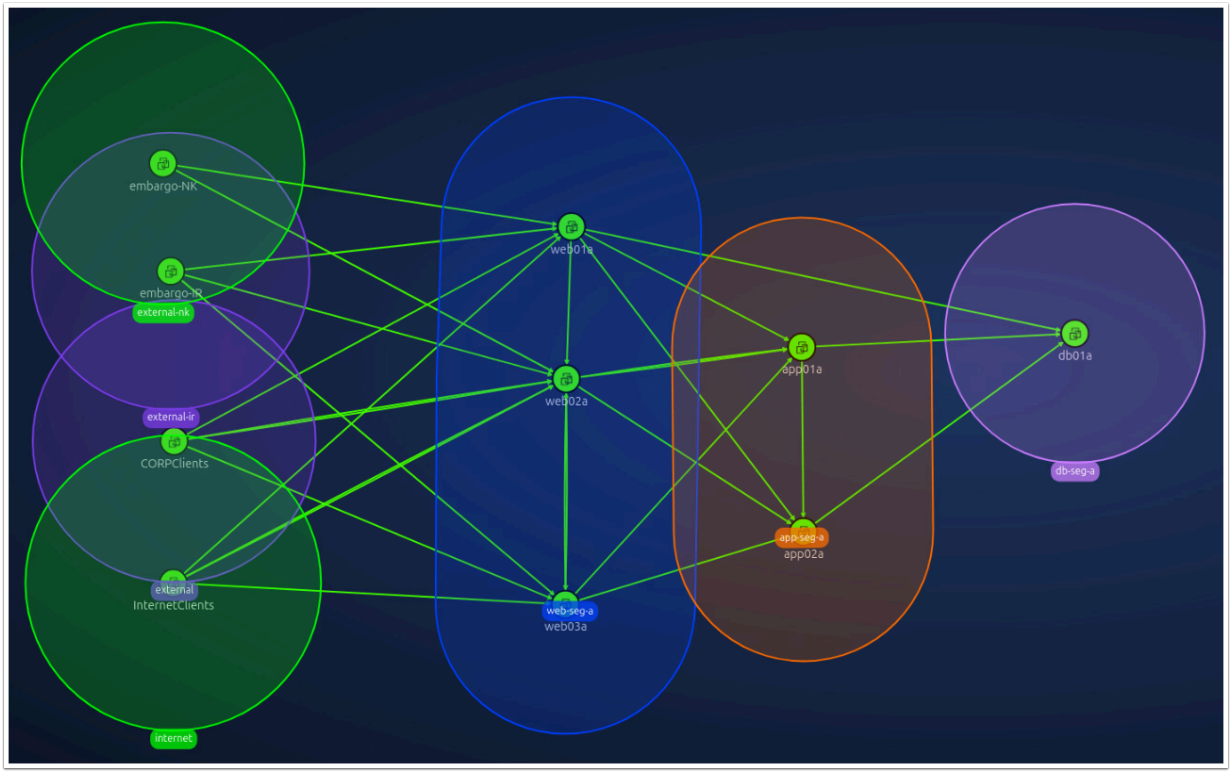
2. In the third column are the app servers (labeled **2** in the diagram). There are inbound flows from the web servers (described in detail above).
   - Between the app servers, there is TCP 22 (SSH) traffic simulated.
   - Outbound from the app servers, there is TCP 3306 (MySQL) traffic to the db server.

3. In the last column is the db server (labeled **3** in the diagram). There are inbound flows from the web and app servers (described in detail above).
4. There are VMs that are sending flows to the web servers (labeled **4** in the diagram):

- embargo-NK and embargo-IR are VMs simulating outside bad actors communicating with the web servers over TCP 443 (HTTPS).
- CORPClients are VMs simulating external corporate users accessing the web servers via TCP 443 (HTTPS) and TCP 80 (HTTP).
- InternetClients are VMs simulating internet users accessing the web servers via TCP 443 (HTTPS) and TCP 80 (HTTP).

5. Click **APPLY** (labeled **5** in the diagram) to deploy the topology.

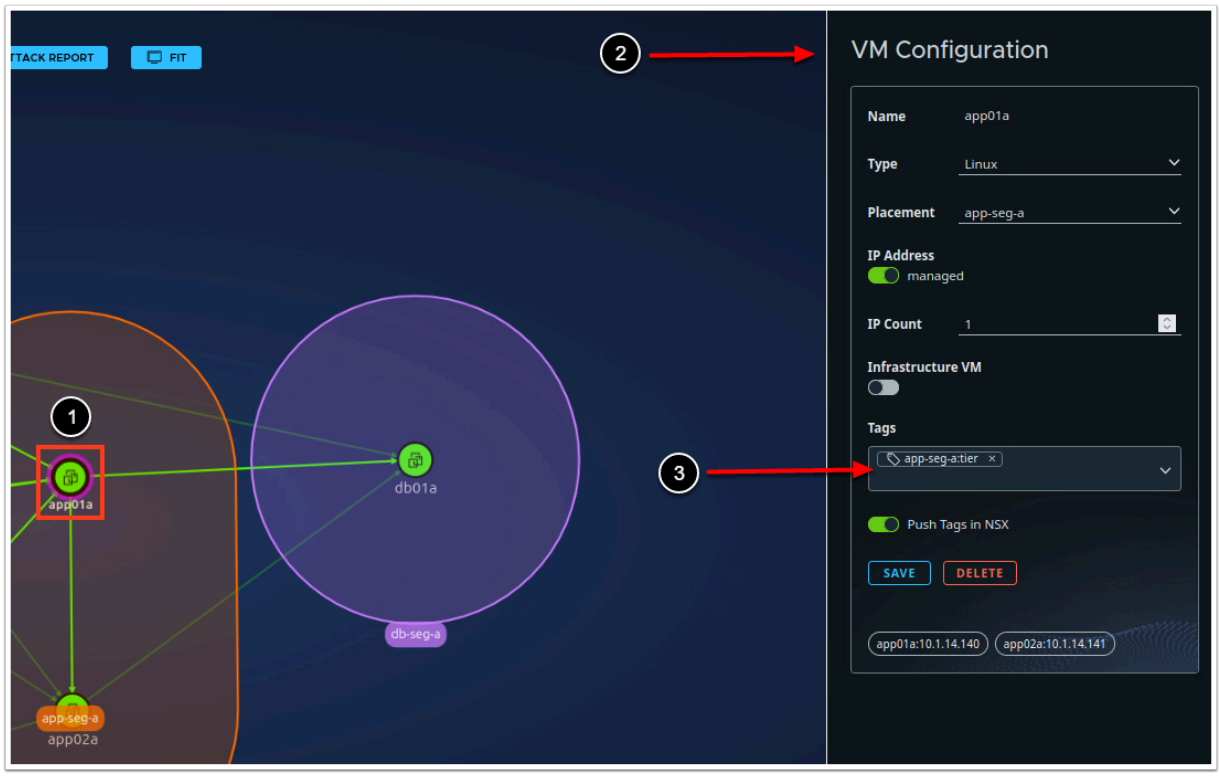
Verify the Application Deployment



The application will take between 1 to 3 minutes to fully deploy. During this time you will notice a progress bar on the top left of the canvas. If the application fails to deploy, click APPLY again.

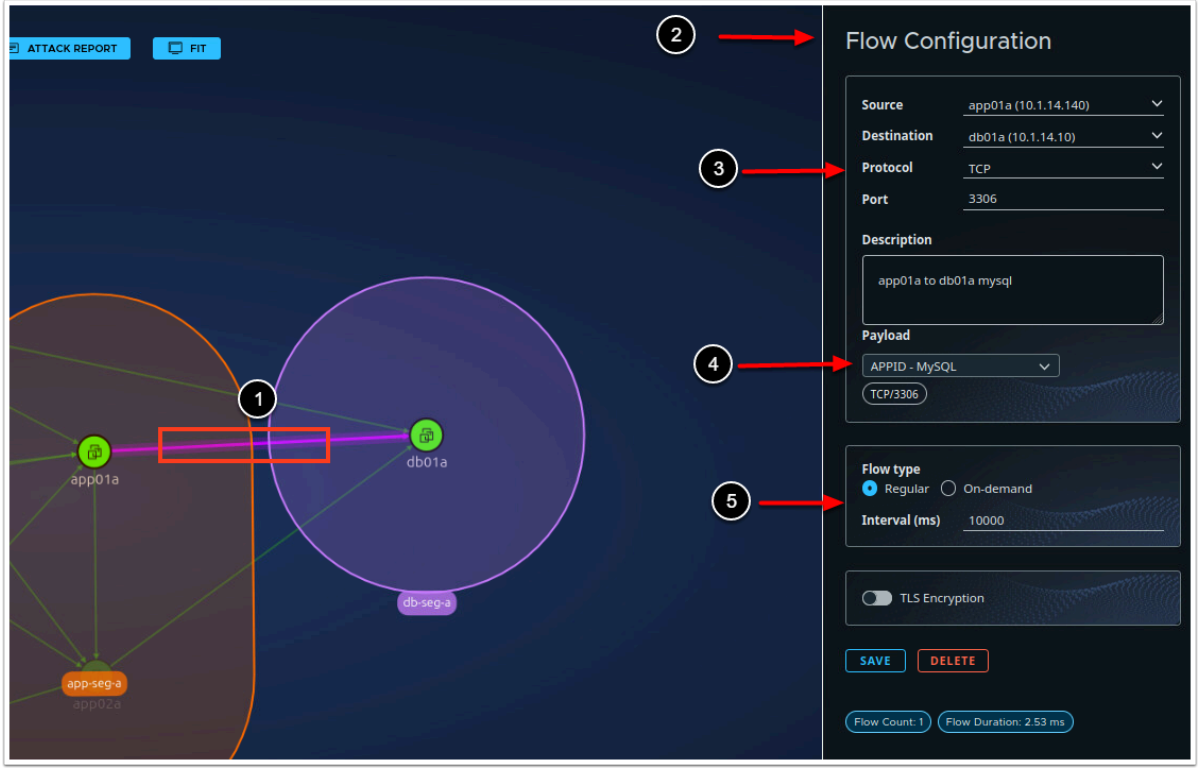
After the application is finished being deployed take note of the VM icons and flow arrows are now showing green. This indicates the application has successfully deployed. Once everything is showing green as indicated in the image above move onto the next step.

Review VM Information



1. Click on the **app01a** VM.
2. Take note of the VM configuration on the right hand side of the page.
3. Take special note of the **Tags** section and the **app-seg-a:tier** tag that is applied. This will be important in the next task.

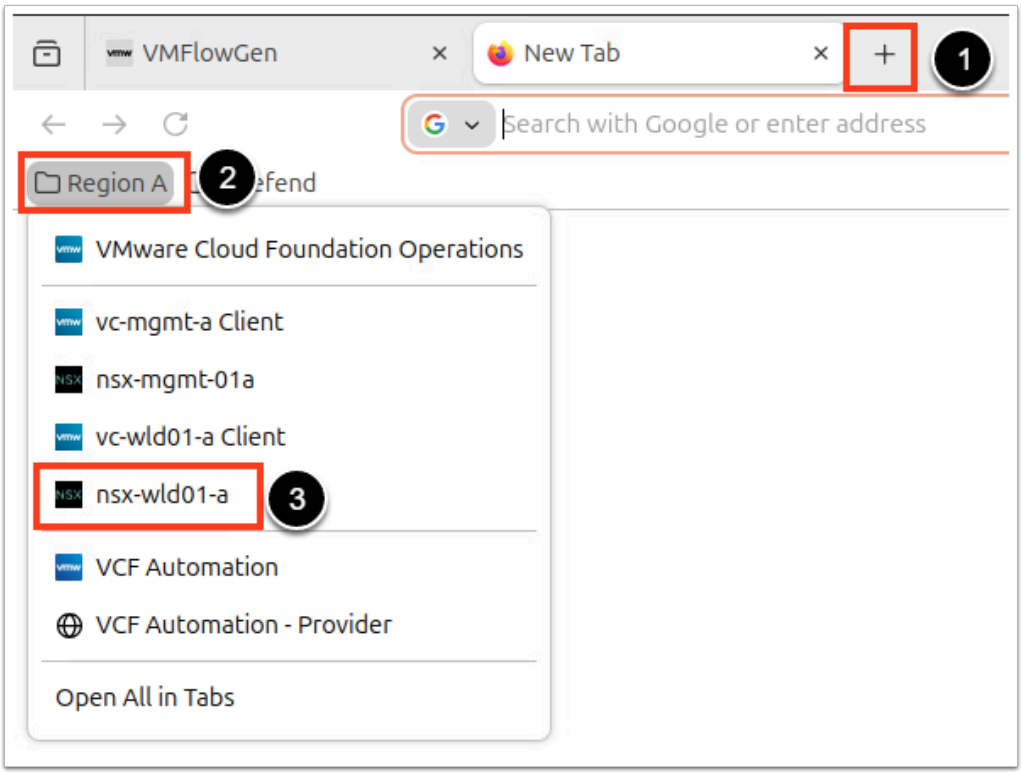
Review Flow Information



1. Click on the arrow between app01a and db01a.
2. Take note of the flow configuration on the right hand side of the page.
3. The top of the first section describes the 4 tuple characteristics of the flow.
4. The bottom of the first section (Payload) provides the Layer-7 App-ID for the flow which will be used in Module 3.
5. The second section details the frequency when the flow is generated.

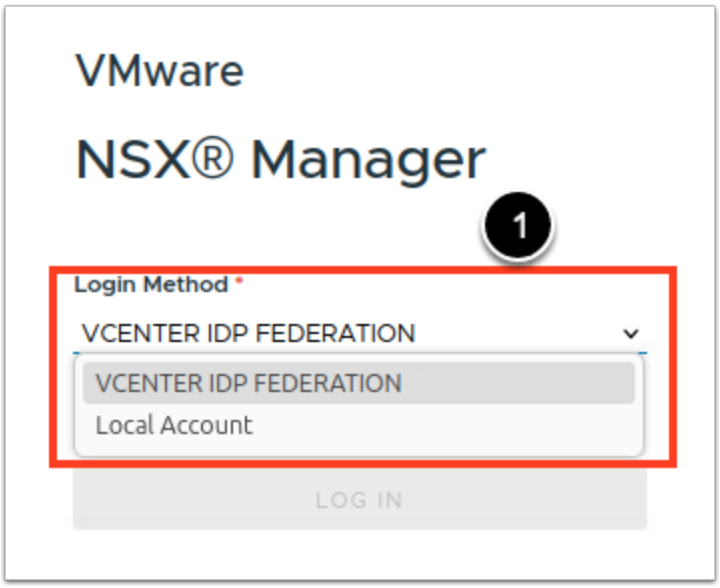
The next step will take you into the NSX manager, if you like take a few moments to familiarize yourself with the rest of the topology.

Log into NSX Manager and Review the Existing Topology



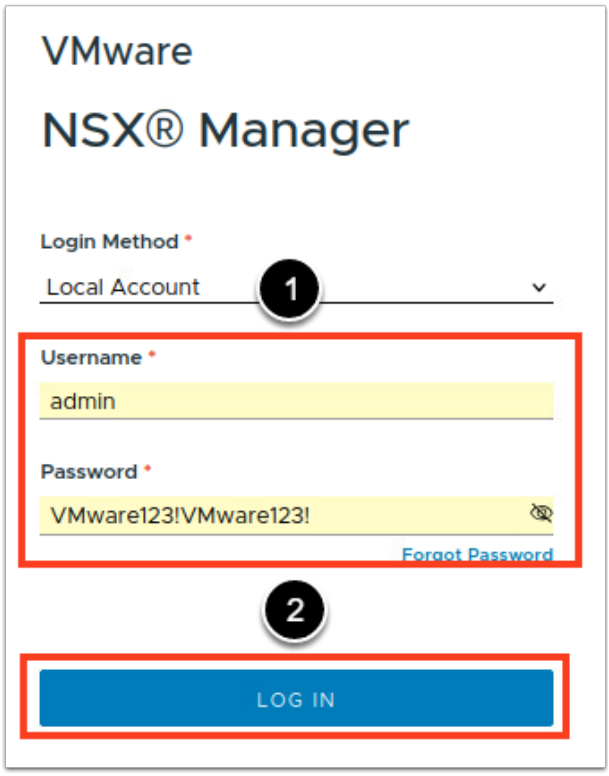
1. Click the **+** to open a new tab in Firefox.
1. Select **Region A** from the Bookmark Bar.
2. Select **nsx-wld01-a** from the drop-down menu.

Select Local Account



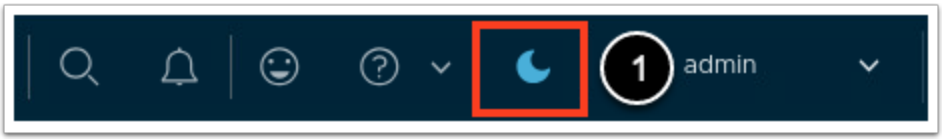
1. Click the **Login Method** drop down menu and select **Local Account.**

Login



1. Verify the Username and Password forms have been filled out. If not use the username **admin** and the password **VMware123!VMware123!**
2. Click **LOG IN.**

(Optional) Switch to Dark Mode

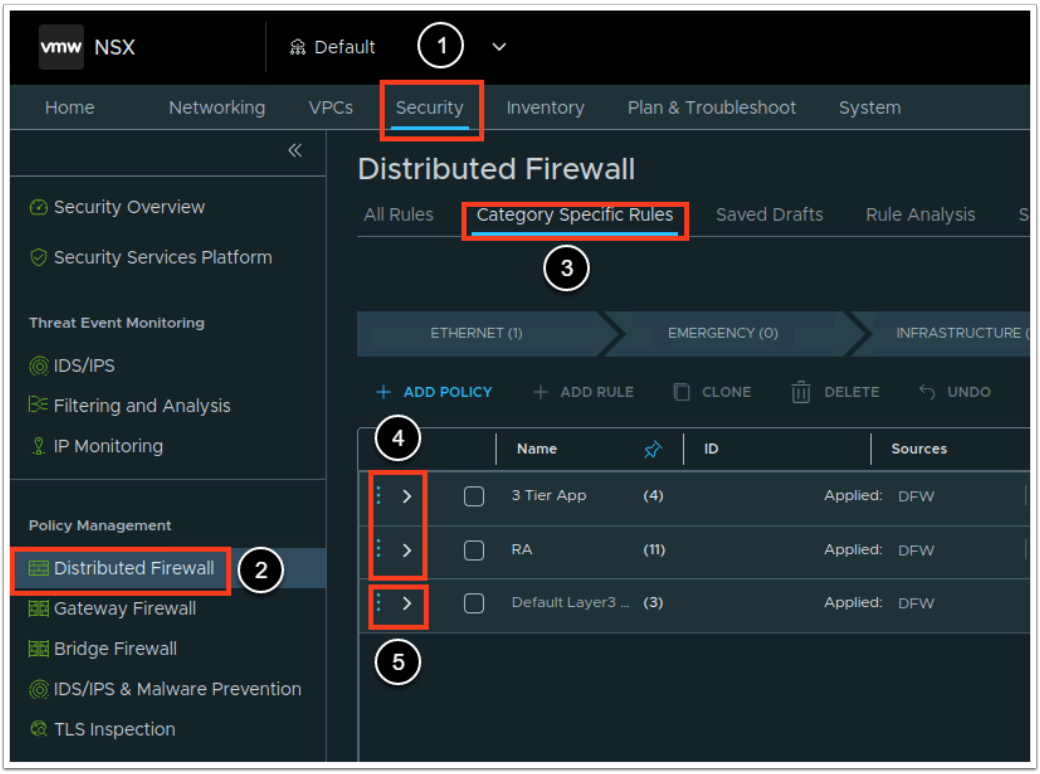


The default mode in the lab is Light Mode, if you prefer to work in Dark mode take the following action, otherwise continue to the next step.

1. Click the **Moon** icon on the top right of the screen to enable Dark mode.

> This manual has been created using Dark mode and all screen caps going forward will be using Dark mode.

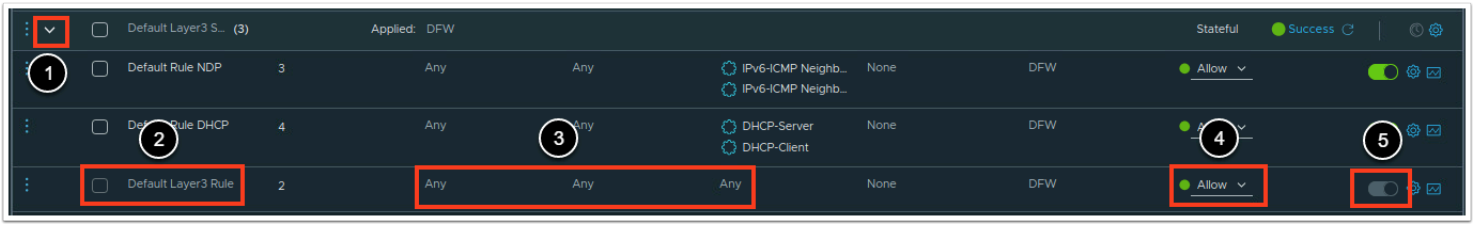
View Current Distributed Firewall Configuration



The first security policy to review is the default configuration of the Distributed Firewall.

1. Click **Security** on the top menu.
2. Click **Distributed Firewall** in the left vertical menu.
3. Ensure that **Category Specific Rules** is selected for the Distributed Firewall menu.
4. Ensure that both the **3 Tier App** and **RA** policies are collapsed as shown.
5. Expand the **Default Layer3 Section** security policy by clicking the arrow.

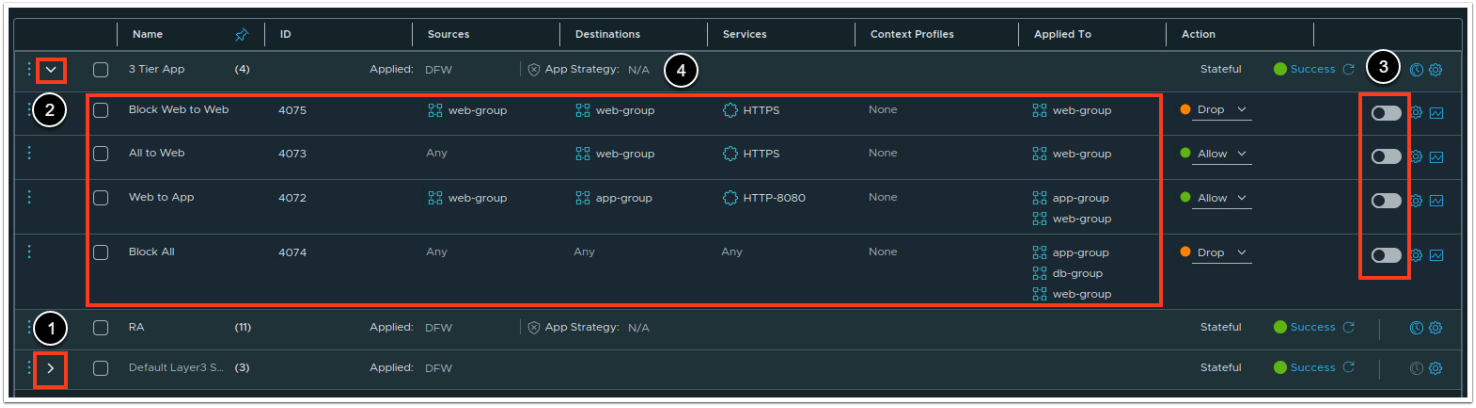
View the Default DFW Rule



1. Expand the **Default Layer3 Section** policy by clicking the arrow.

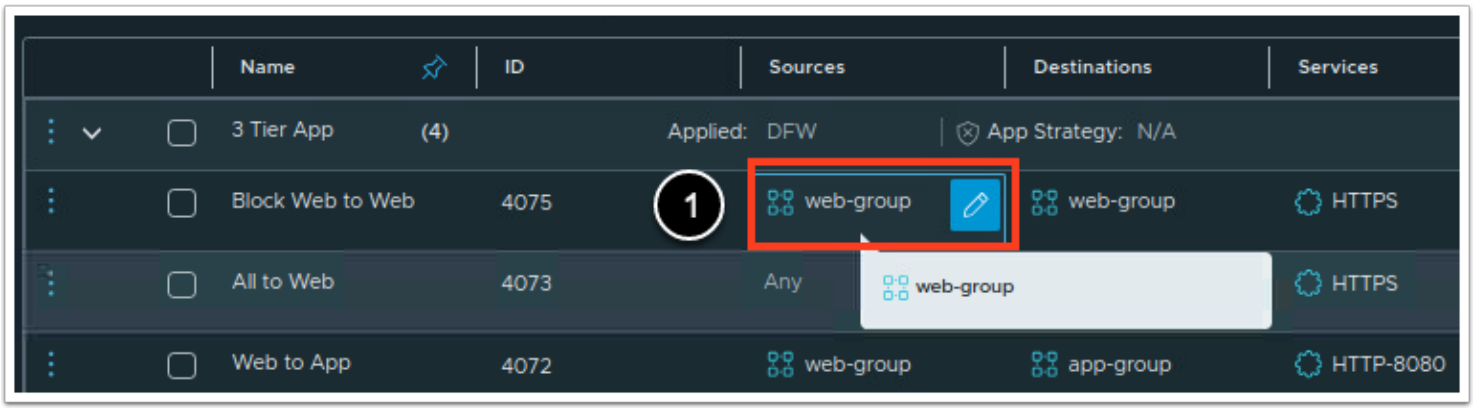
2. This is the Default Rule for the Distributed Firewall. You cannot place any policies or rules below it, so it will determine the DFW default action.
3. The Source, Destination, and Services for this rule are "Any". These fields cannot be changed.
4. The Action is currently set to **Allow** on the default rule. Any traffic flows not matching a rule above the default rule will be processed by the default rule and will be allowed through the distributed firewall.
5. The rule status toggle button is set permanently to an enabled state. It cannot be changed to disabled.

View the Pre-Configured 3 Tier App Policy



1. Collapse the **Default Layer3 Section** policy by clicking the arrow.
2. Expand the **3 Tier App** security policy by clicking the arrow.
3. The rules for the 3 Tier App are built in this section.
   - There are four rules currently configured, which makes this policy incomplete. You will create the final rule and modify the rule set to provide a more refined policy for this application.
4. The rule status toggle button is currently disabled. This means these rules are not active, so none of the traffic flowing through the DFW is currently processed by them.

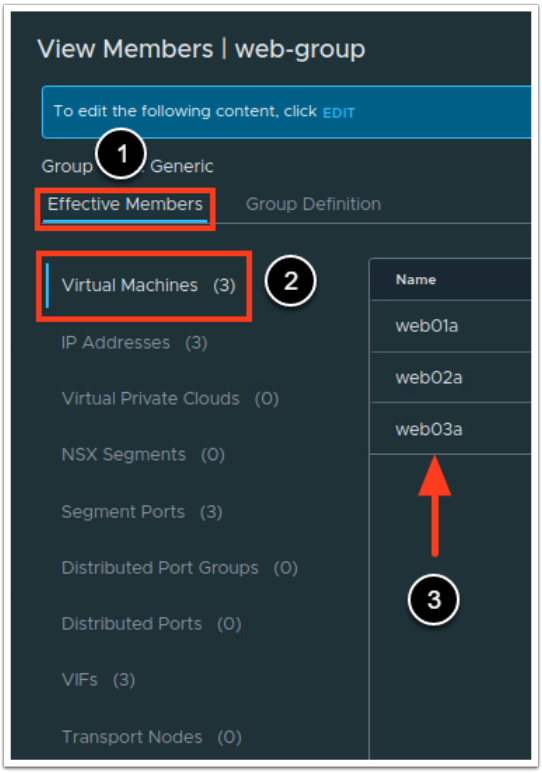
Review Security Group Configuration



When vDefend Distributed Firewall rules are created they rely on groups to identify the source, destination, and scope (Applied To) for that rule. Groups configuration can be checked with the distributed firewall section by clicking on the group you wish to inspect.

1. Click on **web-group** in the Sources column of the first rule in the policy.

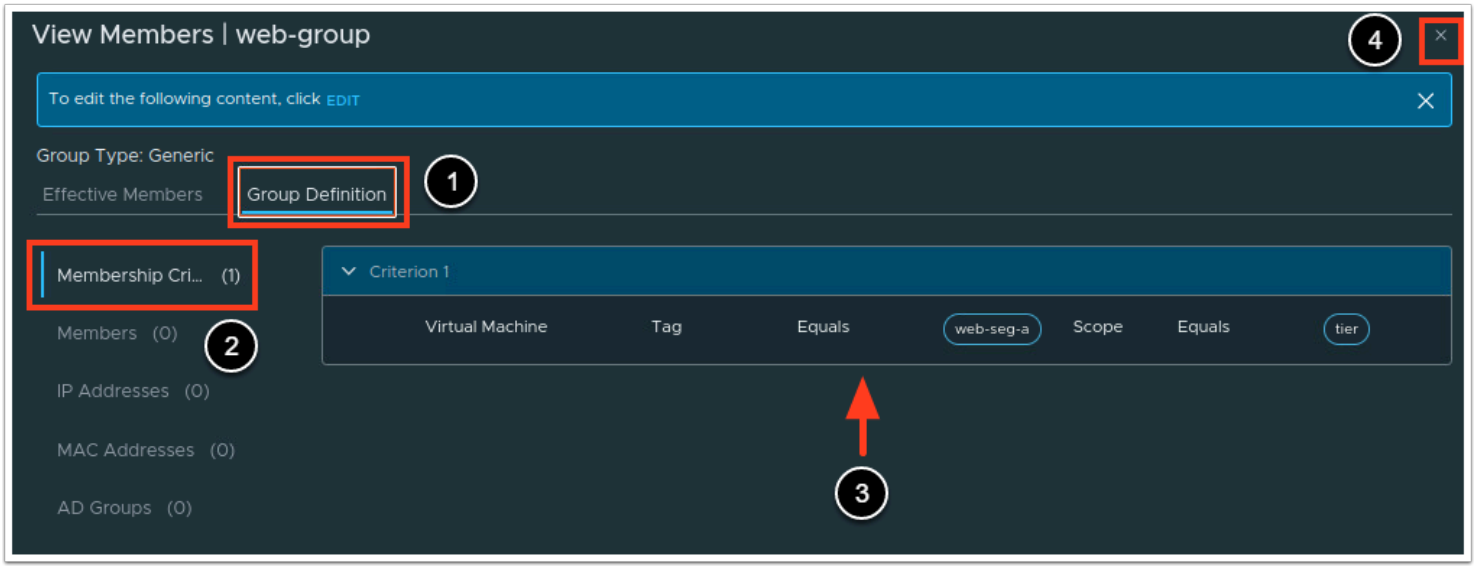
Review the Active Group Members



The Effective Members page shows all of the NSX objects that are part of the security group. This will list VMs and their respective IPs, as well as other objects that are part of the group for the purpose of defining the points of security enforcement.

1. Click on **Effective Members** if not already selected.
2. Click on **Virtual Machines** if not already selected.
3. Take note of the 3 VMs that are part of the group.

Review the Group Definition

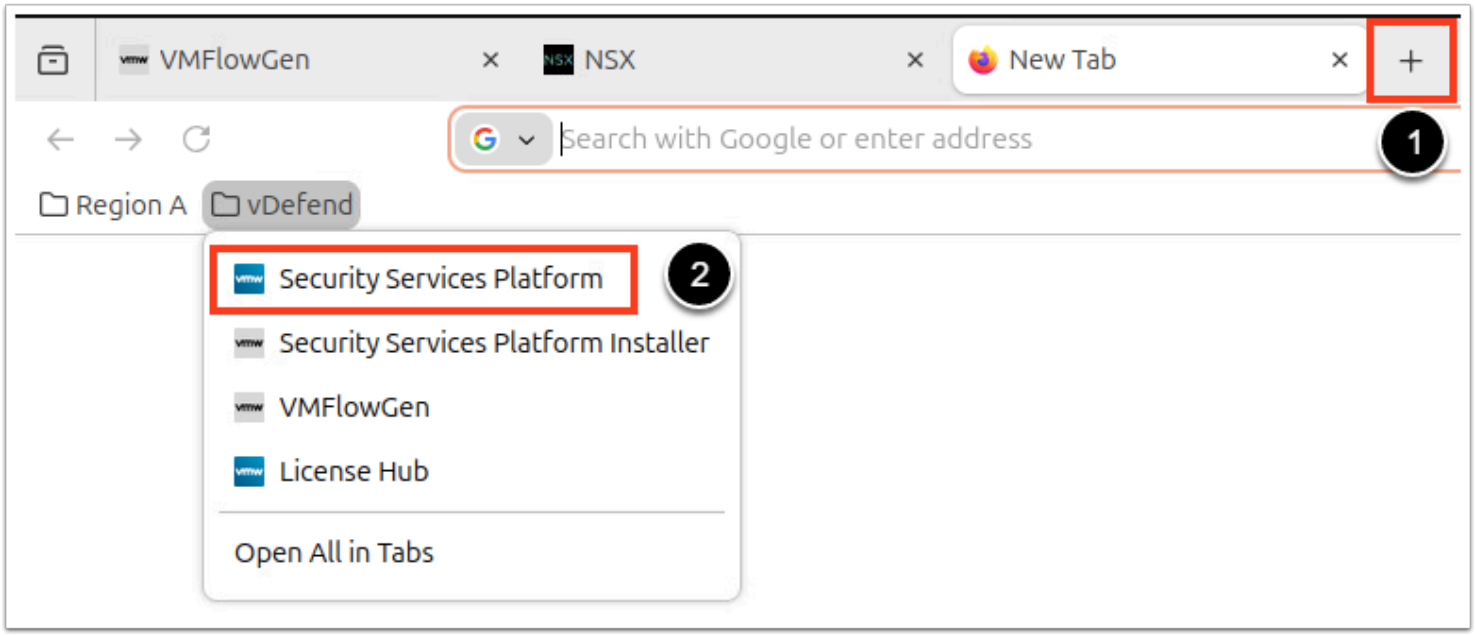


The Group Definition page shows the criteria used to include objects in the security group; the categories on the left-hand side denote the different types of inclusion methods that can be used. For this exercise, you will only be using Dynamic Membership Criteria.

1. Click on **Group Definition** on the menu bar.
2. Click on **Membership Criteria** if not already selected.
3. The dynamic criteria used to define object membership are shown on the right.
4. Click the **X** at the top right to exit out of this view.

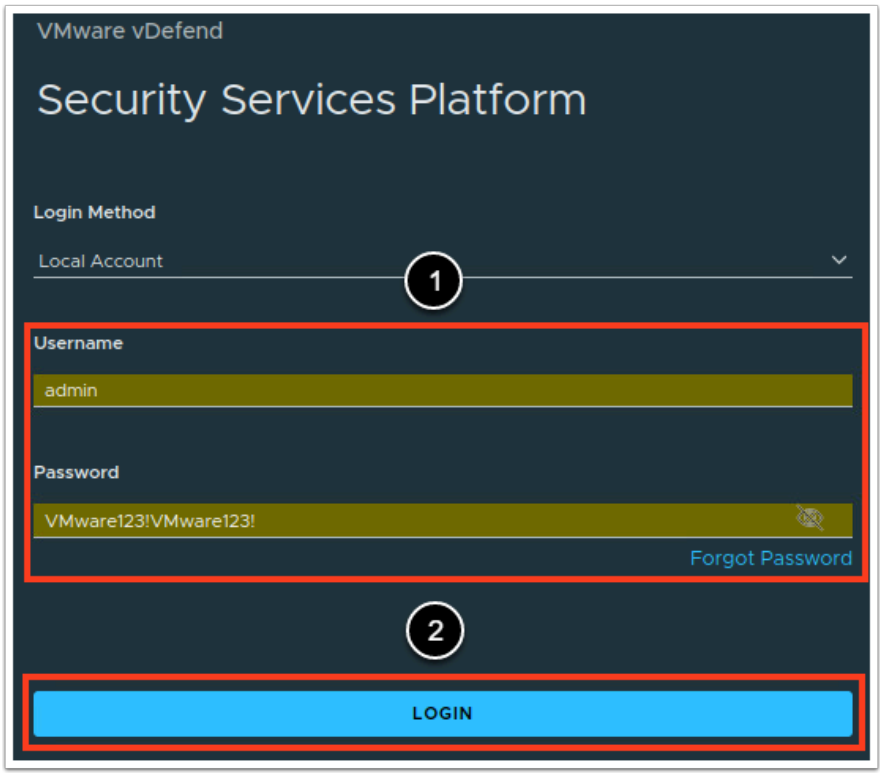
> The next part of this section will be moving Security Intelligence, the primary security monitoring tool for vDefend, before you do so feel free to look at other groups to see how they are constructed and what their members are.

Log Into the Security Services Platform



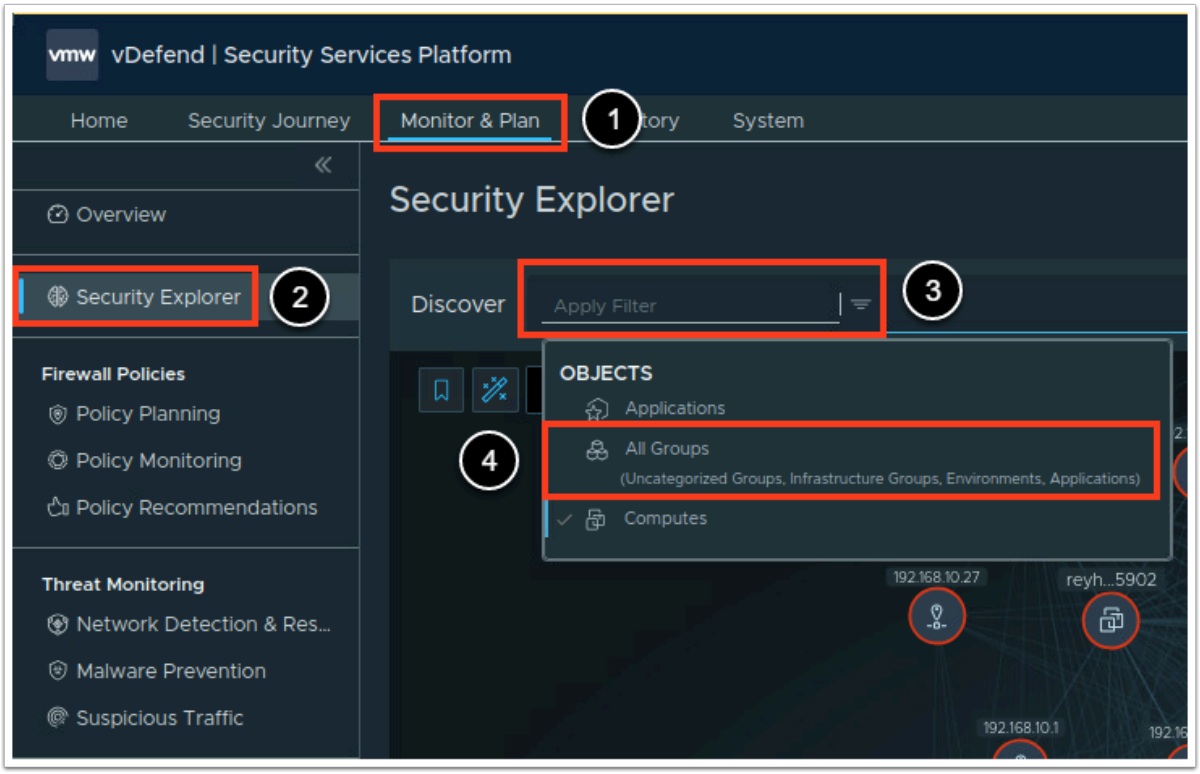
1. Click the vDefend bookmark folder
2. Click Security Services Platform

Log in



1. Verify the Username and Password forms have been filled out. If not use the username **admin** and the password **VMware123!VMware123!**
2. Click **LOGIN**

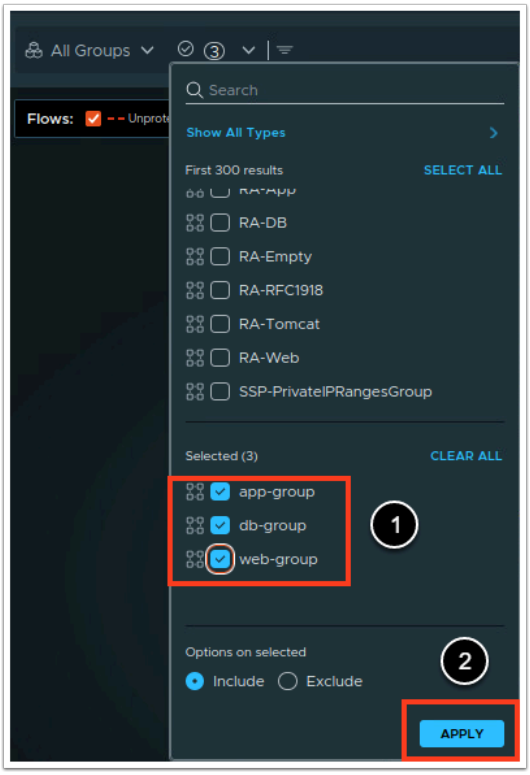
Navigate to Security Intelligence



Security Intelligence is the Flow and Application monitoring tool used within vDefend. It provides a graphical representation of all observed flows and how they interact with the different applications deployed within the environment. You will use Security intelligence throughout this lab to observe the changes in the application's behavior as you develop the security policy applied to that application.

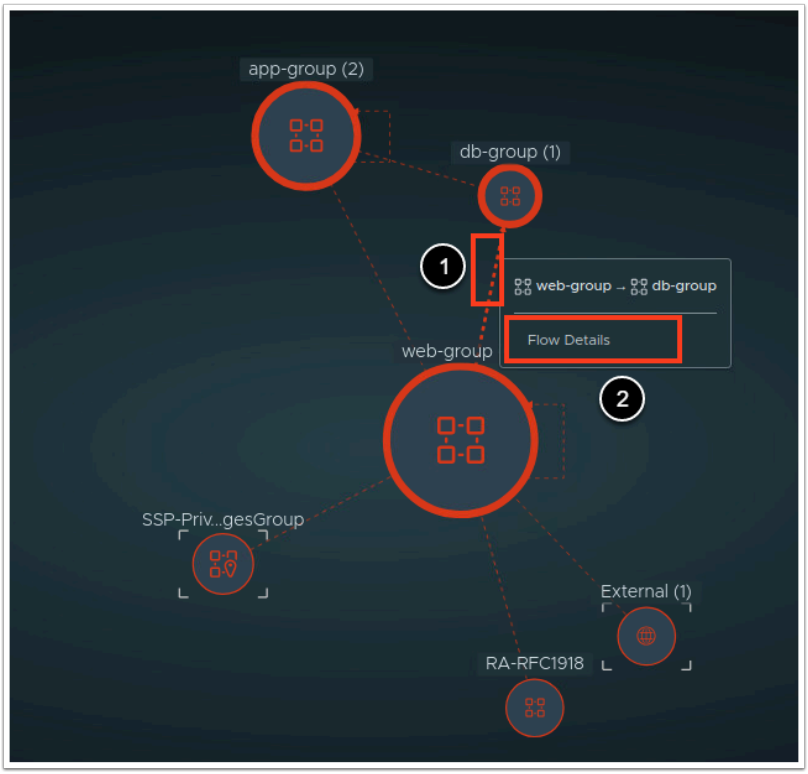
1. Click **Monitor & Plan**
2. Click **Security Explorer**
3. Click on the **Discover** drop down menu
4. Select **All Groups**

Select the 3 Tier App Groups



1. From the list on the top select the following groups:
   - app-group
   - db-group
   - web-group (you will need to scroll down to see this group)
2. Click **APPLY**

Review the Discovered Topology

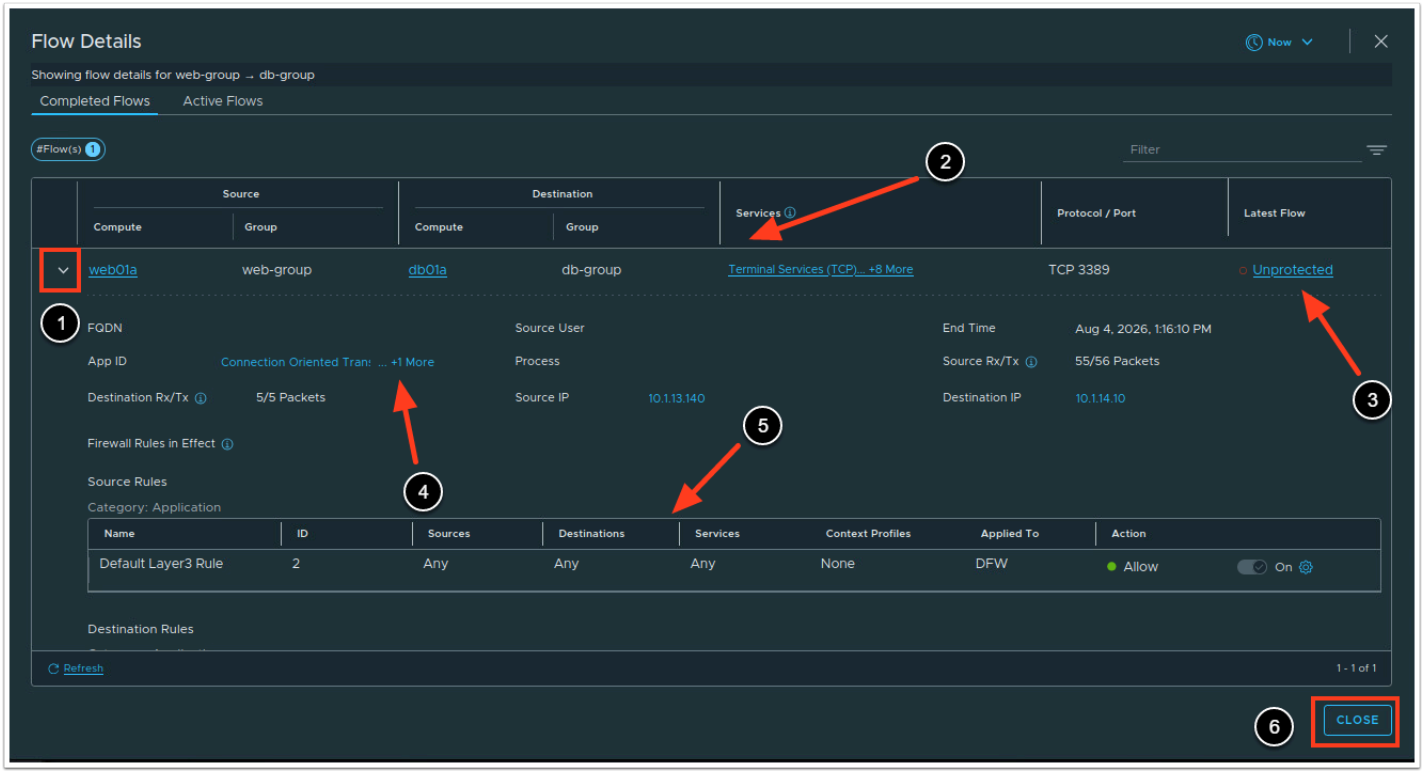


The application view shows the groups selected any other group that interacts with that group. If there isn't a defined group in NSX for the Servers it will default to one of a few different categories such as Internal or External. Each object can be analyzed with this view. In the top right hand corner you will see a key for the color of the flows. All flows at this stage will be unprotected. This means that the flows only have the default rule applied to them.

1. Right Click on the dashed line between **web-group** and **db-group**
2. In the pop-up click **Flow Details**

> The icons shown in the lab may differ from what is seen in this manual. If that is the case please make sure you are clicking on the correct icon.

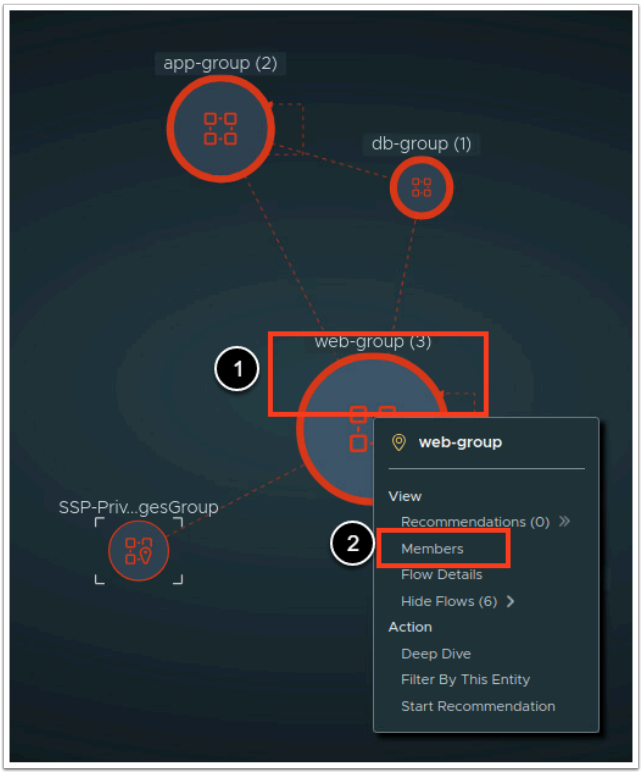
Review Flow Details



This view will show you all of the flows that have been identified between the Web Group and the DB Group. Each flow can be expanded to show the details that have been identified for that flow.

1. Click the Arrow to expand the flow details
2. Take note of the 4 tuple information of the flow (Source, Destination, Port, Protocol)
3. The last flow detected is shown to be unprotected
4. This flow contains an identifiable App-ID for Remote Desktop Protocol
5. The Distributed Firewall Rule that applied to this flow. In this case it is the default permit Any/Any which is why the flow is listed as unprotected.
6. Click **CLOSE**

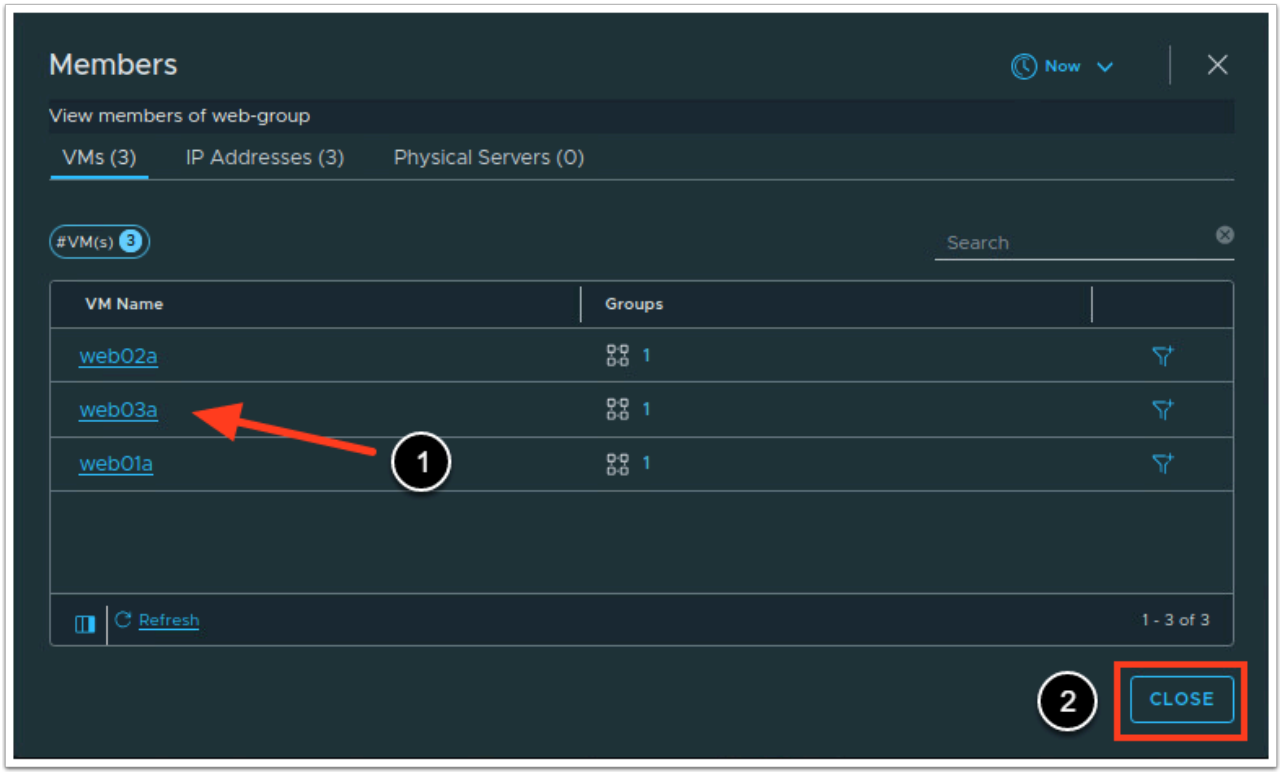
View Group Members



Each group within the canvas will contain a number of objects as members. In most cases these are VMs, but in some cases these could be external entities represented by IP addresses or ranges.

1. Right Click on the **web-group** icon
2. Click on **Members** in the drop-down list

Review Group Members



This view will show the members of the group on the canvas. This will include VMs, IP addresses/ Ranges, and/or physical servers.

1. Note the 3 VMs present here are the same 3 VMs you previously identified in NSX Manager.
2. Click **CLOSE**

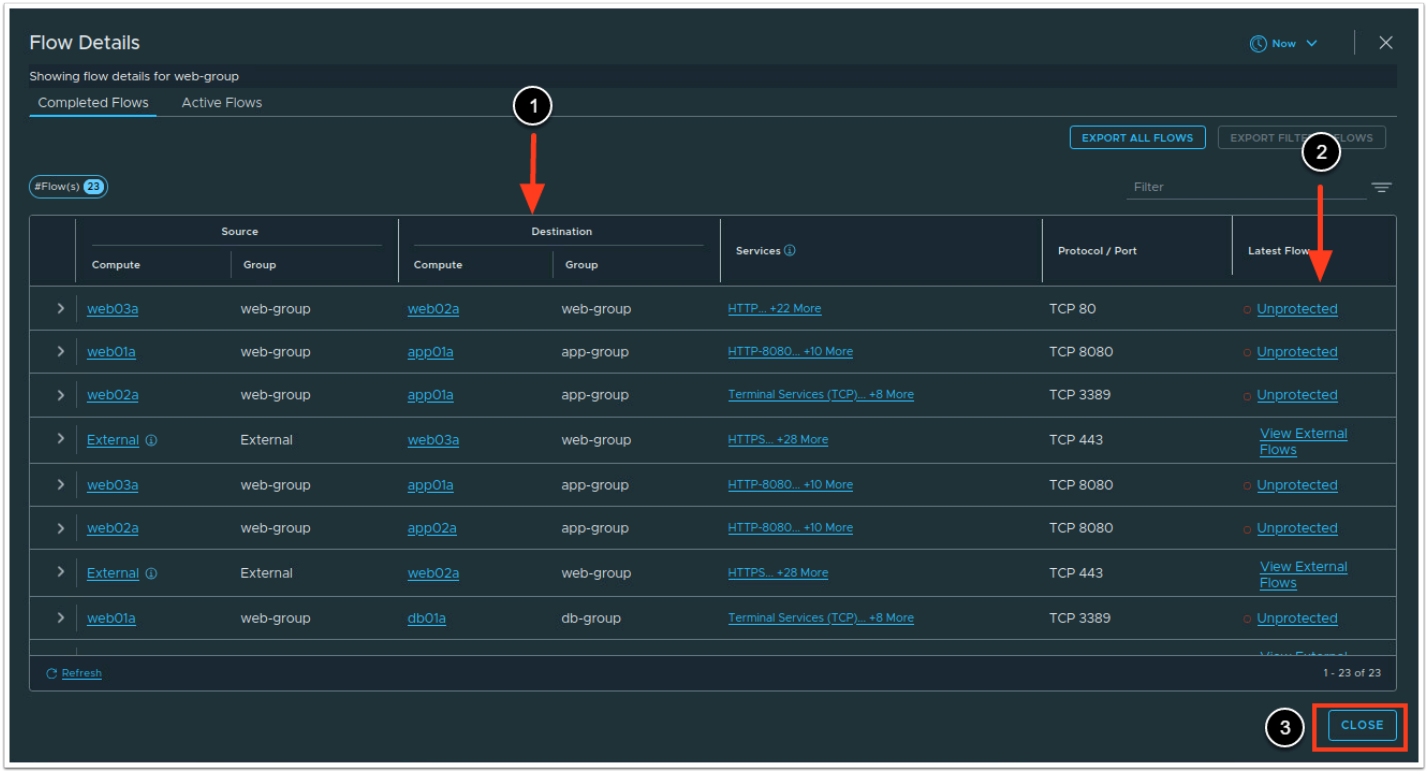
View Group Flows



You are also able to view the flows related to the group. These will be flows that are sourced from or destined to that particular group

1. Right Click on the **web-group** icon
2. Click on **Flow Details** in the drop-down list

Review Group Flows



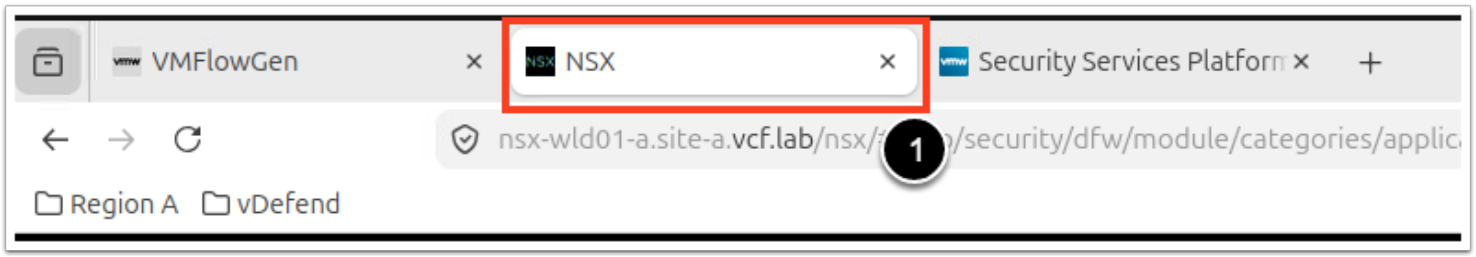
Like the flows reviewed between two groups these flows contain all of the same information. The difference is that the source and destination are not bound to a single connection, but all connection related to the objects within the group.

1. Take some time to review the flows related to the web-group
2. Take note that all of the flows are showing **Unprotected**
3. Click **CLOSE**

Secure 3 Tier App with DFW

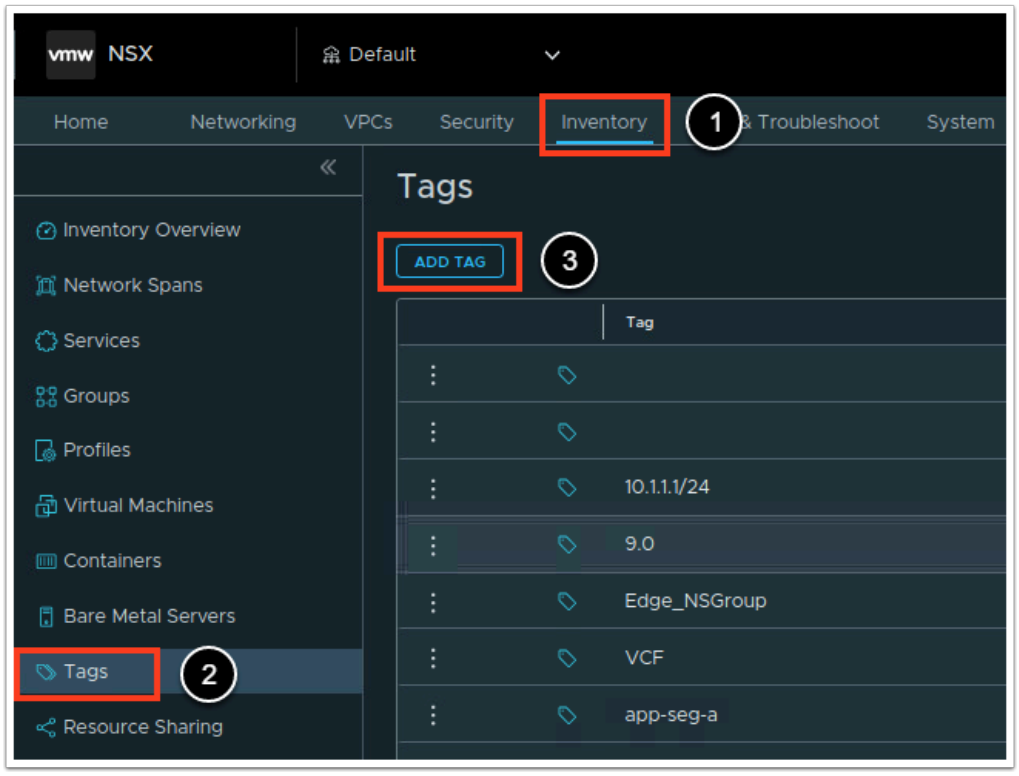
In this task you will first create an application tag of the 3 Tier Application and then use that tag to create an application group. You will then create the missing rule and activate the policy. At the end of this task you will observe the change of the application in Security Intelligence.

Return to NSX



1. Click on the **NSX tab** to return to NSX manager

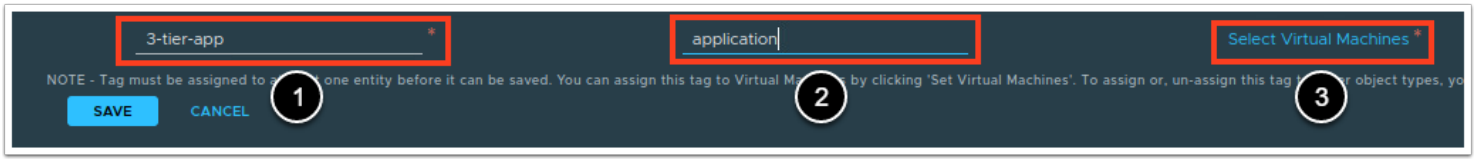
Create the Security Tag



Tags can be added in 2 different ways. The first is directly on an object in NSX, and the second is through the tag inventory section. You will be adding the tag using the latter method.

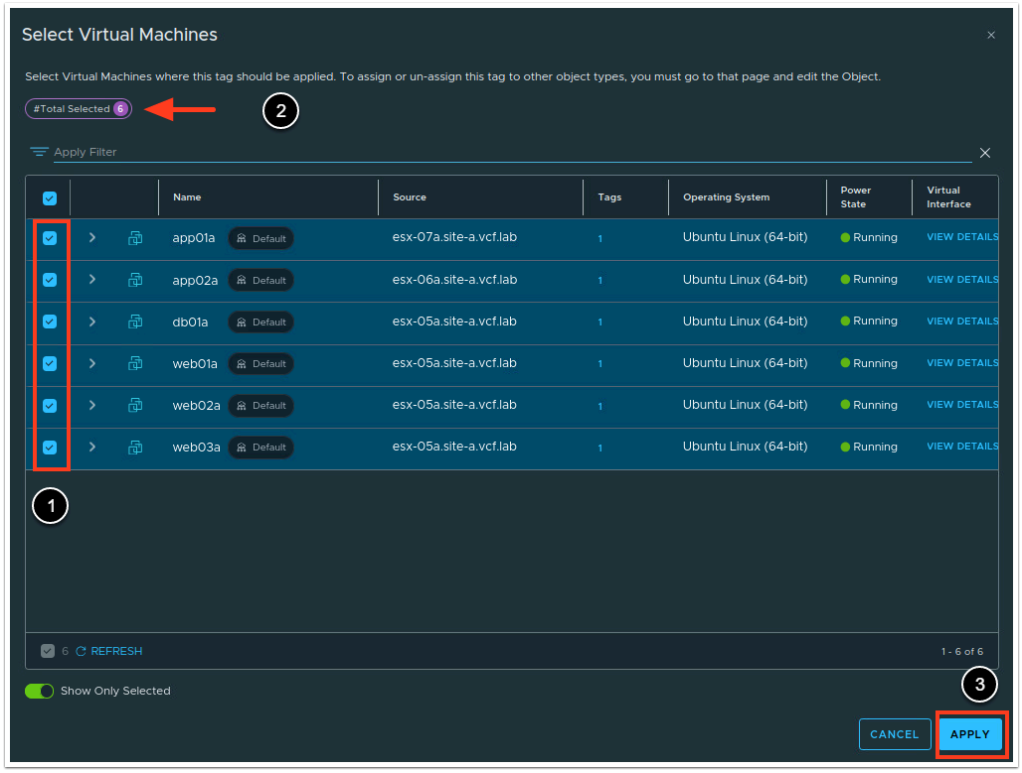
1. Select **Inventory** from the menu on the top
2. Select **Tags** from the menu on the left
3. Click **ADD TAG**

Enter Tag Details



1. Under Enter Name: enter **3-tier-app**
2. Under Enter Scope: enter **application**
3. Click **Select Virtual Machines**

Select Virtual Machines

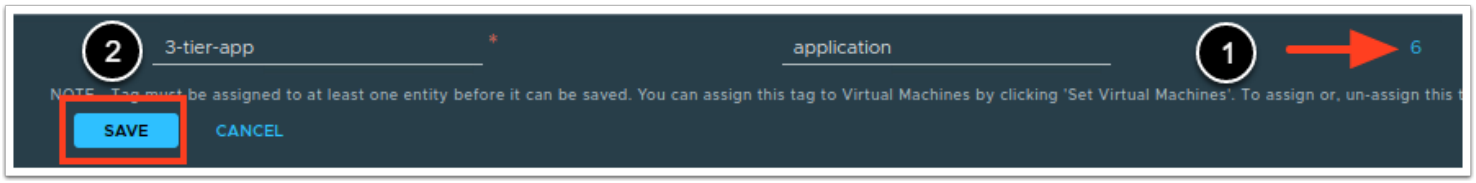


In this task you will select the virtual machines that make up the web, app, and db tiers of the application.

1. Select the following virtual machines:
   - web01a
   - web02a
   - web03a
   - app01a
   - app02a
   - db01a
2. Verify that only **6** VMs are selected
3. Click **APPLY**

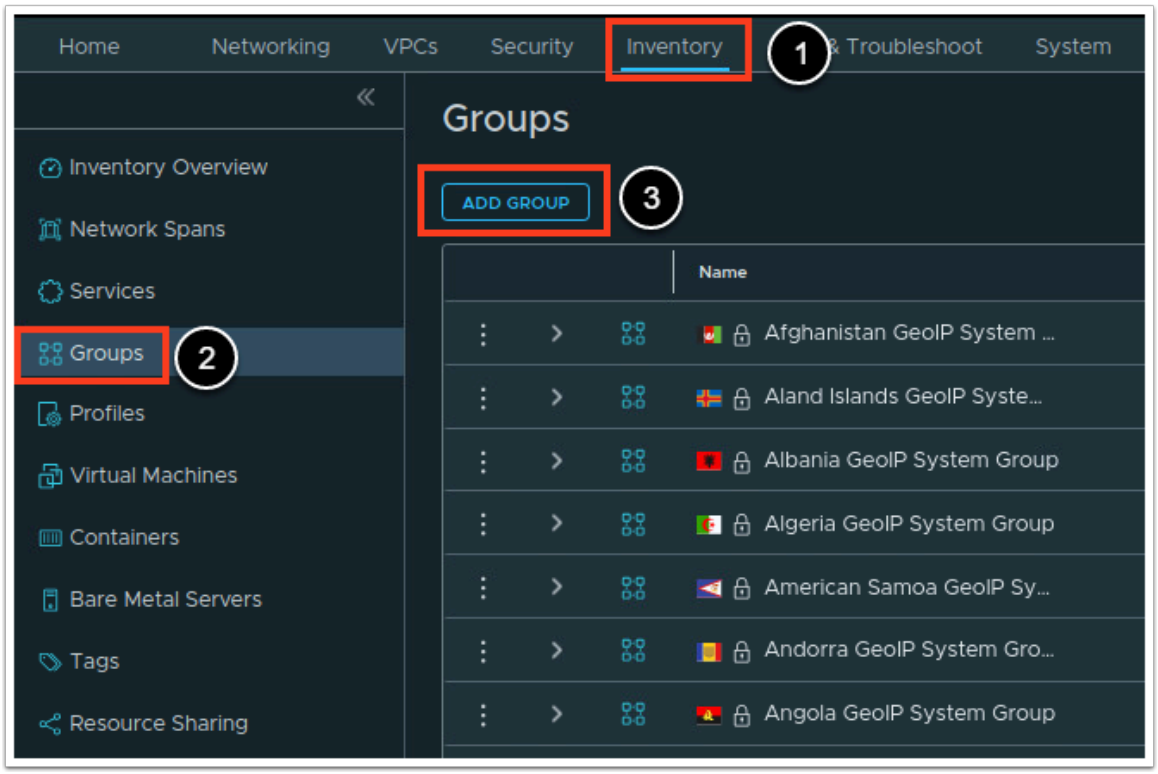
> There will be a number of VMs in this view with only the app and db VMs shown at the top. After selecting those 3 VMs use the filter at the top to find the web VMs by entering **web** into the filter. A total of 6 VMs should be selected.

Verify and Save



1. Verify the number of VMs this tag is assigned to is **6**
2. Click **Save**

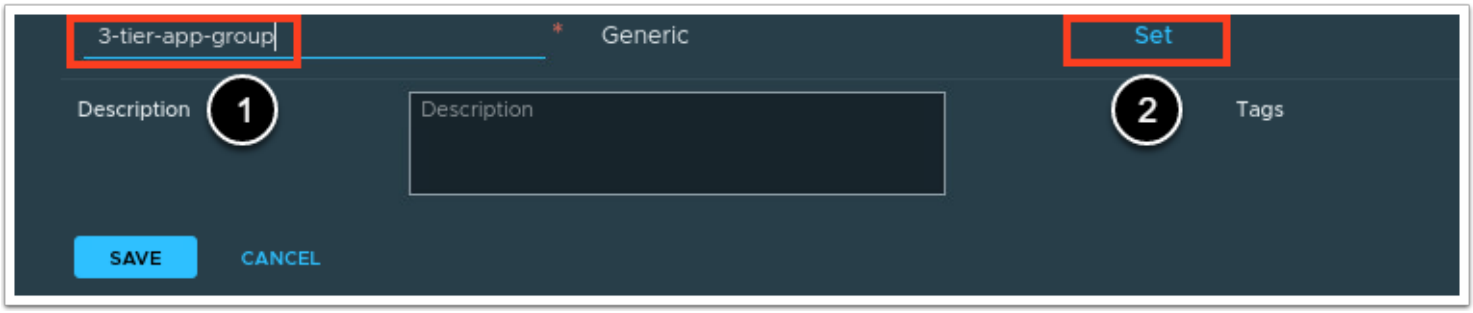
Create the Application Security Group



In this next step you will create a group for all 6 VMs that make up the 3 Tier App.

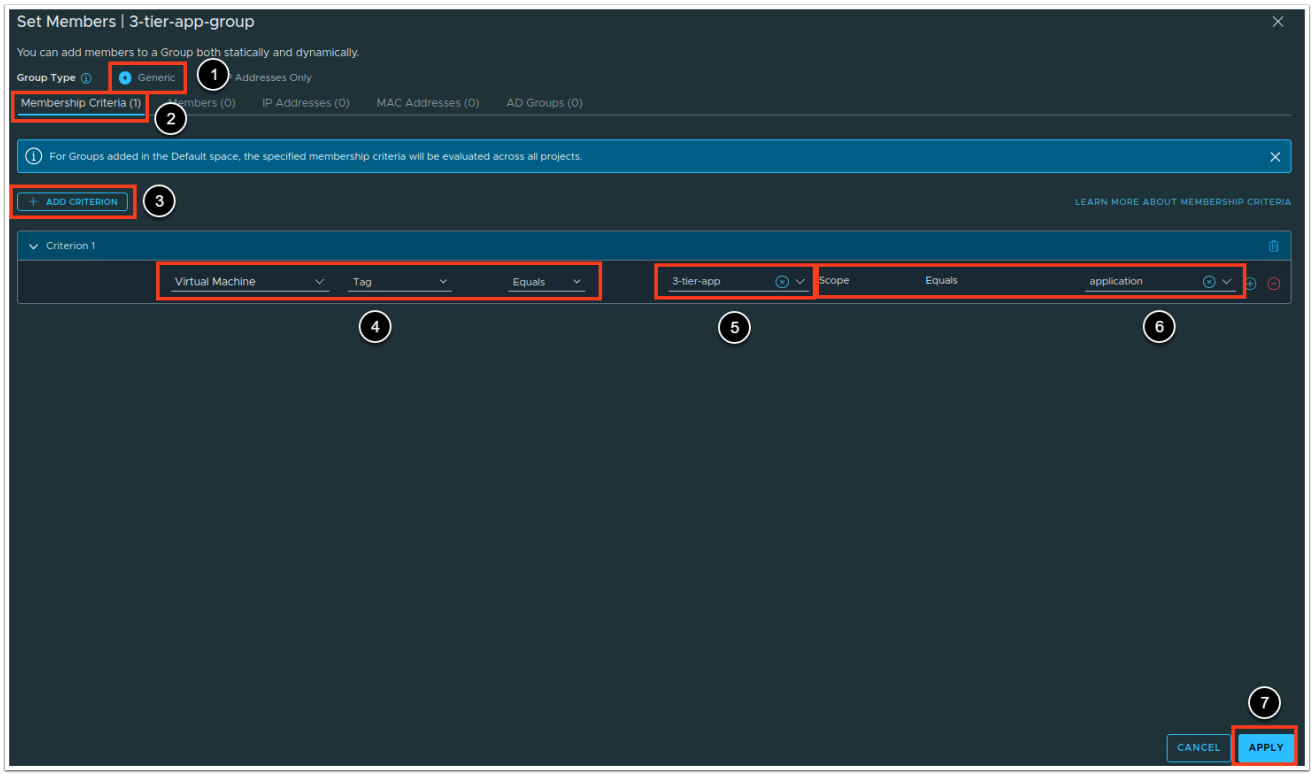
1. Ensure you are still in the **Inventory** section
2. Select **Groups** from the menu on the left
3. Click **ADD GROUP**

Name the Group



1. Under Group Name: enter **3-tier-app-group**
2. Click Set

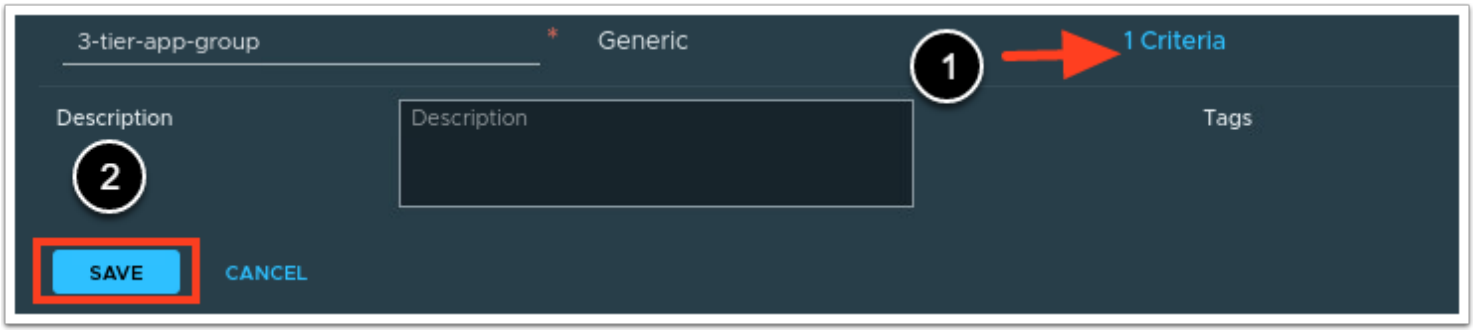
Enter Group Criteria



In this section you have the ability to use a wide variety of criteria to establish a group. For this exercise you will focus on groups made of tags.

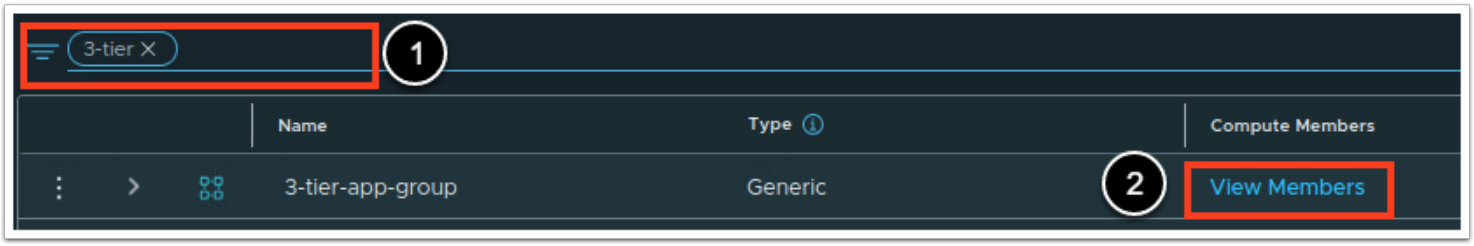
1. Ensure **Generic** is selected for group type
2. Ensure **Membership Criteria** is selected
3. Click **+ ADD CRITERION**
4. In the Criterion 1 form that popped up ensure the object is set to Virtual Machine Tag and the condition is set to Equals
5. Enter **3-tier-app** for the Tag
6. Ensure that **application** shows for Scope
7. Click **APPLY**

Verify and Save



1. Verify there is only **1 Criteria** applied
2. Click **Save**

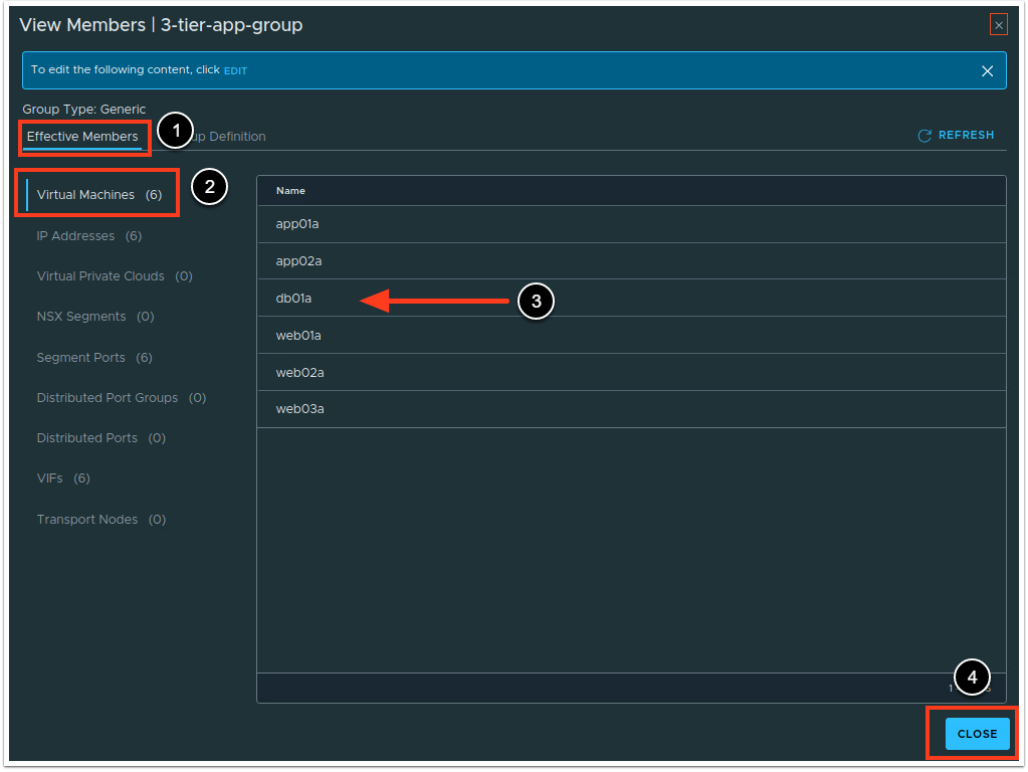
Verify Group Members



Now that the group is created let's validate that the 6 VMs we expect to be part of the group are present in it.

1. In the filter at the top of the section enter **3-tier**
2. Click **View Members**

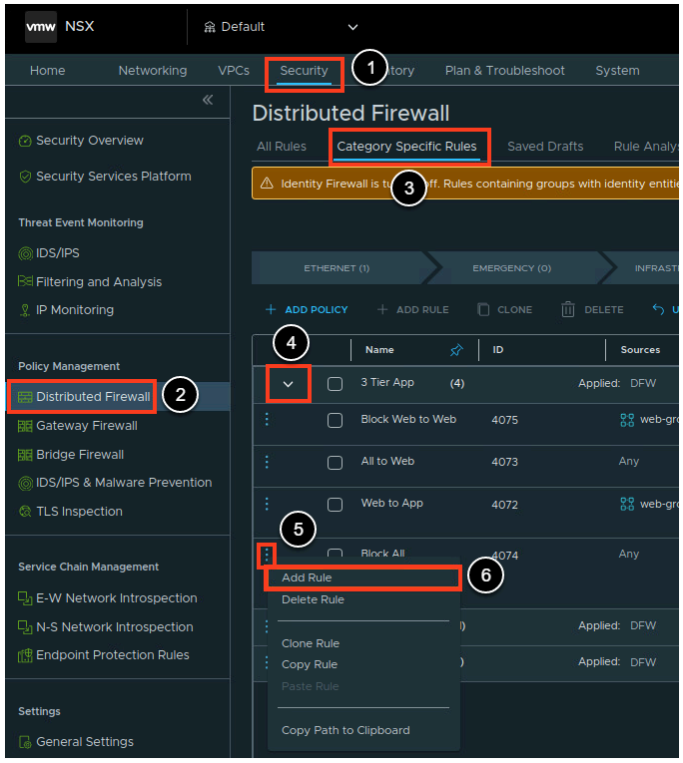
Review the Members



1. Click on **Effective Members**
2. Click on **Virtual Machines**
3. Verify the 6 Virtual Machines listed in the image are present.
4. Click **CLOSE**

> Before closing this window, feel free to look at the other menu items to better understand what makes up the group.

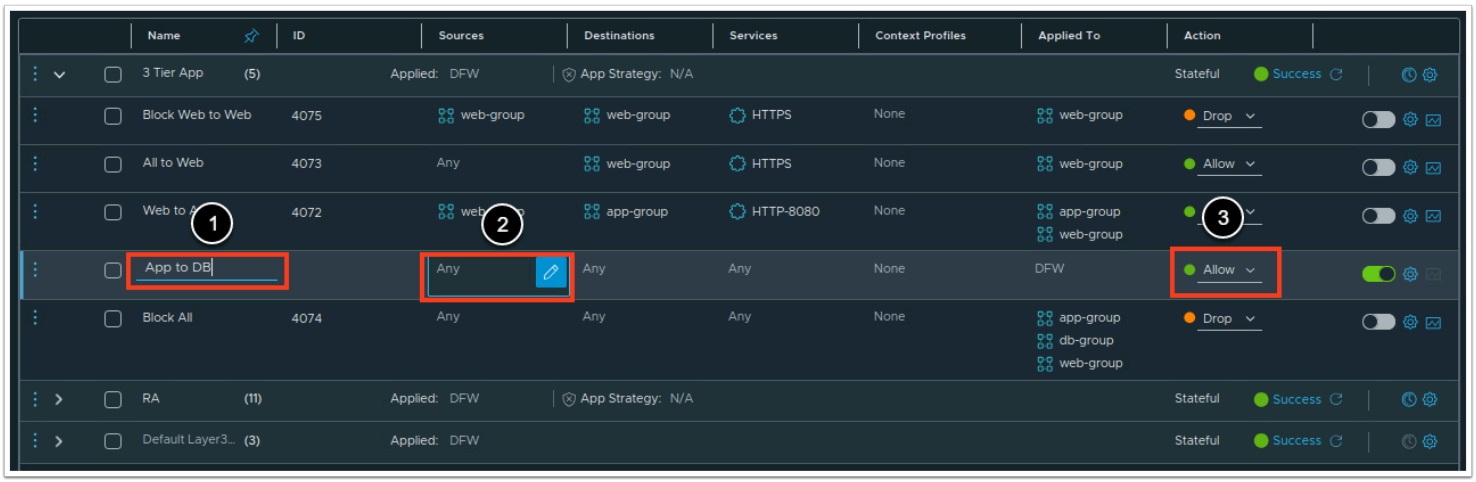
Create the Database Tier Rule



As discussed in the previous task, there is a rule missing. That rule is to lock down the traffic between the app-tier and the db tier. You will create this rule over the next few steps.

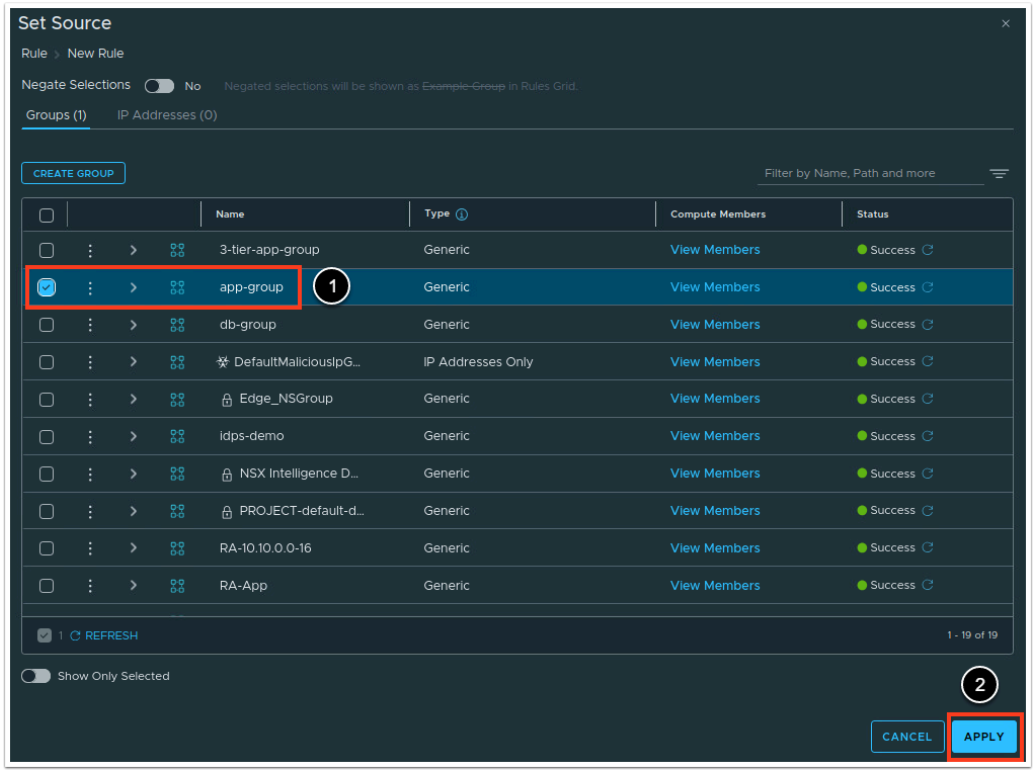
1. Click **Security** from the menu on the top
2. Click **Distributed Firewall** from the menu on the left
3. If you are not already on **Category Specific Rules** click it.
4. Expand the **3 Tier App** policy by clicking on the toggle
5. Click on the **3 Dots** on the rule **Block All**
6. Click **Add Rule**

Name the Rule and Add a Source



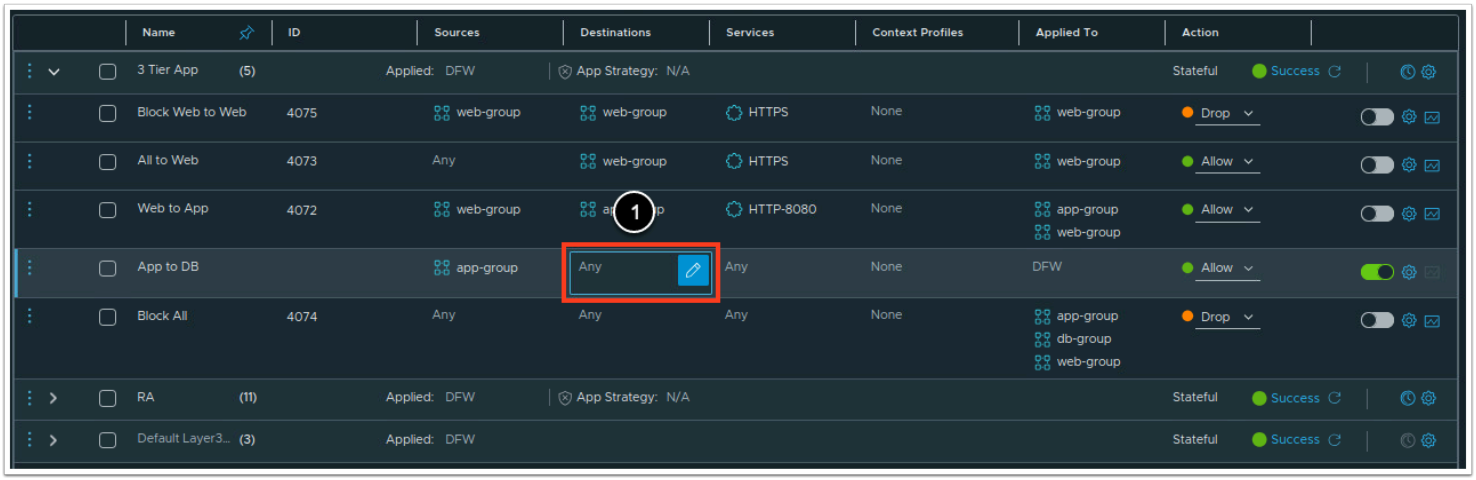
1. Enter **App to Db** for the rule Name
2. Click the **Pencil** next to Any for Sources to edit the source groups
3. The Action should be set to **Allow**

Add Source Group

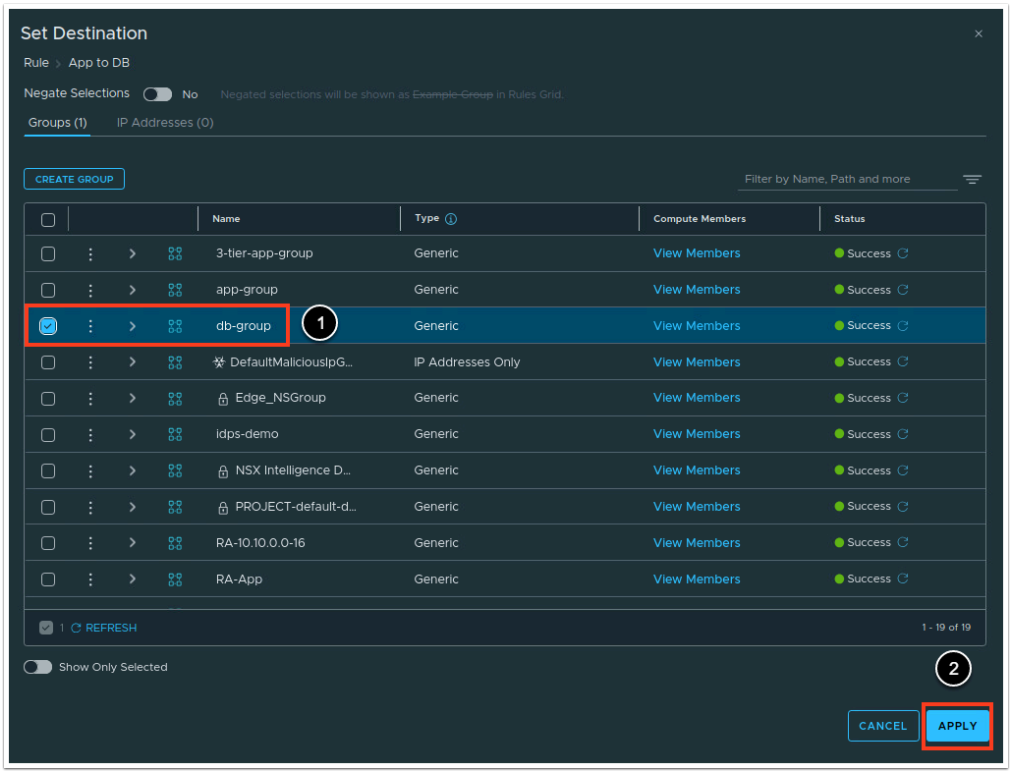


1. Select **app-group** to be the source group
2. Click **APPLY**

Add Destination Group

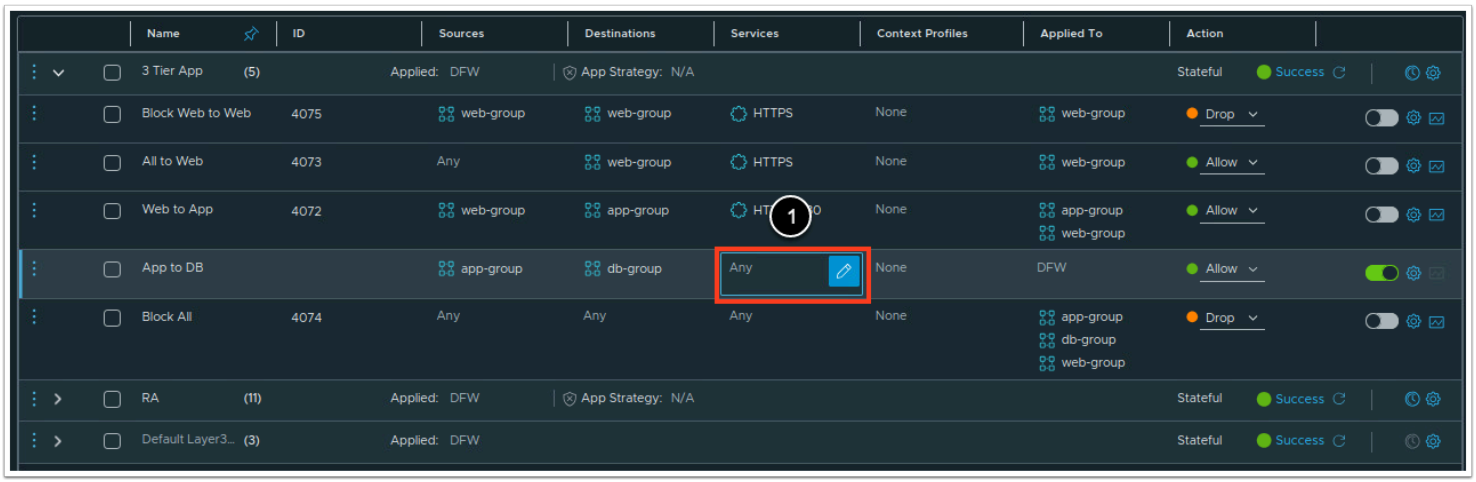


1. Click the **Pencil** next to Any for Destinations to edit the Destination groups



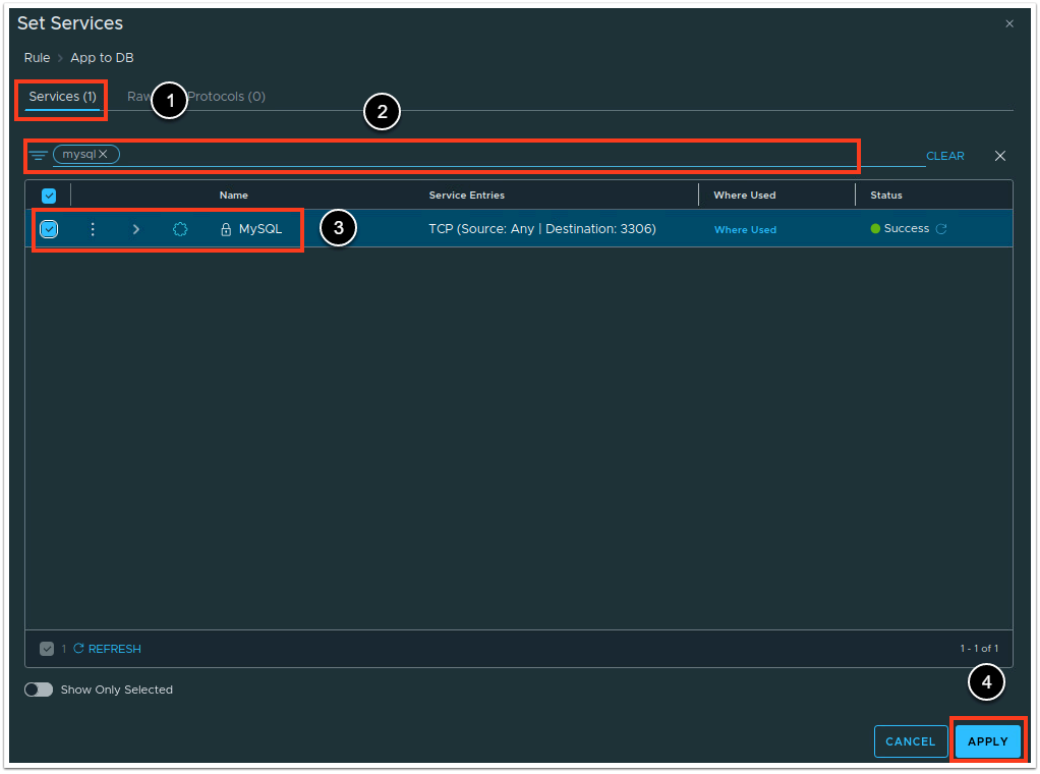
1. Select **db-group** to be the destination group
2. Click **APPLY**

Add Services



1. Click the **Pencil** next to Any for Services to edit the service groups

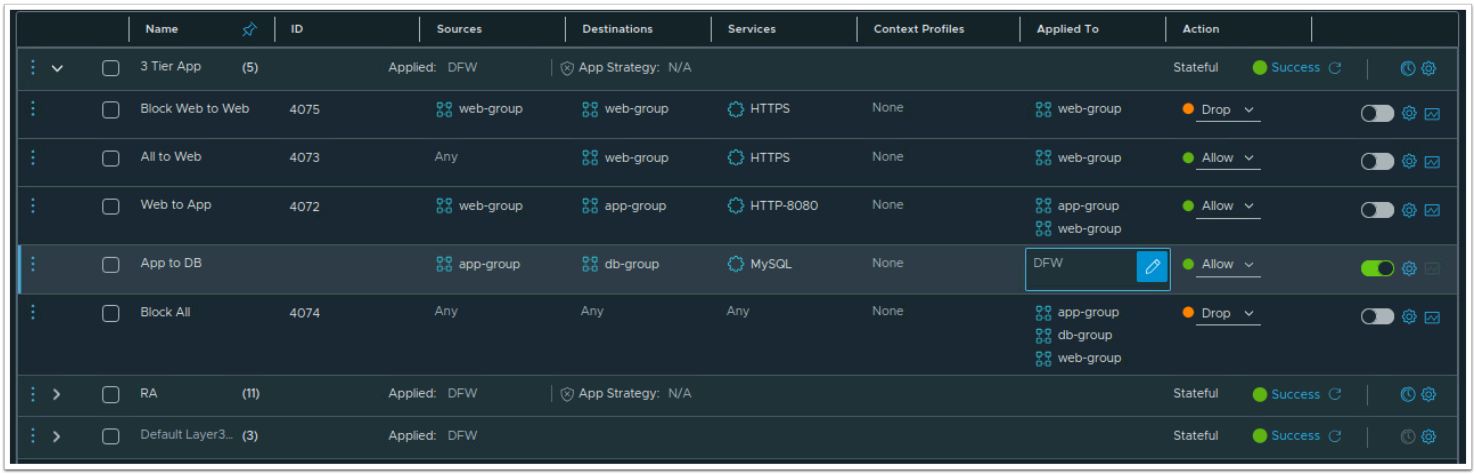
Add the MySQL Service



1. Ensure you are on the **Services** section
2. In the filter enter **mysql**

3. Select the **MySQL** Service
4. Click **APPLY**

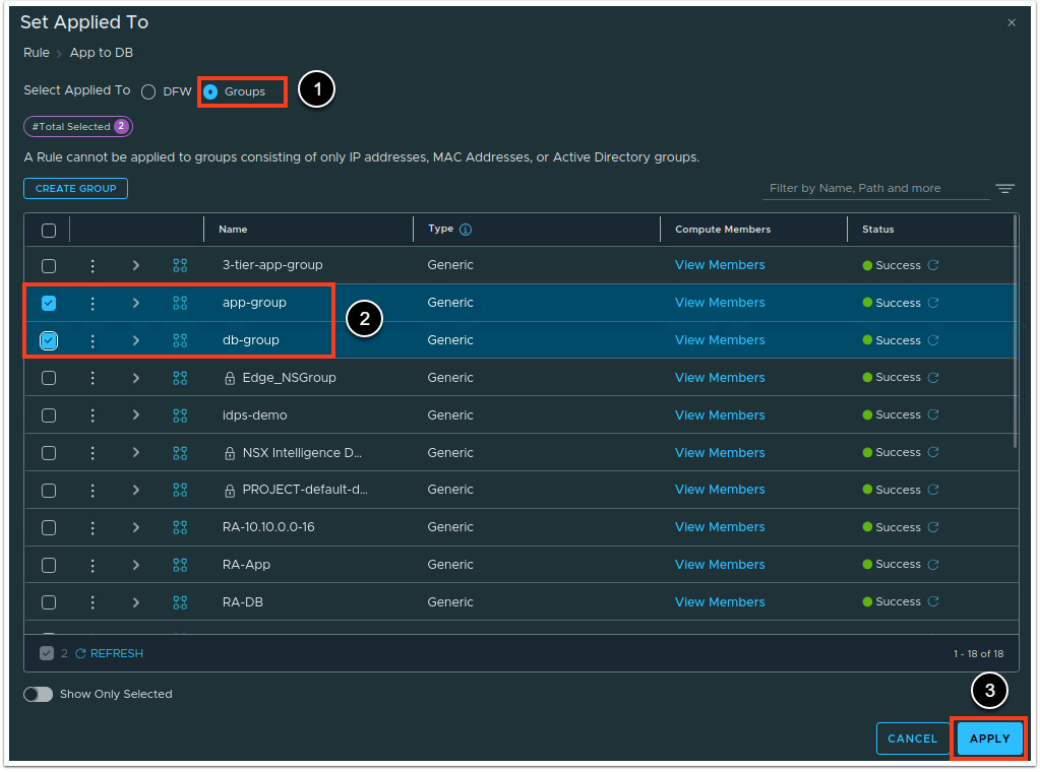
Set the Applied To



The last action you will need to take for this rule is to set the **Applied To** field. This is a critical component which sets the scope of the rule to the groups listed in the Applied To field. If set to DFW, this rule will be evaluated for all rules in the Distributed Firewall table. By limiting its scope in this manner, the rule will only be evaluated for the VMs that are part of the Applied To field.

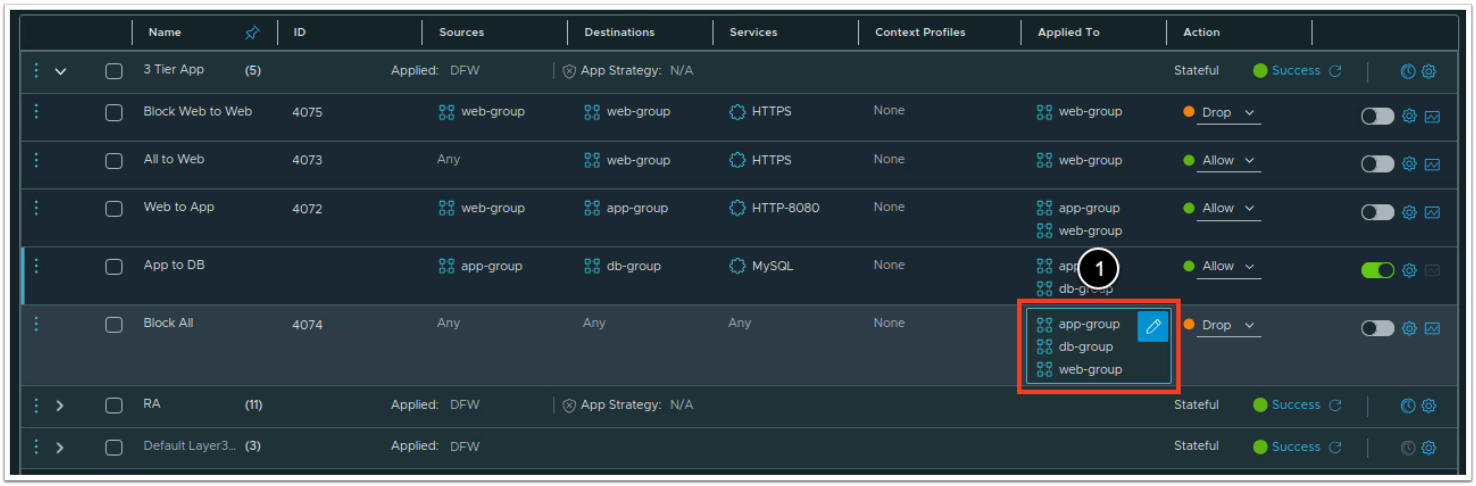
1. Click the **Pencil** next to Any for Applied To to edit the Applied To groups

Apply the App and DB Groups



1. Select **Groups**
2. Select **db-group** to be the destination group
3. Click **APPLY**

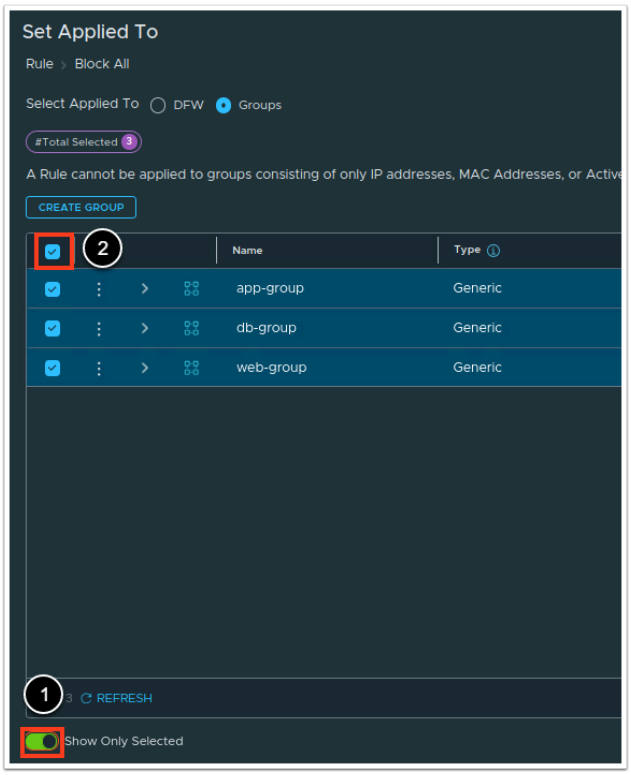
Update the Block All Applied To Field



With the final rule in place let's take a moment to clean up the Applied To field for the Application Default Rule, **Block All**, to use the group you created earlier in this task

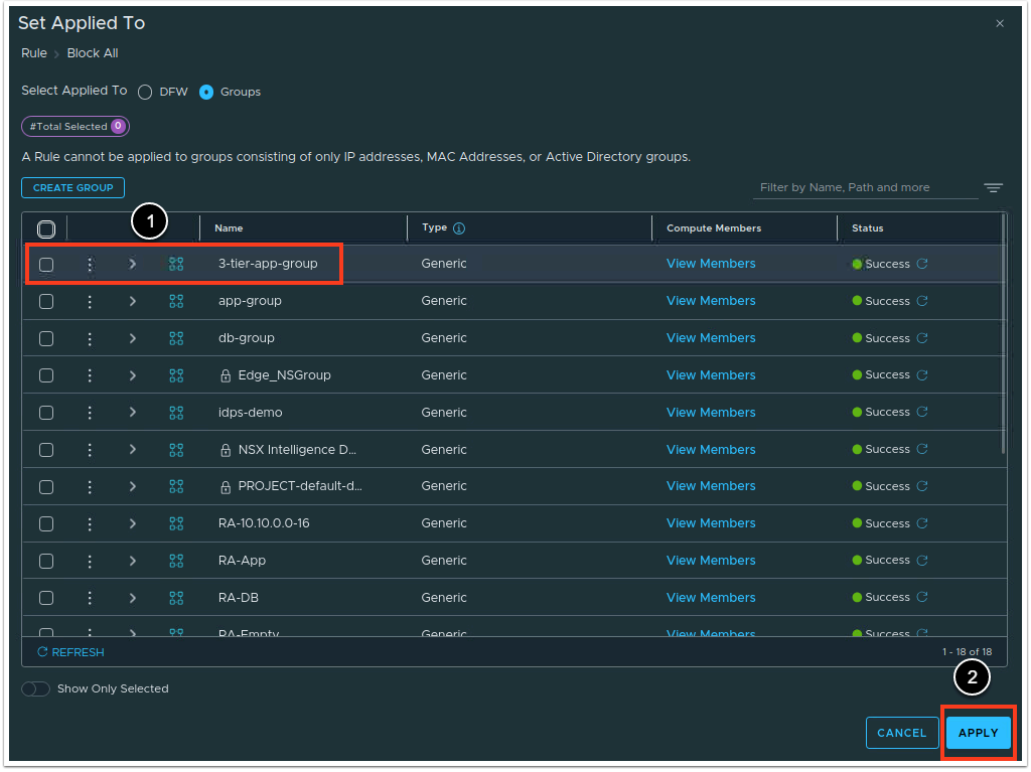
1. Click the **Pencil** next the existing 3 groups for Applied To to edit the Applied To groups

Remove the Existing Groups



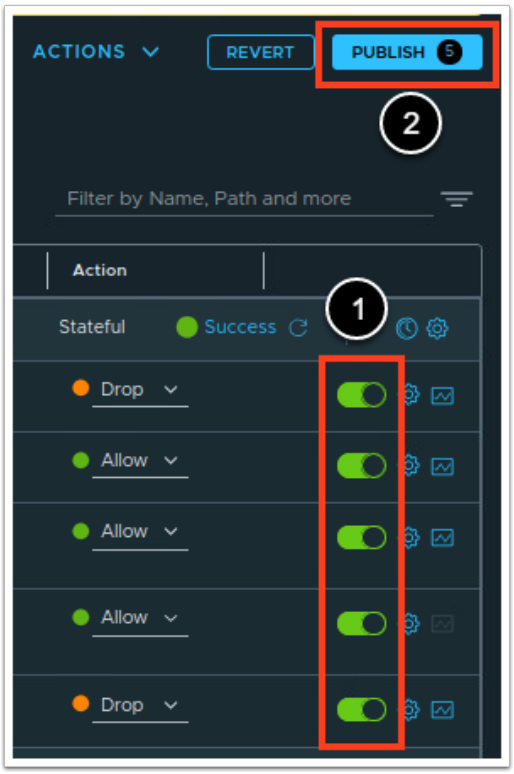
1. Click the **Show Only Selected** toggle
2. Uncheck the Select All box as shown. This will immediately result in all of the groups being shown but none selected.

Select the 3-tier-app-group Group



1. Select **3-tier-group** to be the destination group
2. Click **APPLY**

Enable Remaining 3 Tier App Rules and Publish



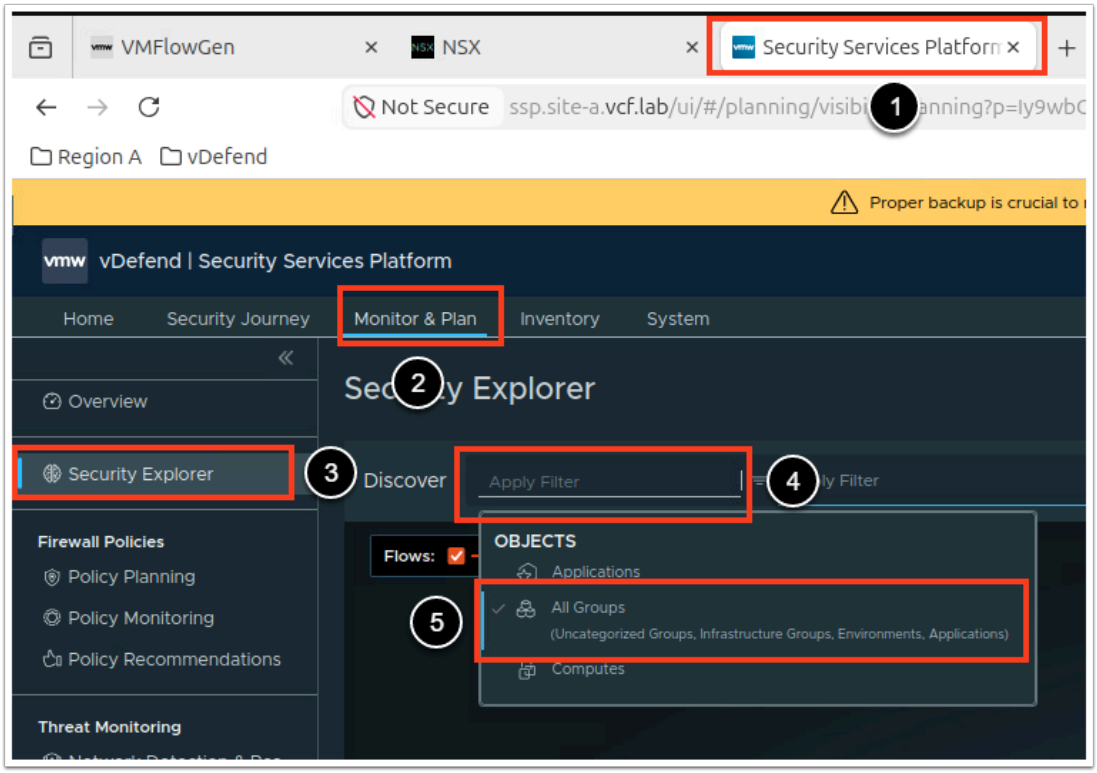
1. On the right hand side of the rules click the toggles for each rule so that they are showing activated (green).
2. Click **Publish** to save and apply the DFW rule changes.

Verify 3 Tier App Flows

With the policy enabled you will now return to Security Inteligence to verify the the changes in the policy.

If you are still logged into to Security Intelligence skip the next 2 steps.

Navigate back to Security Intelligence



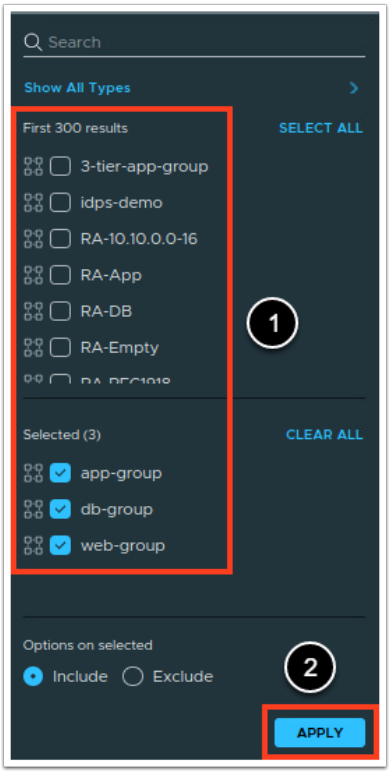
If you have been logged out, please log back in using the credentials that have been saved to the browser

Now you will return to security Intelligence to observe how the application monitoring in the Security Explore has changed.

1. Select the **Security Services Platform** tab in Firefox
2. Select **Monitor & Plan** from the top menu

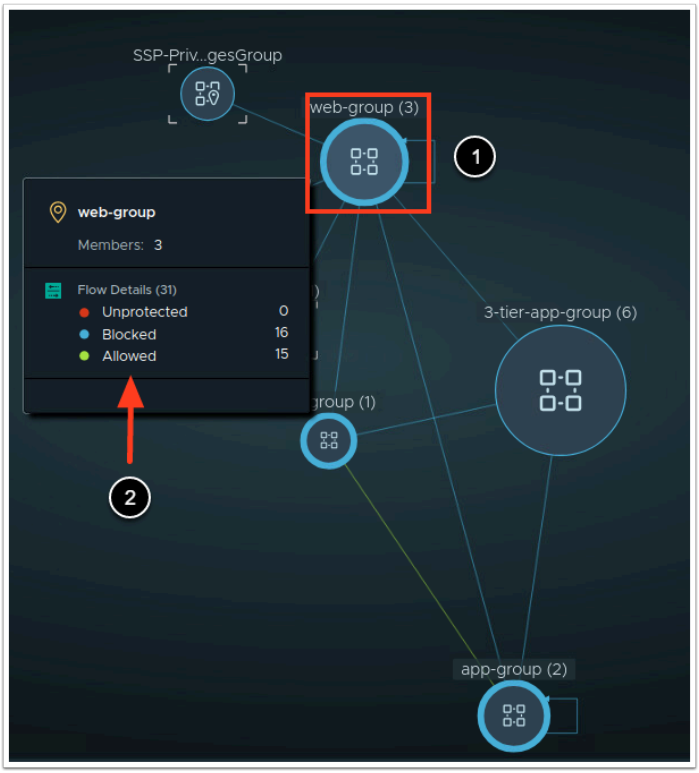
3. Select **Security Explorer** from the menu on the left
4. Click on the **Discover** filter drop down
5. Select **All Groups**

Apply Filter for the 3 Tier Application



1. From the list on the top select the following groups:
   - app-group
   - db-group
   - web-group (you will need to scroll down to see this group)
2. Click **APPLY**

View 3 web-group flows

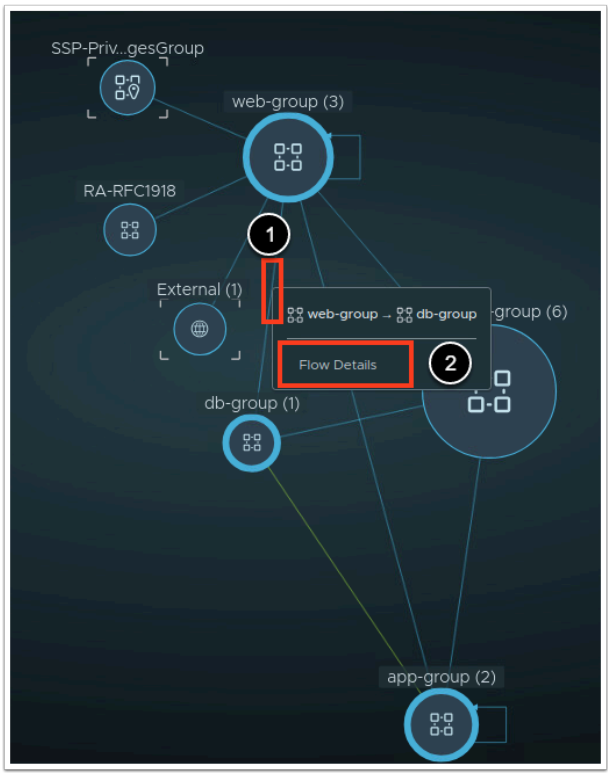


This next step will show you how many flows a blocked, allowed, or unprotected for a security group

1. **Hover** your mouse over the **web-group**.
2. A quick view of the flow details will appear. We can see that now we have 0 unprotected flows, 16 blocked, and 15 allowed. (The number of blocked and allowed flows may vary)

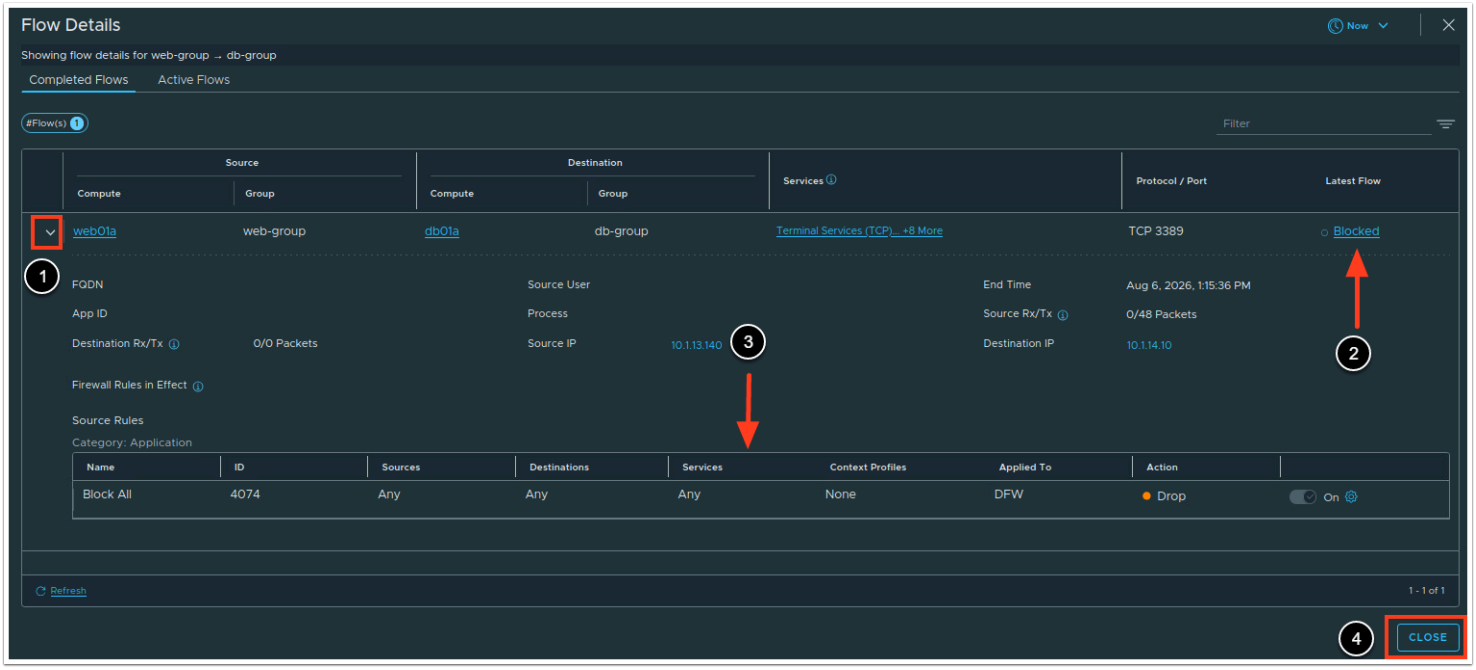
> Security Intelligence needs at least 5 minimum for the allowed flows to be updated, you might need to wait a little and refresh the view by clicking the refresh icon in the top right of the canvas. You may need to do this several times to show everything being updated.

Review Flows Between web-group and db-group



1. Right Click on the link between **web-group** and **db-group**
2. In the pop-up click **Flow Details**

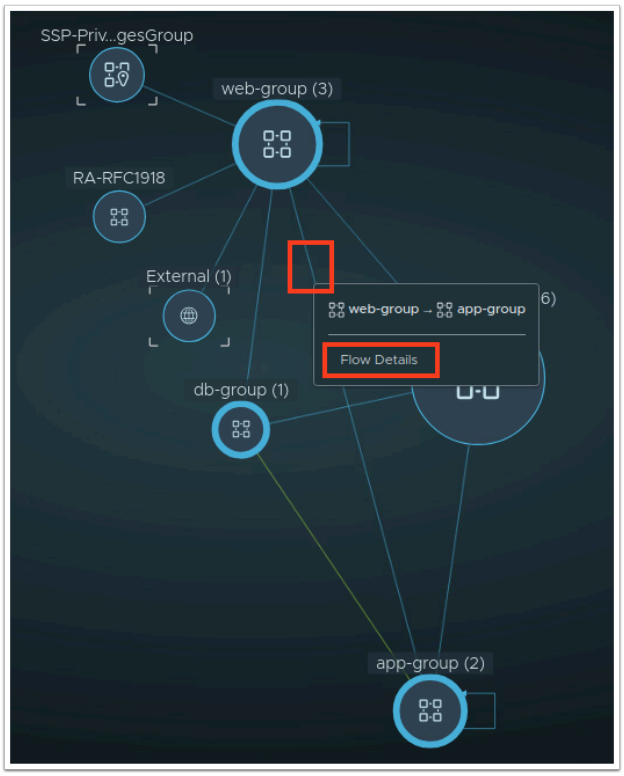
Review Flow Details



Now that the Flow is being enforced, you see that it is prohibited by the Block All rule listed at the bottom. In the first task you saw this flow as Unprotected.

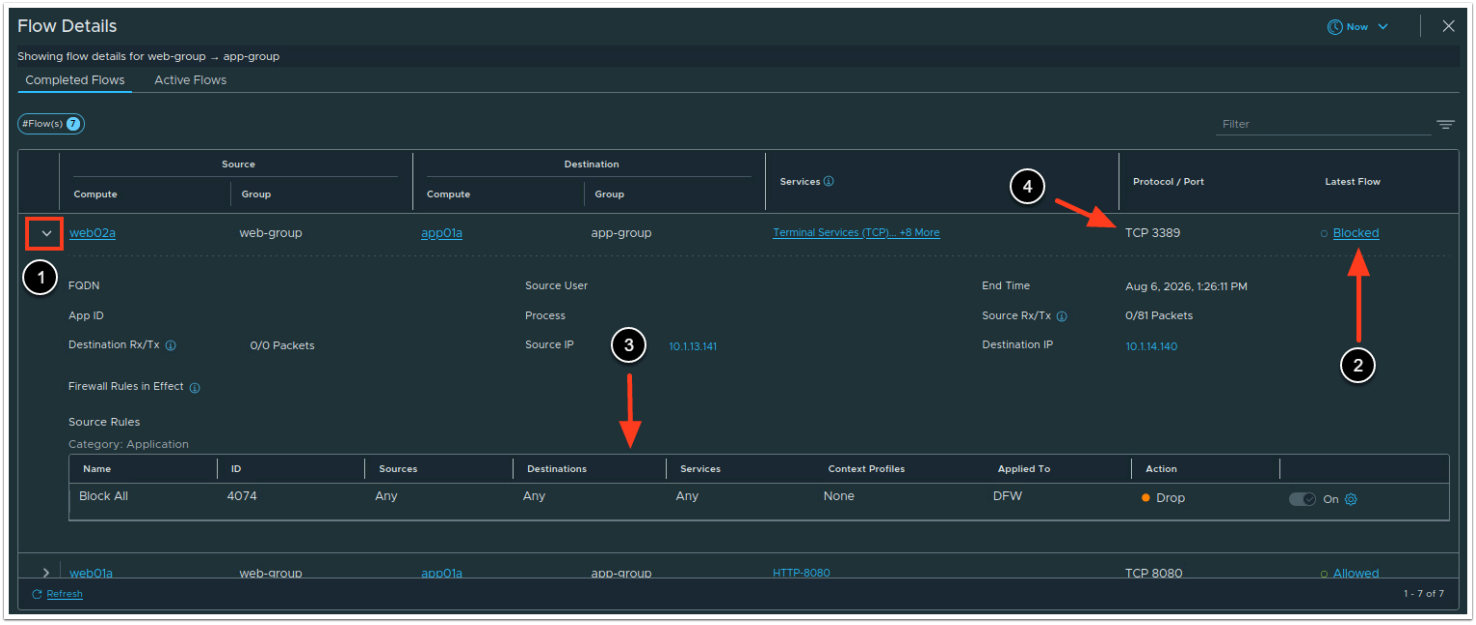
1. Click the toggle to expand the flow details.
2. Take note that the flow is listed as **Blocked**
3. Review the Rule that is blocking the flow
4. Click **CLOSE**

Review Flows Between web-group and app-group



1. Right Click on the link between **web-group** and app**-group**
2. In the pop-up click **Flow Details**

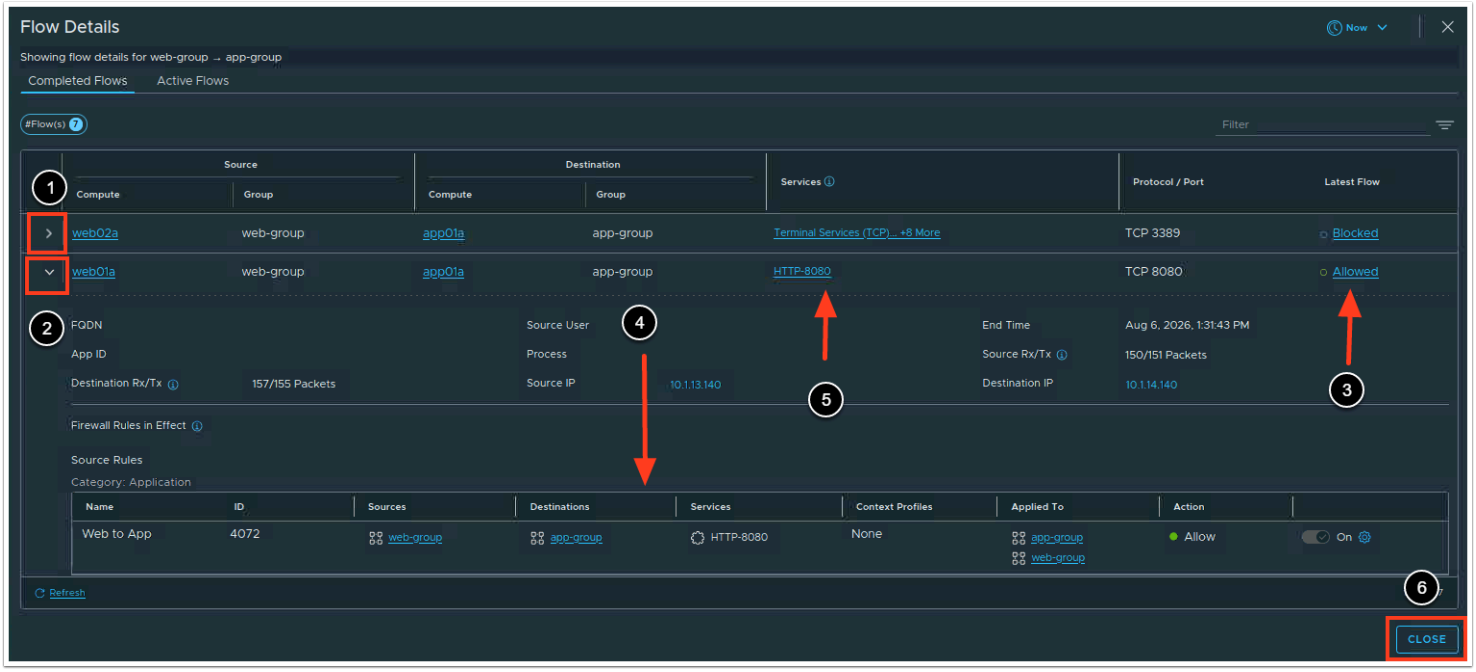
Review Blocked Flow



1. Click the toggle to expand the flow details for the first flow.
2. Take note that the flow is listed as **Blocked**
3. Review the Rule that is blocking the flow
4. The port being used here is **TCP 3389** or RDP.

Since there is no rule permitting TCP 3389 from communicating between web-group and app-group it will be dropped with the default rule for the application, Block All.

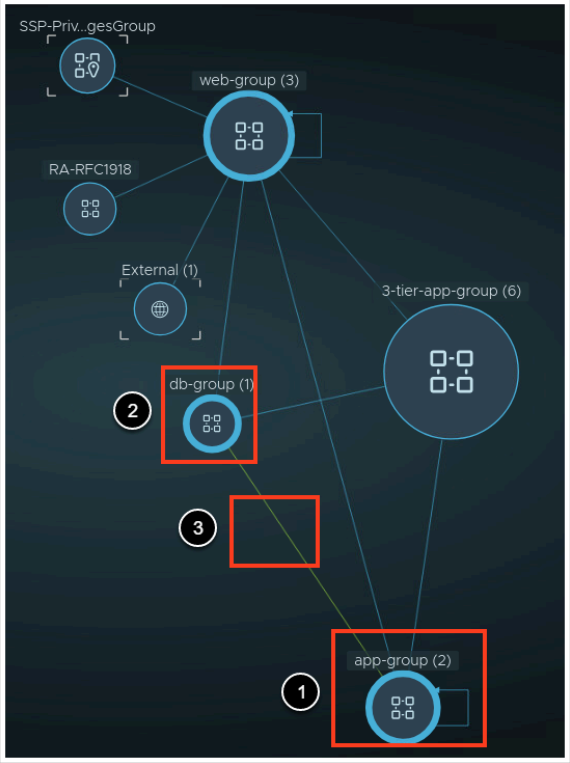
View the Allowed Rule



1. Click the toggle to collapse the flow details for the first flow.
2. Click the toggle to expand the flow details for the second flow.
3. Take note that the flow is listed as **Allowed**
4. Review the Rule that isAllowing the flow
5. The port being used here is **HTTP-8080**.
6. Click **Close**

The Web to App rule allows communication between the web-group VMs and the app-group VMs over HTTP-8080 only.

Review Other Rules



Take some time to review some of the other objects on the canvas. Some suggestions would be to:

1. Hover your mouse over **app-group** to see the flows associated with that group
2. Hover your mouse over **db-group** to see the flows associated with that group
3. Left click on the flow between **app-group and db-group** to inspect the flows there.

Before moving onto the next step take some time to familiarize yourself with function of the canvas. You cannot make any changes or break anything in the lab by playing around here.

Conclusion

Conclusion

In this module, you successfully implemented the distributed firewall to secure our three tier application. You learned how to create and apply tags to virtual machines and then use those tags to create groups to be used in vDefend Firewall Policies.

By applying micro-segmentation and enforcing granular security policies at the VM level, you effectively controlled and isolated traffic between the tiers based on defined rules. You used vDefend Security Intelligence and VCF Operations for Logs to confirm that only legitimate, policy-allowed traffic was permitted while all other traffic was correctly blocked.

From here you can:

- Continue with the next lab module.
- Click [vlp:table-of-contents|Show Table of Contents] to jump to any module or lesson in this lab.
- End your lab and return in the future.

Module 3 - Provide Enhanced Application Security With Layer 7 Firewall Rules (20 Minutes) Advanced

Introduction

Layer 7 Application Identity

While traditional Layer 4 objects match the port specified in the TCP/UDP header of a flow, Layer 7 objects are port-independent and instead use signatures to match content in the payload of a flow. This allows a firewall to distinguish for instance HTTP from MYSQL traffic, even if both services run on port 80. vDefend comes with a set of Layer 7 context profiles. These map with the services customers commonly run within the data center and enable port-independent enforcement. These can be used on the distributed firewall (DFW) and the Tier-1 gateway firewall (GFW). Layer 7 context-profiles can be used in combination with a static L4 port-based service object for port-independent enforcement. Just a handful of packets are required for the NSX firewalls to fingerprint the APP-ID a flow is generated to/from, which means there is no throughput penalty for enabling this functionality.

Customers should use Layer 7 context-profiles in their intra-application micro-segmentation policies along with L4 services objects to reduce the attack surface by ensuring the protocol traveling across the open firewall port is indeed the intended protocol. In addition, by writing firewall policies that are Layer-based instead of port-based, the appropriate set of firewall rules for new applications that are being rolled out can be determined quicker; for instance, a single firewall rule can be used to allow MySQL across any port coming into an MySQL server, regardless of what port the application developers decided to run that service on.

In this module, we will enable a Layer 7 Context Profile on an existing Distributed Firewall Rule to further secure our environment. We will then validate that traffic that does not match our Layer 7 Context Profile is blocked.

Adjust MySQL Flows in the 3-tier-app

Return to VMFlowgen

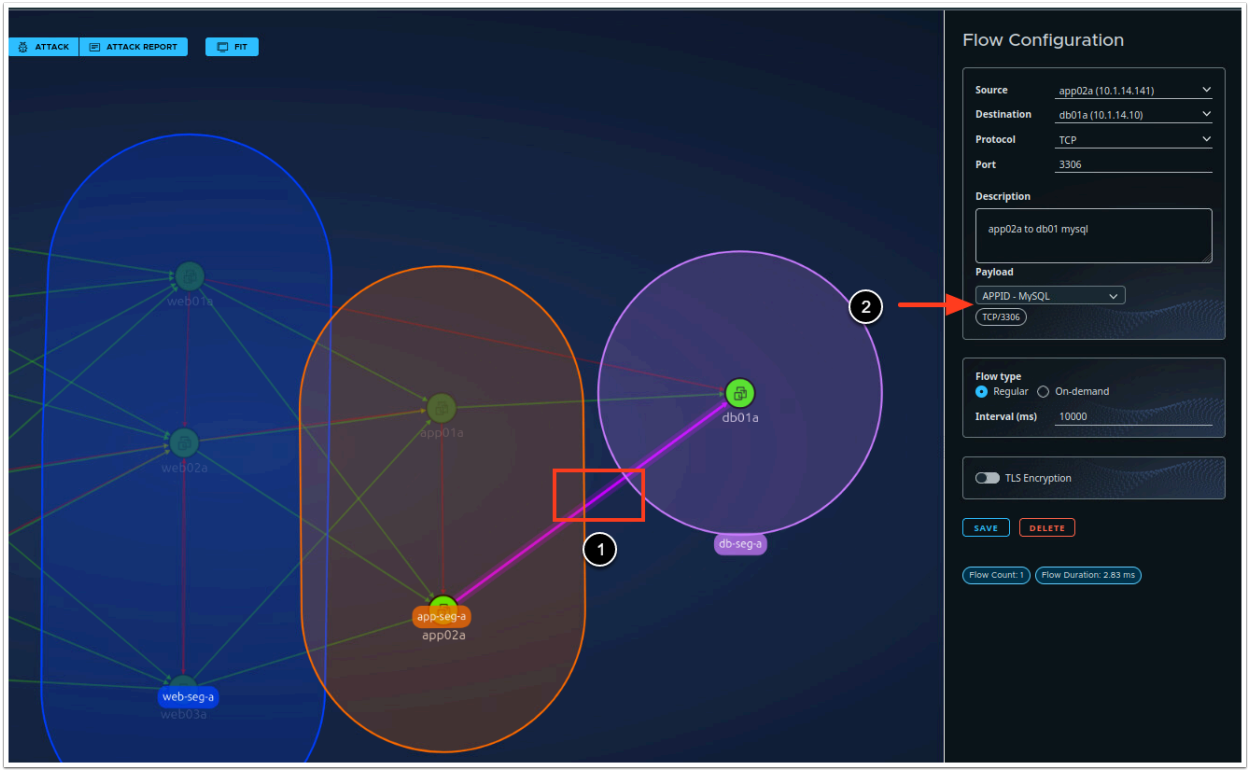


You will return to VMFlowgen to make an adjustment to the communication between the app02a VM and the db01a VM to illustrate the capabilities of L7 firewall

1. In Firefox select **VMFlowGen** to return to FlowGen.

> If you are logged back out, log back in using the un/pw of admin/VMware123!VMware123!
>
> After logging in go to Topologies -> vdefendfirewalldemo

View the MySQL Flow For app02a



Flow gen provides you with ability to not only change the layer 4 characteristics of a flow, but can also apply layer 7 context to a flow, called an APPID. An APPID is a fingerprint in the application layer of the packet that allows firewalls to recognize the application being used.

1. Click the **line between app02a and db01a**.
2. The payload for this flow is using **APPID-MySQL**

Modify the Flow to Oracle

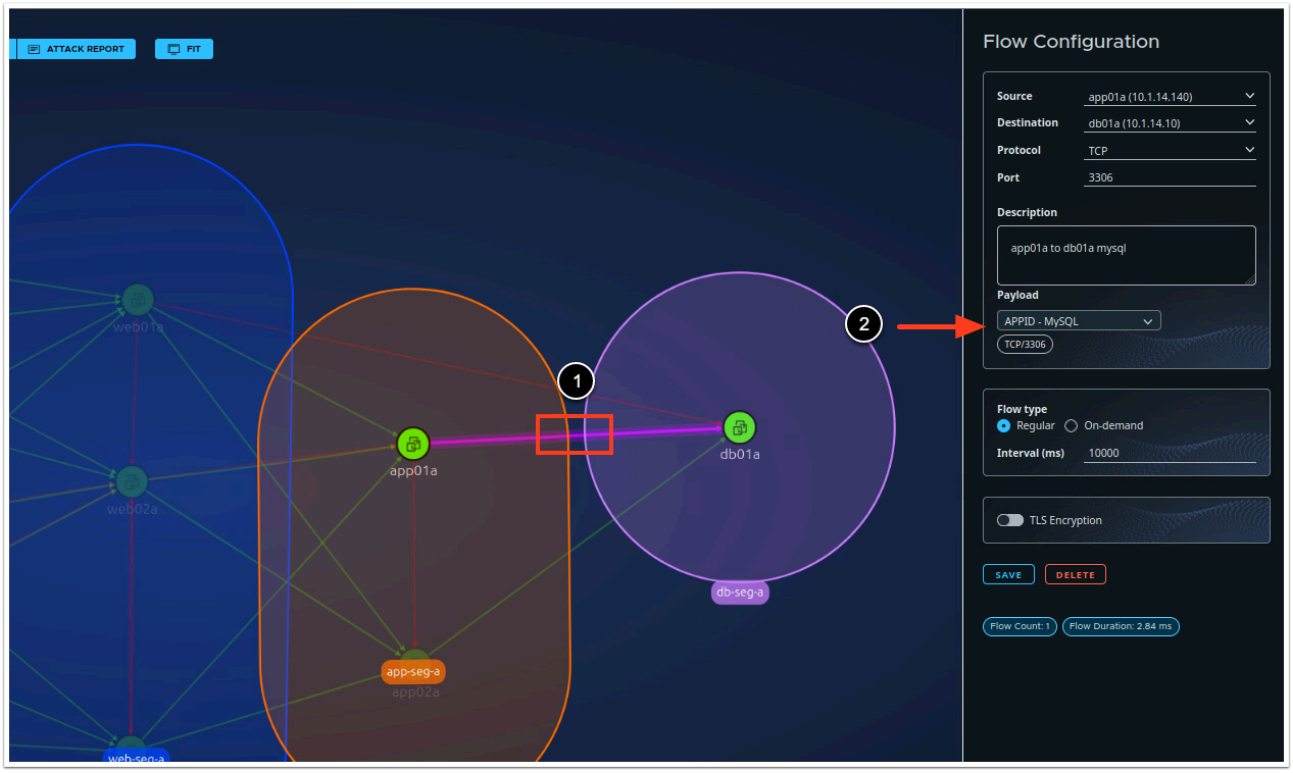


For this module you will be adjusting the APPID for one of the two flows between the app VMs and the db VM. You will be using Oracle as the APPID simulating an improperly deployed app server using a MySQL port but Oracle as the protocol.

1. Click the **Payload** drop down
2. Select **APPID-ORACLE** from the drop down list

> **Note:** The lab has been built so that TCP3306 is used by both of the app servers when talking with the db server. By changing the APPID from MySQL to Oracle you should be able to the Layer 7 Firewall at work.

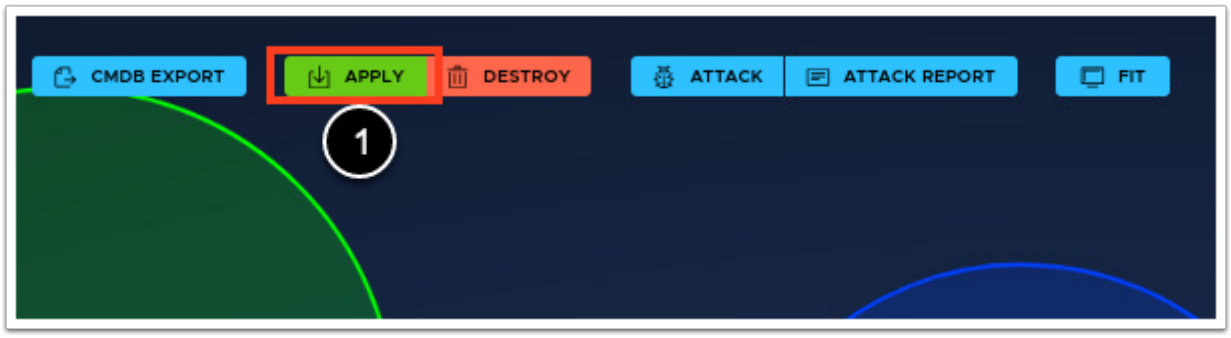
Inspect the app01 to db01 Flow



The flow between app01a and db01a will remain unchanged

1. Click the **line between app01a and db01a**.
2. The payload for this flow is using **APPID-MySQL**

Apply the VMFlowGen Config



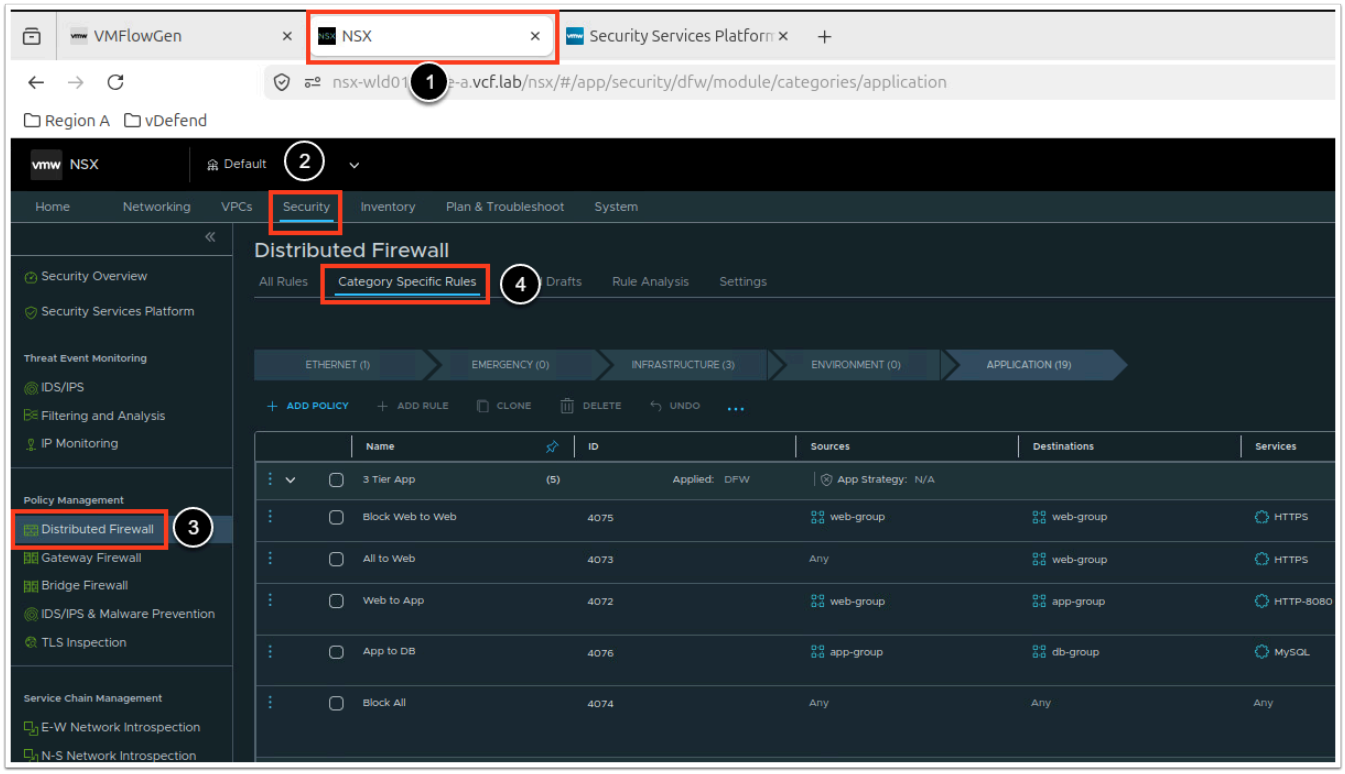
Now that the configuration is set you will need to apply it to the FlowGen VMs that have been deployed.

1. On the menu click **APPLY**

Apply Layer 7 Context Profile to the App to DB Rule

Now that the Flows have been updated you will need to apply the L7 context profile to the App to DB rule

Return to NSX Manager

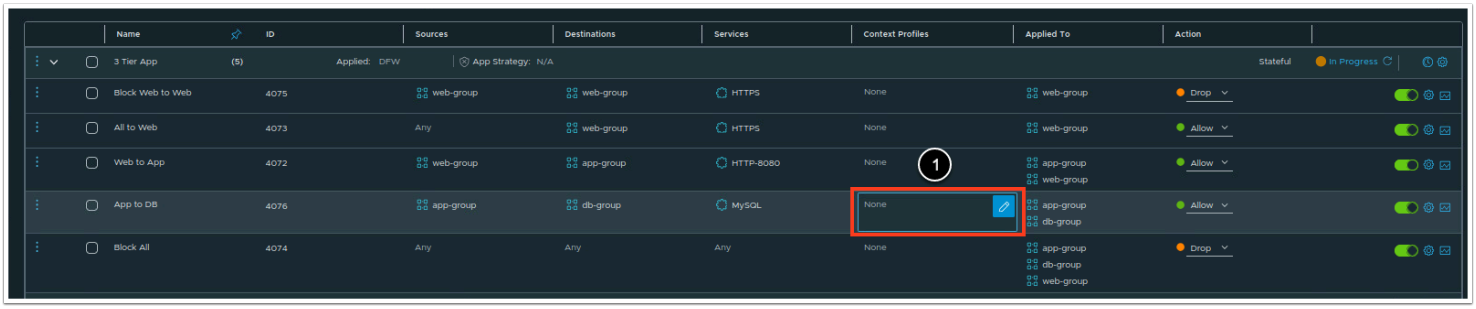


You will now return to the 3 Tier App police you had worked on in the previous module

1. Select the **NSX** tab in Firefox
2. Select **Security** from the top menu
3. Select **Distributed Firewall** from the menu on the left
4. Select **Category Specific Rules** from the Distributed Firewall menu

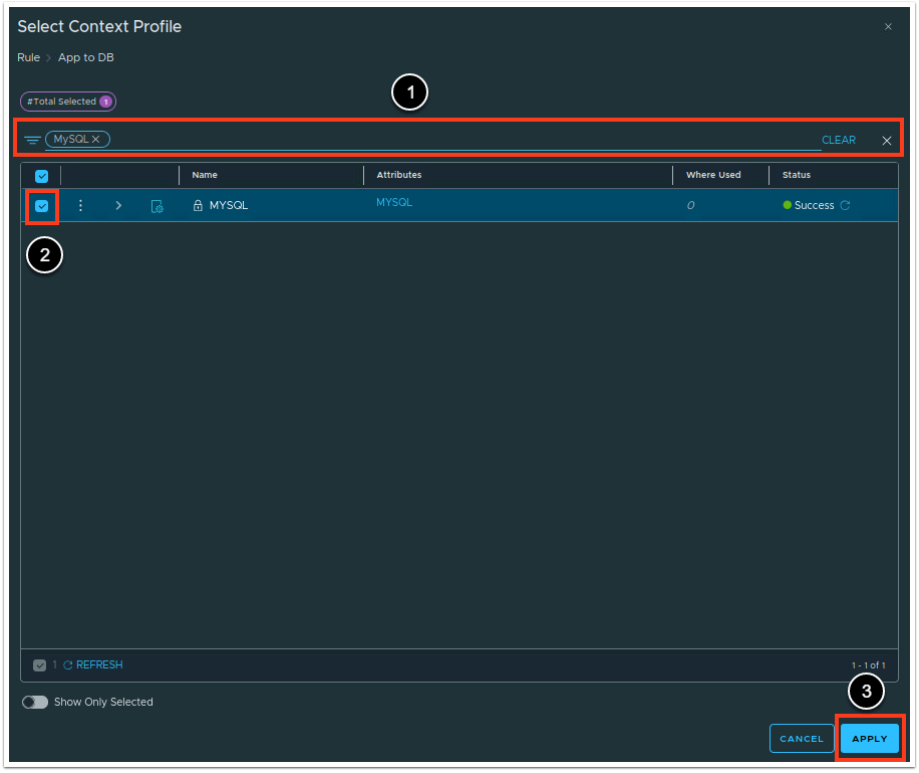
> In the event you have been logged out of NSX use prefilled username **admin** and prefilled password **VMware123!VMware123!** select Login.

Apply the MySQL Context Profile



1. Click the **Pencil** for Context Profiles for the App to DB policy

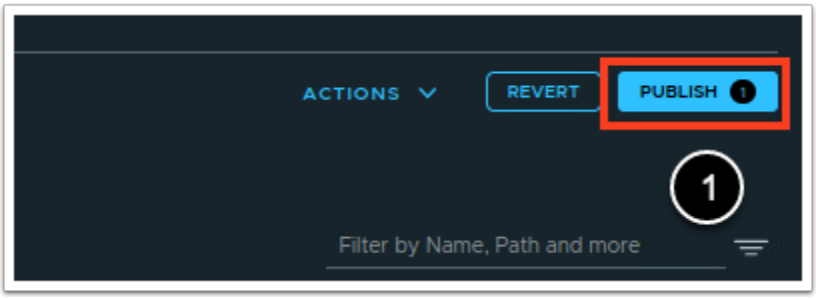
Select the Context Profile



Now select and apply the MySQL Context Profile

1. Type **MySQL** in the filter field
2. Select the **MYSQL** Context Profile
3. Click **APPLY**

Publish the Policy



Now that the Context Profile is set the updated policy will need to be pushed

1. In the top right of the policy table, click **Publish**

Validate the MySQL Flow

To validate the change you will head back into the Security Services Platform and view the flows in the Security Explorer.

Switch to Security Services Platform

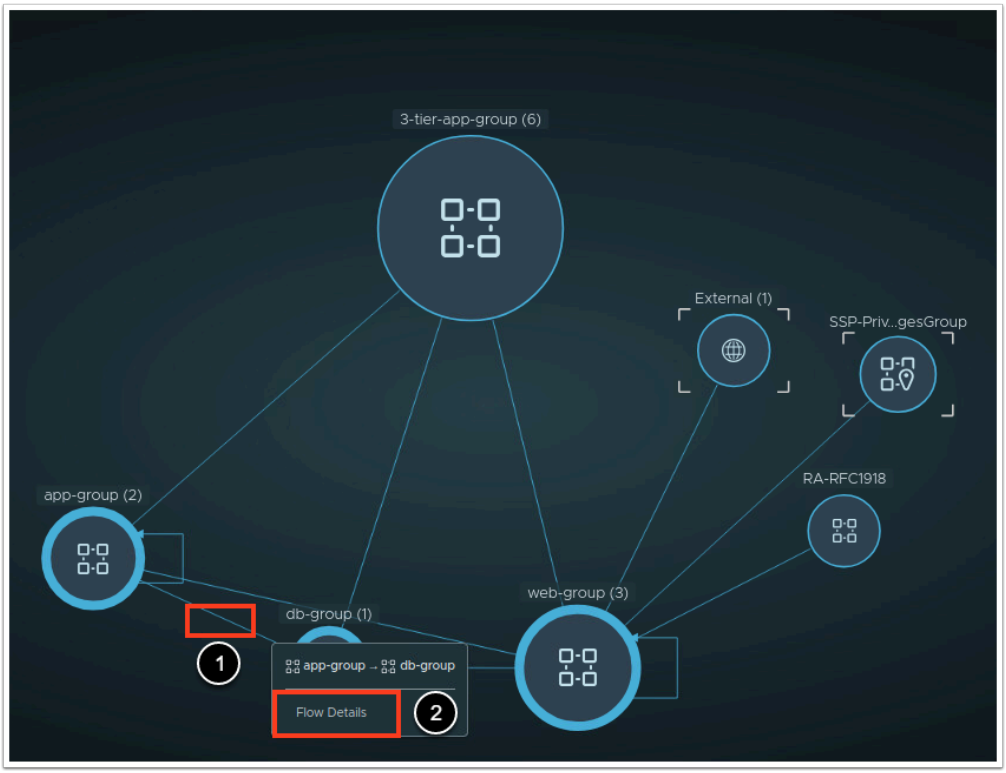


The Security Services Platform will be the location you will go to monitor the application.

1. Click on the **Security Services Platform** tab to return to Security Explore

> If you are logged out the Security Services Platform, log back in using the un/pw or admin/VMware123!VMware123! if not already prepopulated. Navigate to Monitor & Plan -> Security Explorer -> All Groups and filter for the following 3 groups:
>
> - app-group
> - db-group
> - web-group

View Flows in Security Explorer

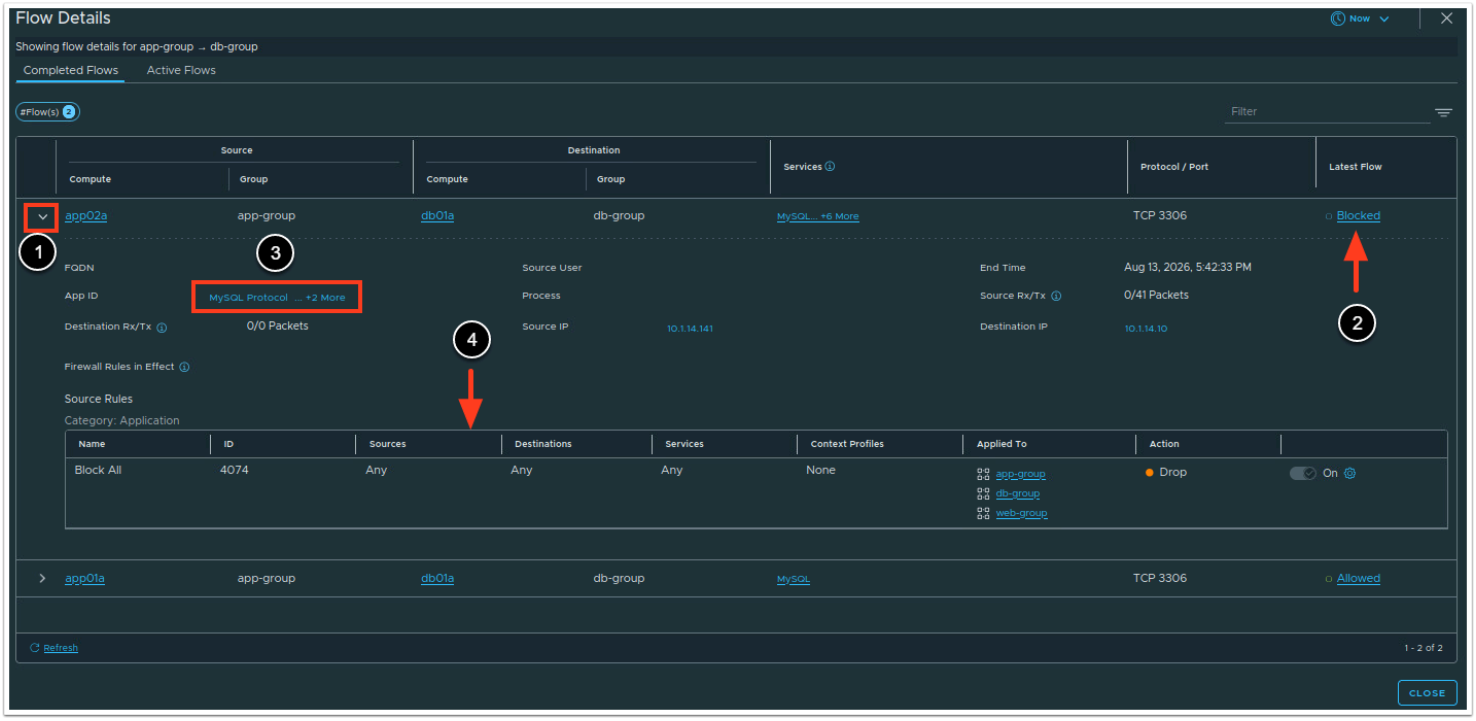


After a few minutes the flows will be computed for the change you have made.

1. Right click on the flow between app-group and db-group
2. Select **Flow Details**

> This may take up to 5 minutes to complete. During that time click the refresh button on the top right of the canvas until the green line turns blue.

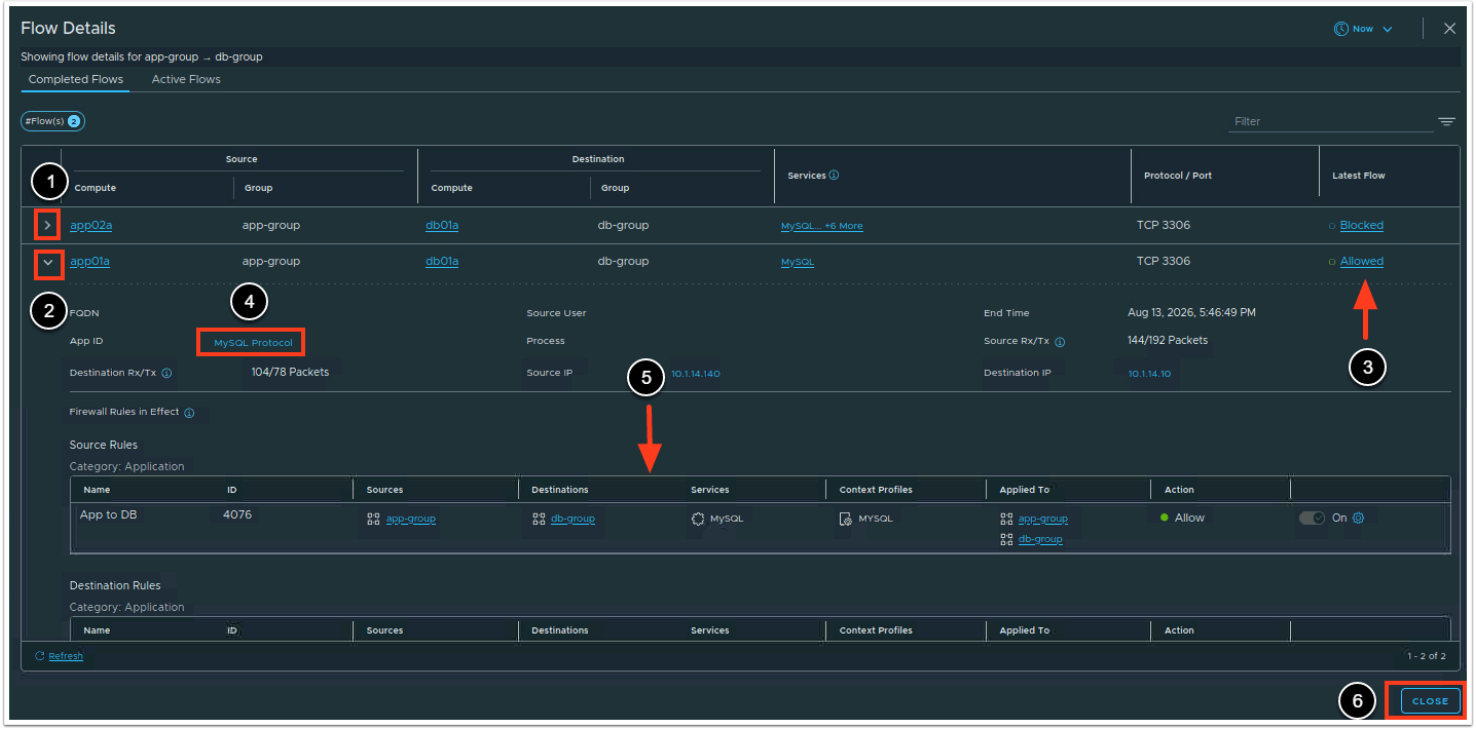
Review the Updated Flows



You will see both the blocked flow from app02a and the allowed flow from app01a. Examine the blocked flow first, then move onto the allowed flow

1. Click on the arrow to expand the app01a flow
2. The flow is now showing as **Blocked**
3. Click on the **App ID** shown to look at the details on the app ID
4. Note the **Block All** rule is blocking the flow, which is the lock down rule for the application

Examine the Flow from app01a to db01a



The allowed flow from app01a to db01a only has the MySQL protocol listed for its App ID

1. Click on the arrow to collapse the app02a flow
2. Click on the arrow to expand the app01a flow
3. The flow is still showing as **Allowed**
4. Click on the **App ID** shown to look at the details on the app ID
5. Note the **App to DB L7** rule is allowing the traffic to go through.
6. Click **Close**

If you like to can examine the available App IDs by selecting the context profile and browsing through what is available.

Conclusion

Conclusion

In this lab module, we demonstrated the configuration and implementation of Layer 7 App ID within a distributed firewall rule. By adding the Layer 7 App ID via a context profile to our distributed firewall rule, we enhanced the ability to control traffic based on specific applications rather than relying solely on port information. This additional layer of control significantly improves security posture, enabling more precise enforcement of organizational policies.

From here you can:

- Continue with the next lab module.
- Click [vlp:table-of-contents|Show Table of Contents] to jump to any module or lesson in this lab.
- End your lab and return in the future.

Module 4 – vDefend Gateway Firewall (15 Min)

Introduction

Introduction

vDefend Gateway Firewall is a software-only, layer 3-7 firewall that can be deployed on an NSX Edge node, which can be a virtual machine or a physical server. It can be used to implement zone-based controls within the private cloud, securing traffic entering and exiting the network.

vDefend Gateway Firewall complements the Distributed Firewall to provide enhanced protection for your environment, and offers a range of features such as URL filtering, FQDN Analysis, TLS inspection and more. In this module, we will explore Geo-Fencing feature introduced in vDefend Firewall 9.0 to protect your data center workloads for malicious foreign actors residing in specific countries.

Explore Gateway Firewall

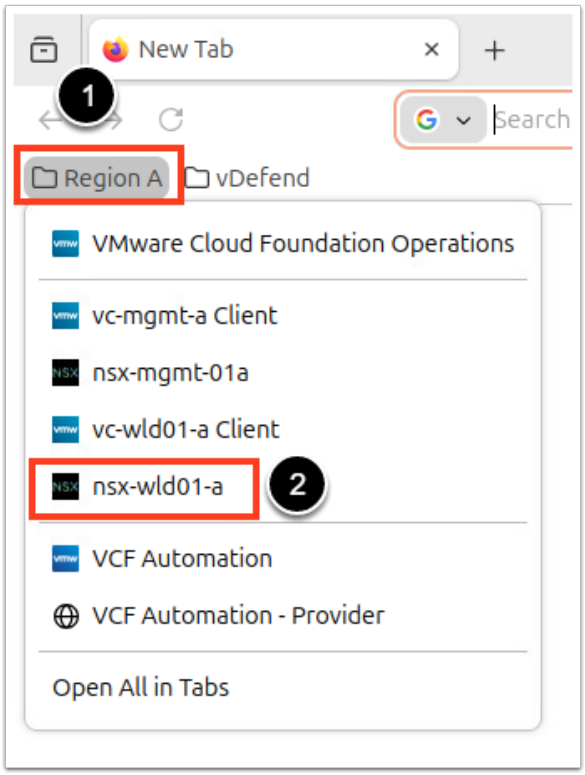
Launch Firefox



If you are continuing from the previous model you can skip the next few steps as you are already logged into NSX Manager. If you are starting the lab on this module please follow to next 4 steps.

1. Launch Firefox by clicking on the Firefox icon in the lower left-hand corner of the console

Open NSX Manager



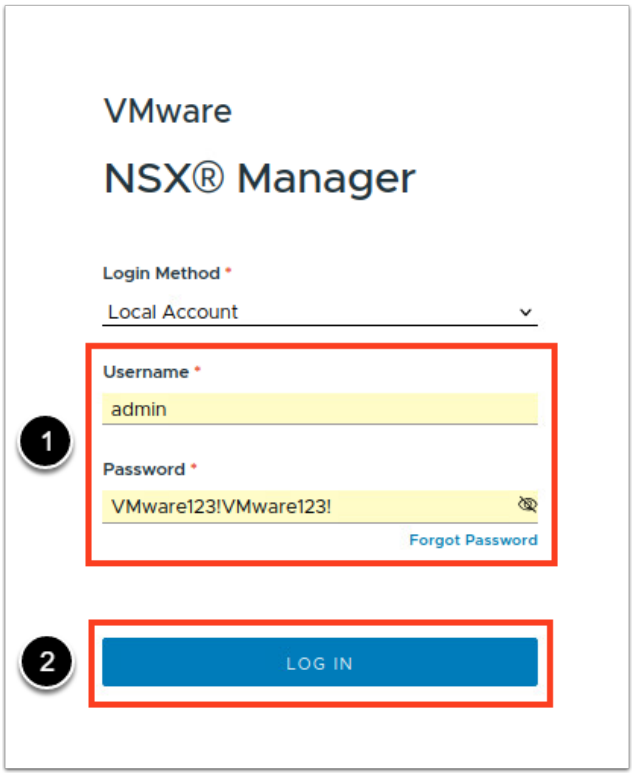
1. Select Region A from the Bookmark Bar
2. Select **nsx-wld01-a** from the drop-down menu

Change Login Method for NSX Manager



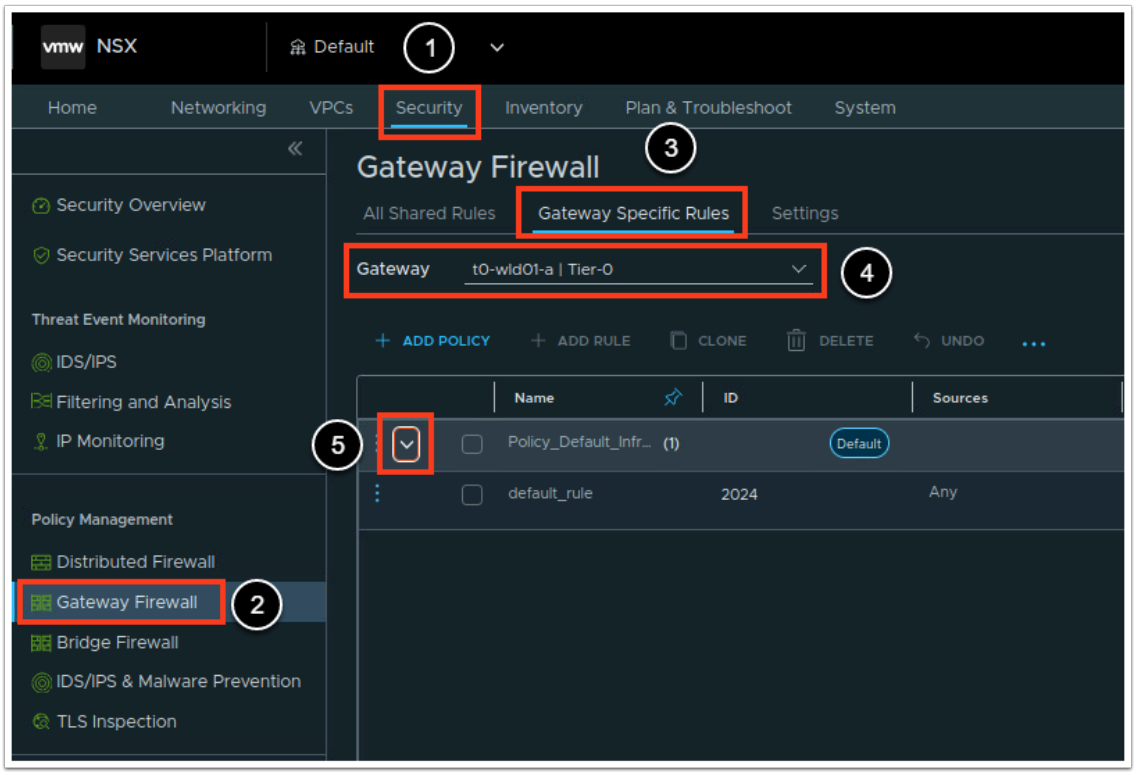
1. Click the arrow to open Login Method drop-down menu
2. Select Local Account

Login to NSX Manager



1. Verify the username **admin** and password **VMware123!VMware123!** have been prefilled
2. Click **LOG IN**

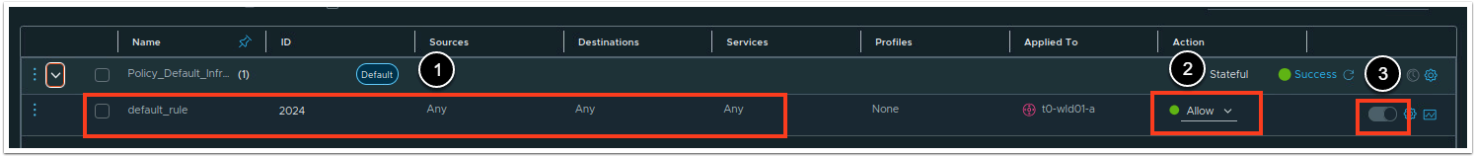
View Gateway Firewall



The Gateway Firewall section provides you with the ability to create firewall policies and rules on a specific (Gateway Specific Rules) or broadly (All Shared Rules) depending on the scope and need of that policy. A Gateway Firewall rule differs from a Distributed Firewall Rule in that it only applies to traffic that transits a T0 and/or T1 Gateway.

1. Select **Security** from the menu on the top
2. Select **Gateway Firewall** from the menu on the left
3. Select **Gateway Specific Rules** from the Gateway Firewall menu
4. The t0-wld-a | Tier-0 Gateway is currently selected. You can click the **drop-down arrow** to view other gateways and their rules
5. Expand the Default policy by clicking on the toggle to the left of its name

View Gateway Firewall Default Rule



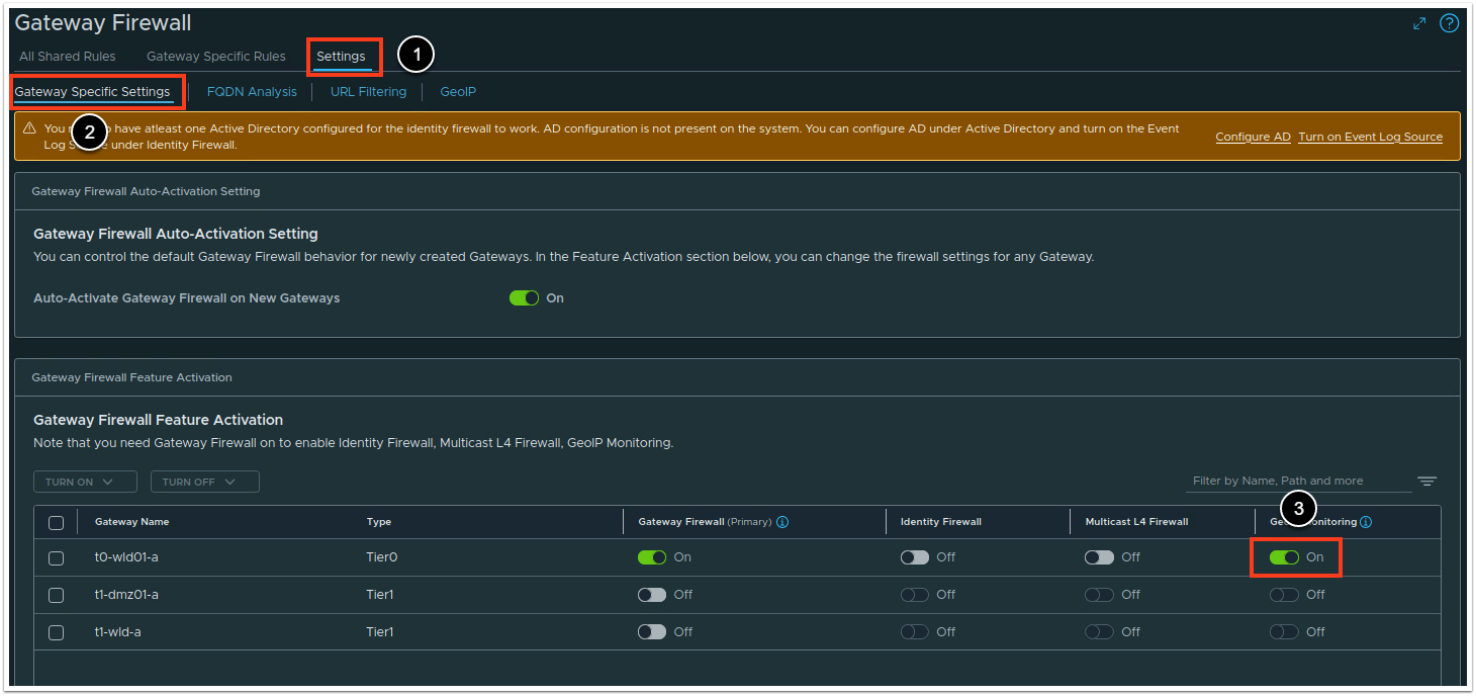
Currently there are no other firewall policies configured for the T0 router

1. This is the Default Rule for this Gateway Firewall.You cannot place any policies or rules below it, so it will determine this Gateway Firewall default action. The Source, Destination, and Services for this rule are “Any”. These fields cannot be changed.
2. The action is currently set to Allow on the default rule. Any traffic flows not matching a rule above the default rule will be processed by the default rule and will be allowed through this gateway firewall.
3. The rule status toggle button is set permanently to an enabled state. It cannot be disabled.

Gateway Firewall Geo-Fencing

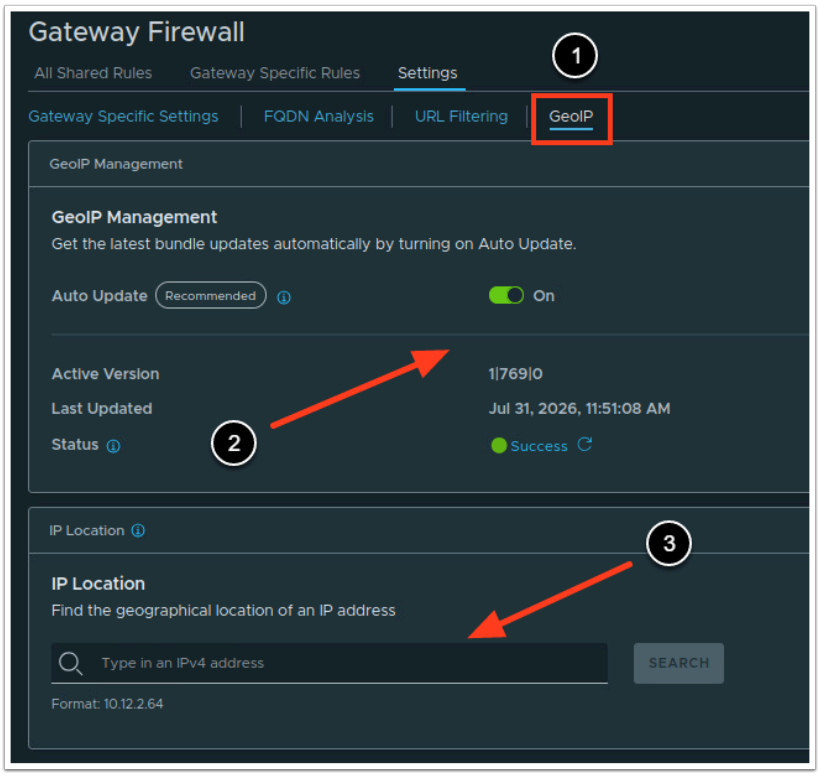
The first Gateway Firewall policy you will configure will be the Geo-Fencing. You will use Geo-IP based filtering to protect your application from malicious users in 2 geographical regions.

Enable Geo IP Monitoring



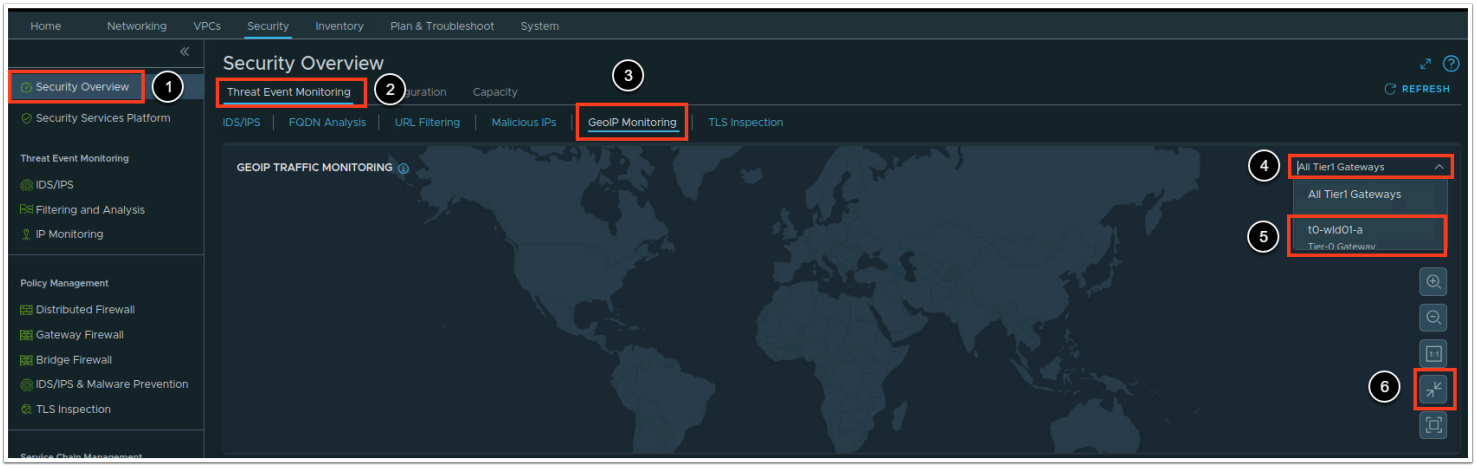
1. Select **Settings** from the Gateway Firewall menu
2. Select **Gateway Specific Settings** from the Settings menu
3. Enable GeoIP Monitoring by clicking the toggle for t0-wld01-a

Review GeoIP Settings



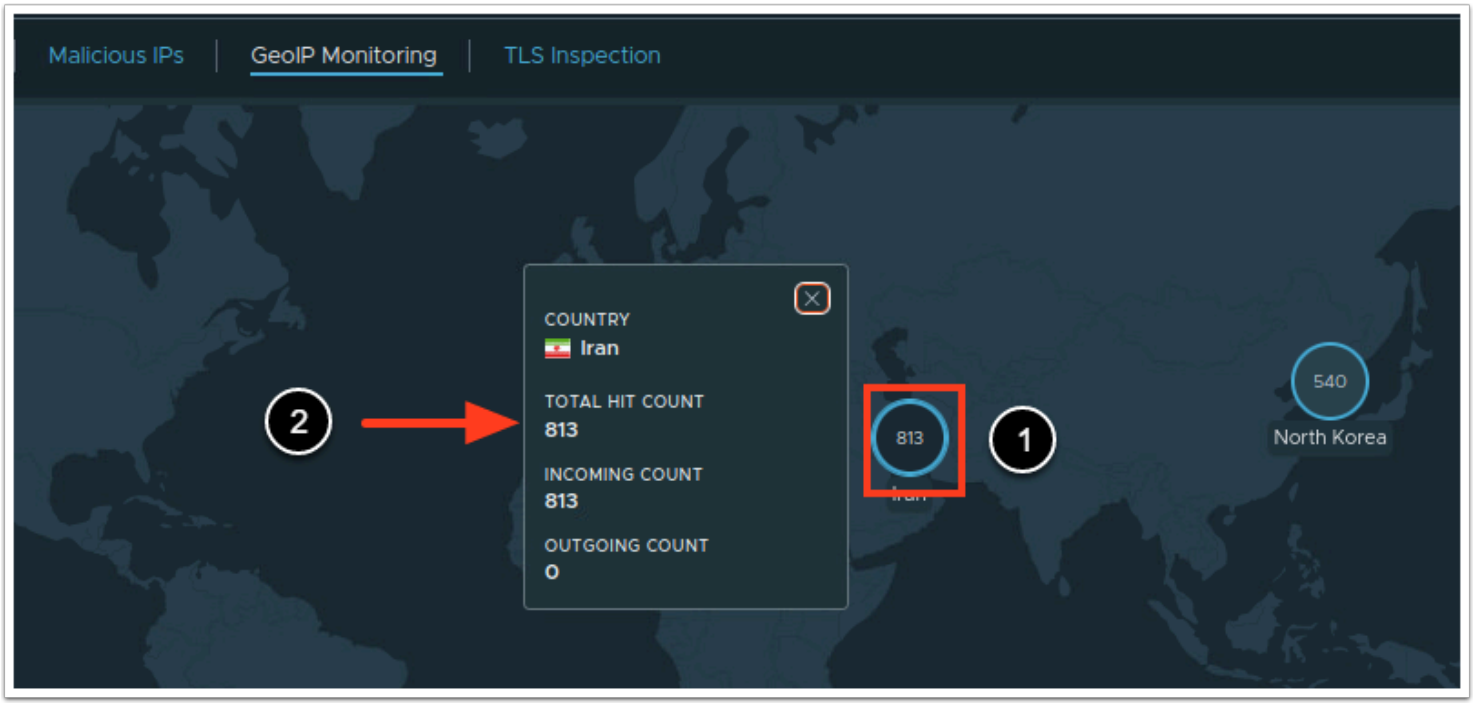
1. Select **GeoIP** from the Settings menu
2. Take note of the Settings for GeoIP Management, you can chose to have the IP database updated automatically or not.
3. In the IP Location section you can search to get the GeoIP of any publicly registered IP address.

Navigate to Geo-Fencing Monitoring



1. Select **Security Overview** from the menu on the left
2. Select **Threat Event Monitoring** from the Security Overview Menu
3. Select **GeoIP Monitoring** from the Threat Event Monitoring Menu
4. Click the **drop-down arrow** to view list of gateways
5. Select **t0-wld01-a** from the list
6. Click the **Fit to View button**

Explore GeoIP Traffic Monitoring Map

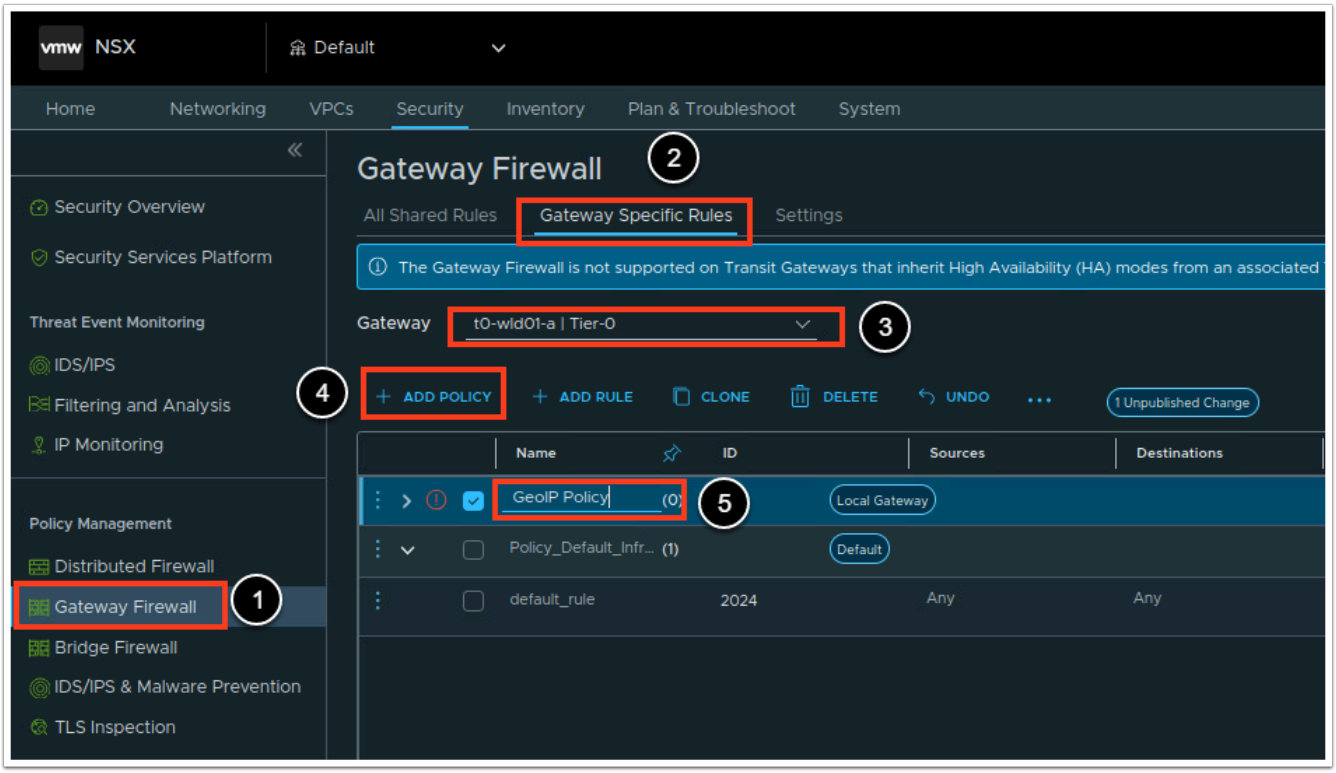


There are two countries that have a hit count number displayed, North Korea and Iran.

1. Click the **Iran** icon on the map to see the hit counts for flows originating in Iran
2. Hit count details for Iran display in the viewing canvas

> Repeat the above for North Korea as well.

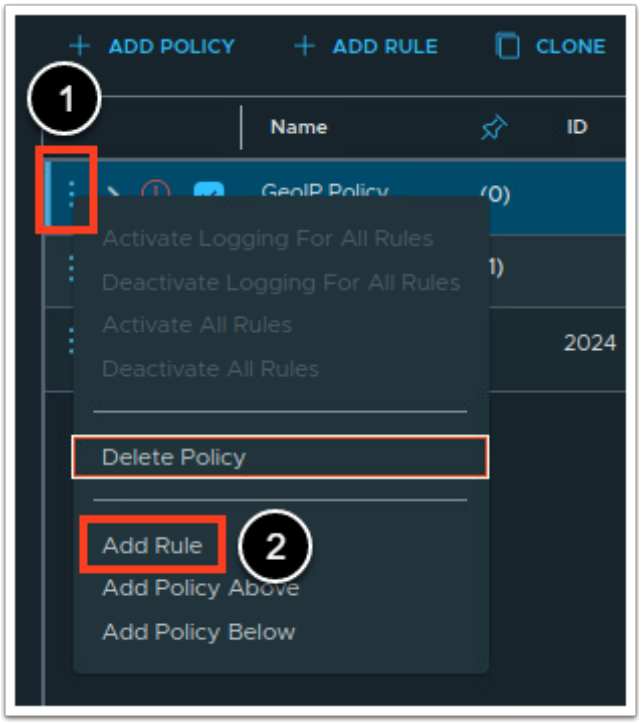
Create a GeoIP Policy



Now that you have seen that flows are originating from 2 foreign regions you will create a policy to block traffic from that region to your applications.

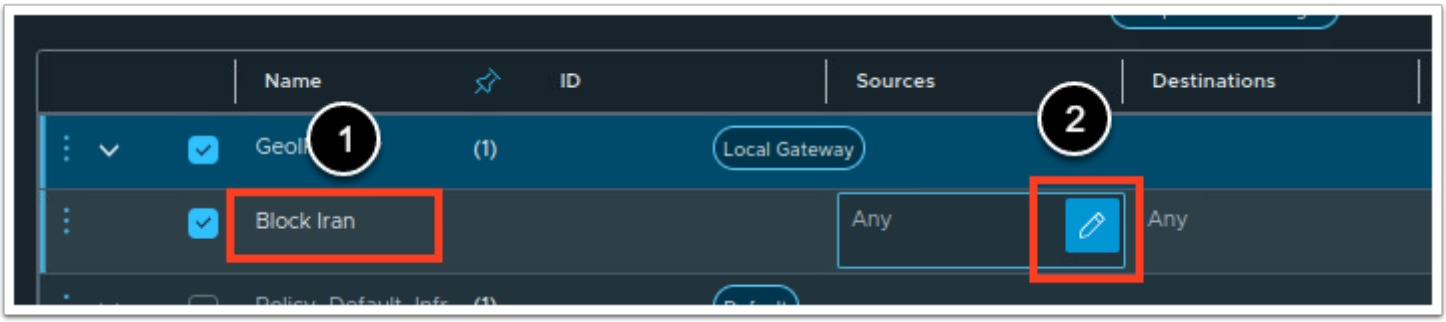
1. Select **Gateway Firewall** from the menu on the left
2. Select **Gateway Specific Rules** from the Gateway Firewall Menu
3. Ensure to-wld01a | Tier-0 is selected for **Gateway**
4. Click **+ ADD POLICY**
5. Enter **GeoIP Policy** for the policy's name and click

Add the Block Iran Rule



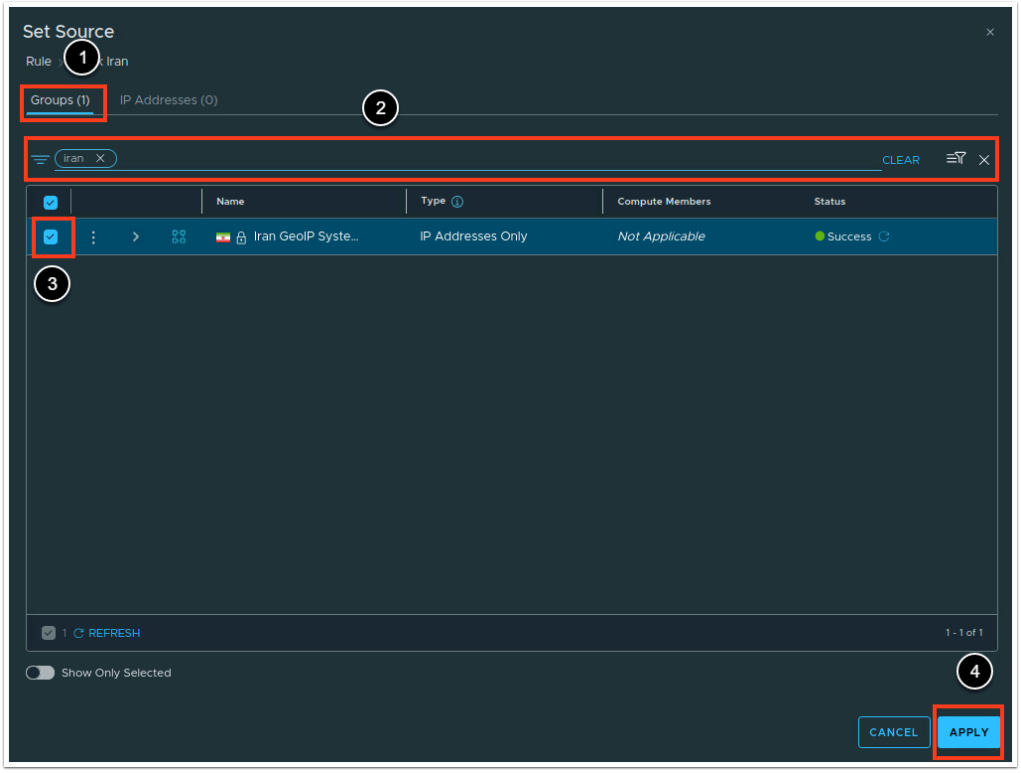
1. Click on the 3 dots on the left of the GeoIP policy to show the menu
2. Select **Add Rule**

Name the Rule and Define the Source



1. Enter **Block Iran** for the rule name
2. Click on the **Pencil Icon** under sources

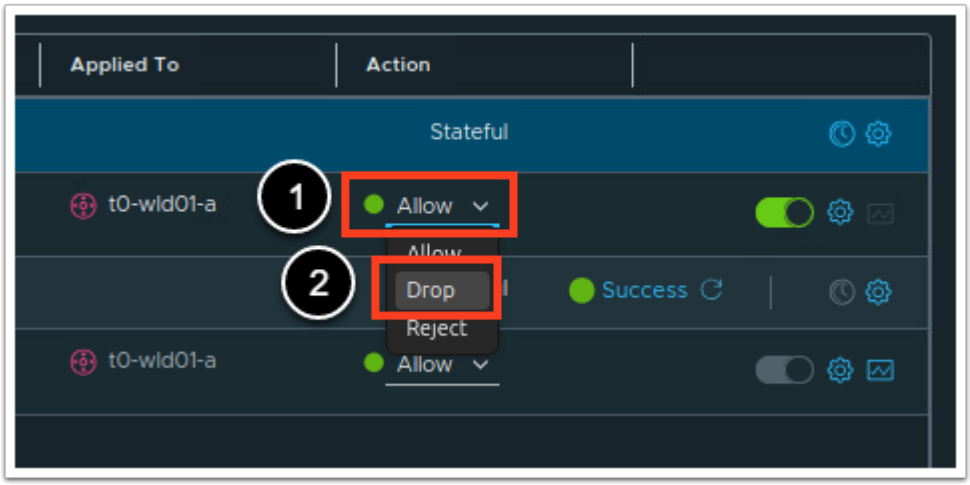
Select the Iran GeoIP group



Each GeoIP region has its own group in NSX. These groups are updated automatically as shown earlier in this task. You will select the Iran GeoIP region here.

1. Ensure you are in the **Groups** view
2. Enter **Iran** in the search bar to filter out all the other regions
3. Select the **Iran GeoIP** group
4. Click **Apply**

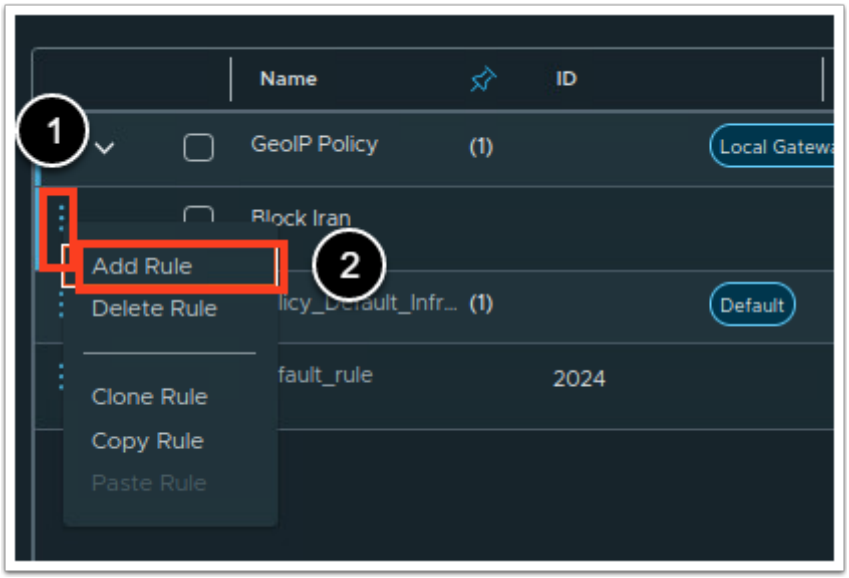
Change Action to Drop



The final step will be to drop all of the traffic sourced from Iran.

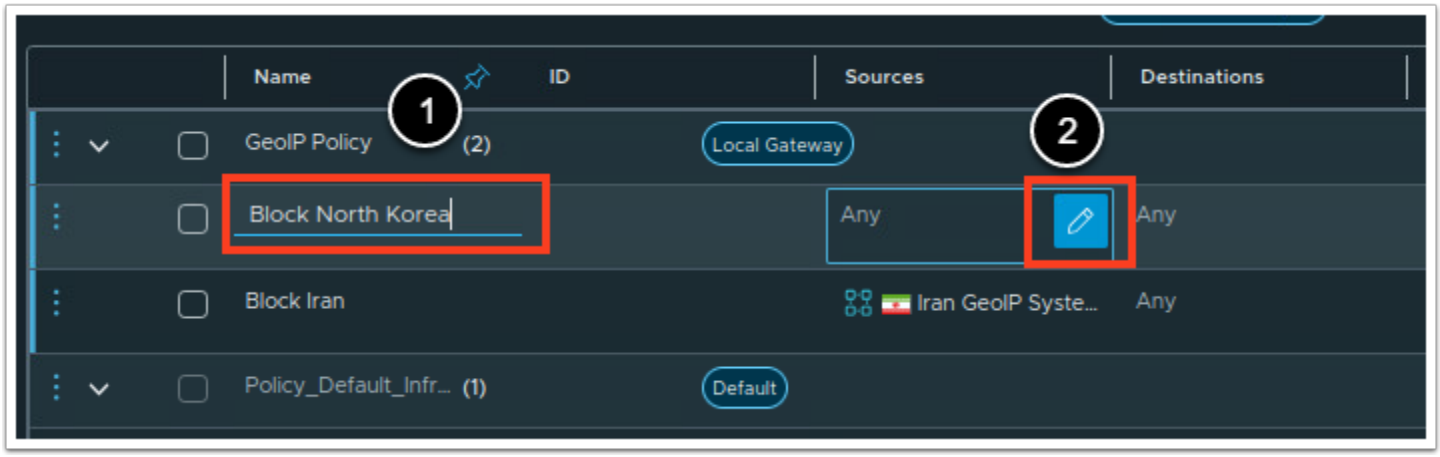
1. Click on the Action for the rule (should say allow)
2. Select **Drop** from the drop down menu

Add the Block North Korea Rule



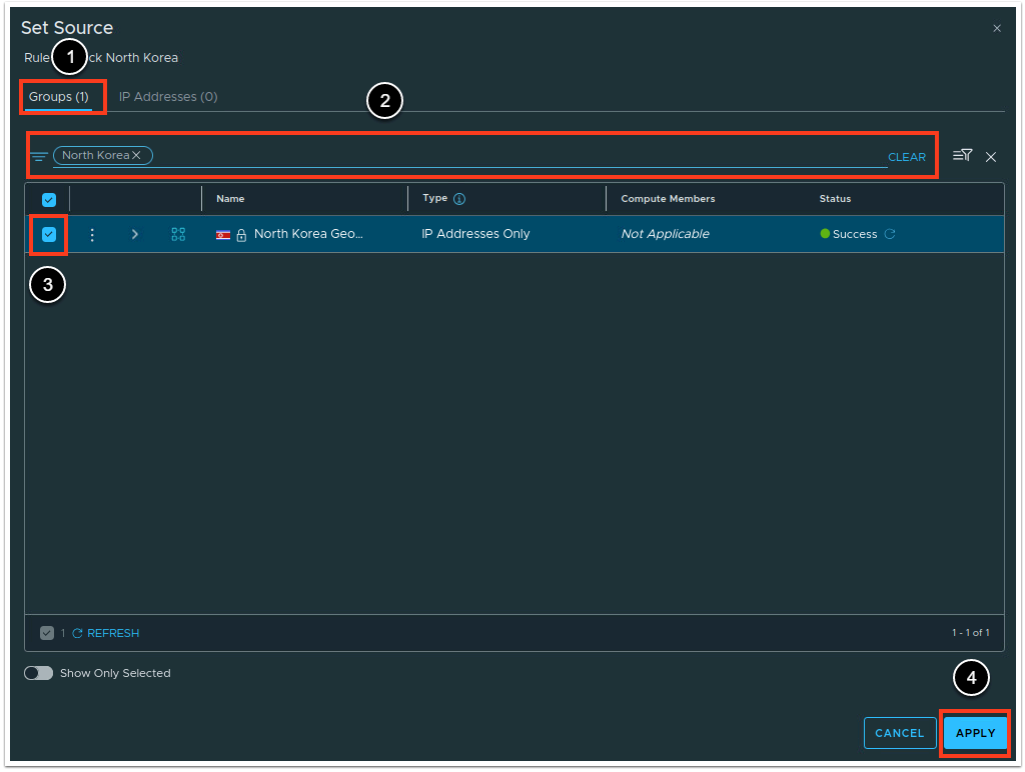
1. Click on the 3 dots on the left of the Block Iran rule to show the menu
2. Select **Add Rule**

Name the Rule and Define the Source



1. Enter **Block North Korea** for the rule name
2. Click on the **Pencil Icon** under sources

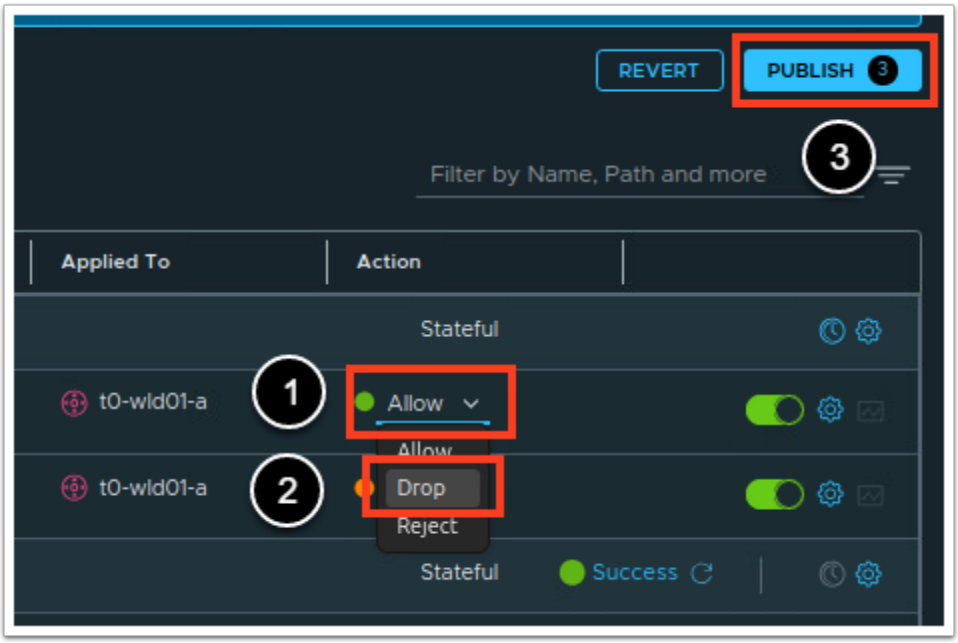
Select the Iran GeoIP group



Each GeoIP region has its own group in NSX. These groups are updated automatically as shown earlier in this task. You will select the Iran GeoIP region here.

1. Ensure you are in the **Groups** view
2. Enter **North Korea** in the search bar to filter out all the other regions
3. Select the **North Korea GeoIP** group
4. Click **Apply**

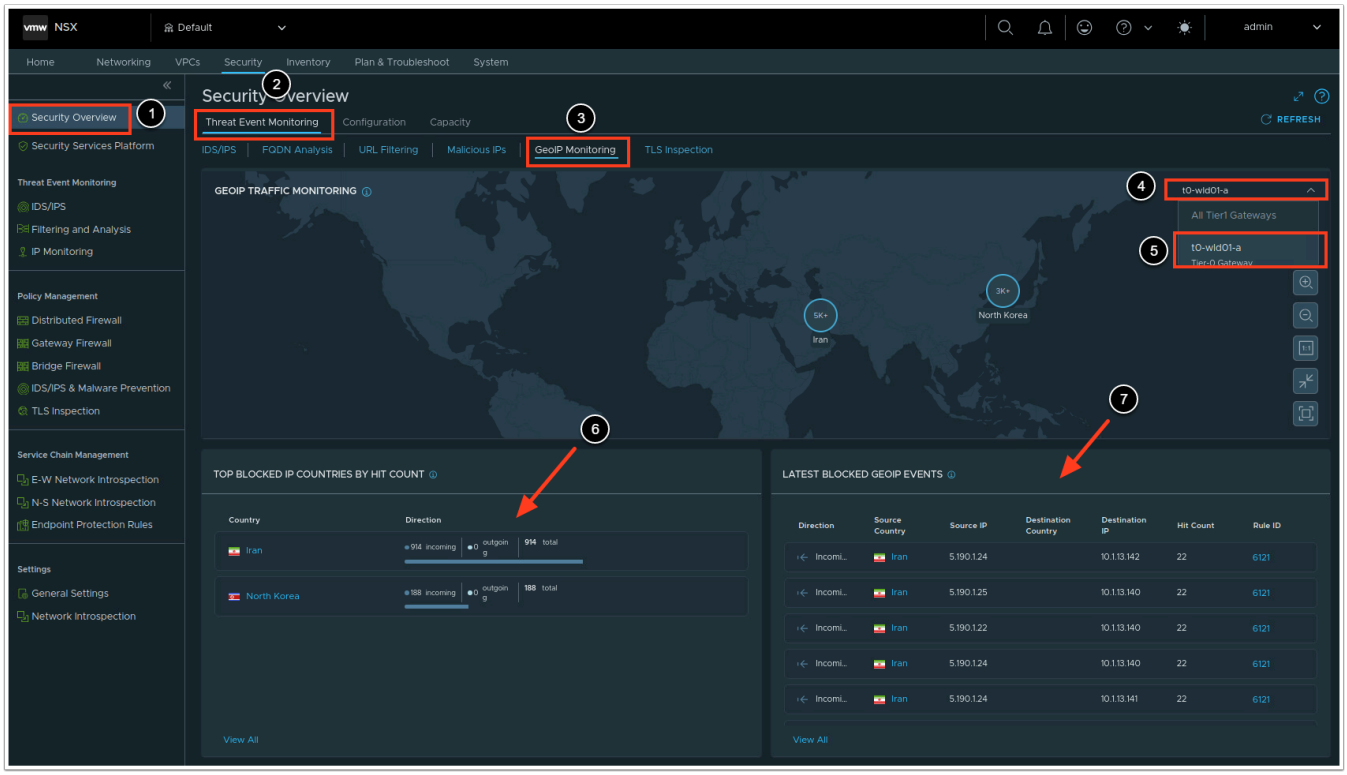
Change Action to Drop and Publish



The final step will be to drop all of the traffic sourced from Iran.

1. Click on the Action for the rule (should say allow)
2. Select **Drop** from the drop down menu
3. Click **Publish**

Validate GeoIP Rules are Enforced



Now that the rules have been published you will return to the GeoIP Security Overview section to see the impact of the rules.

1. Select **Security Overview** from the menu on the left
2. Select **Threat Event Monitoring** from the Security Overview Menu
3. Select **GeoIP Monitoring** from the Threat Event Monitoring Menu
4. Click the **drop-down arrow** to view list of gateways
5. Select **t0-wld01-a** from the list
6. On the bottom left take note of the top countries with IPs blocked by hit count
7. On the bottom right take note of the most recent IPs beling blocked

Conclusion

Conclusion

In this lab module we successfully demonstrated the use of Geo IP filtering and monitoring features for vDefend Gateway Firewall. By configuring the Geo IP filtering and monitoring on the Gateway Firewall, we were able to monitor and restrict traffic based on the origin country of IP addresses. This allowed us to simulate real-world scenarios where organizations may want to block traffic from high-risk regions or limit access to specific geographic locations for compliance or security purposes. Geo IP filtering and monitoring adds an additional layer of security to vDefend Gateway Firewall to help organizations adapt to ever-evolving threat landscapes.

From here you can:

- Continue with the next lab module.
- Click [vlp:table-of-contents|Show Table of Contents] to jump to any module or lesson in this lab.
- End your lab and return in the future.

Module 5 – vDefend DFW Rule Analysis (10 Min)

Rule Analysis Overview

Introduction

In complex firewall environments, managing a vast number of rules can lead to inefficiencies, contradictions, and potential security gaps. The Rule Analysis feature is designed to address these challenges, ensuring a robust and efficient security posture within your distributed firewall infrastructure.

This feature helps you identify critical issues such as rule contradictions - where conflicting policies might lead to unintended traffic behavior, as well as redundancy - where duplicate or unnecessary rules may clutter configurations and impact performance.

The results of the Rule Analysis are presented on an intuitive anomaly dashboard, enabling you to easily visualize and address the detected issues. This approach simplifies rule management and ensures that distributed firewall rules reflect your desired security posture.

With Rule Analysis, you can:

1. **Streamline distributed firewall policies**: Evaluate your distributed firewall rules for potential issues and optimization opportunities. Assess the configuration anomalies that are flagged and make relevant changes.
2. **Manage detected anomalies**: Use the intuitive anomaly dashboard to visualize, categorize, and address the detected anomalies.
3. **Set an analysis schedule**: Customize your analysis as a scheduled task to assess your complete distributed firewall configuration or run the analysis on demand for immediate insights.
4. **Simplify rule management**: A cleaner and more consistent rule set is easier to manage. You can quickly query and navigate to the distributed firewall rules associated with the identified anomalies, allowing for prompt and targeted action.

Understanding Rule Analysis

The Rule Analysis feature helps identify issues such as rule contradictions, redundancies, and irrelevance, among other anomalies. The feature checks for overlaps across a 6-tuple: source, destination, applied to, services, context profiles, and direction, in conjunction with the specified action.

When two or more rules exhibit an overlap across one or more of these fields, the rules may be classified into one of the following anomaly categories, depending on their actions, precedence, and the extent of the overlap. Similarly, rules that are permissive and irrelevant are also flagged.

Initiate Rule Analysis

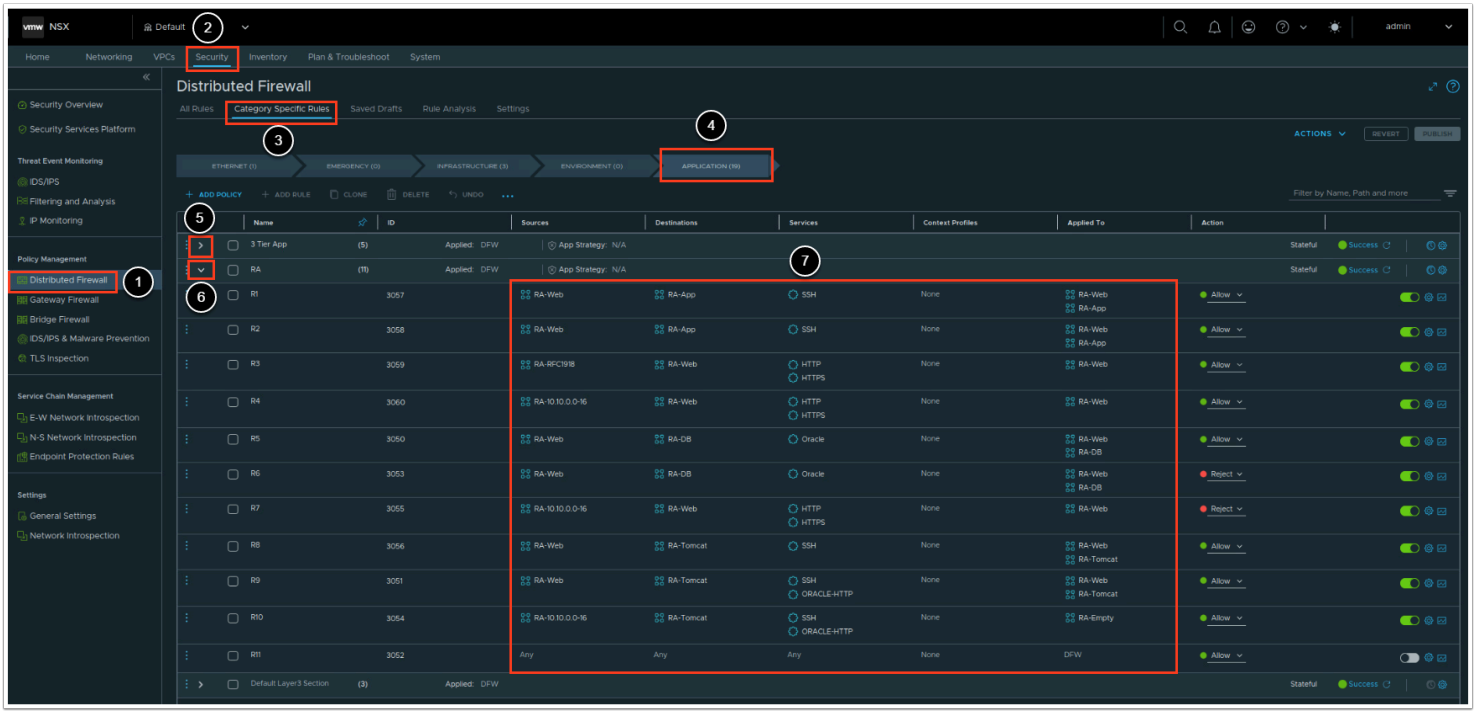
Rule Analysis is done inside NSX Manager as opposed to the Security Services Platform

Return to NSX Manager



1. In Firefox select **NSX**

Review Security Policies in the NSX Manager

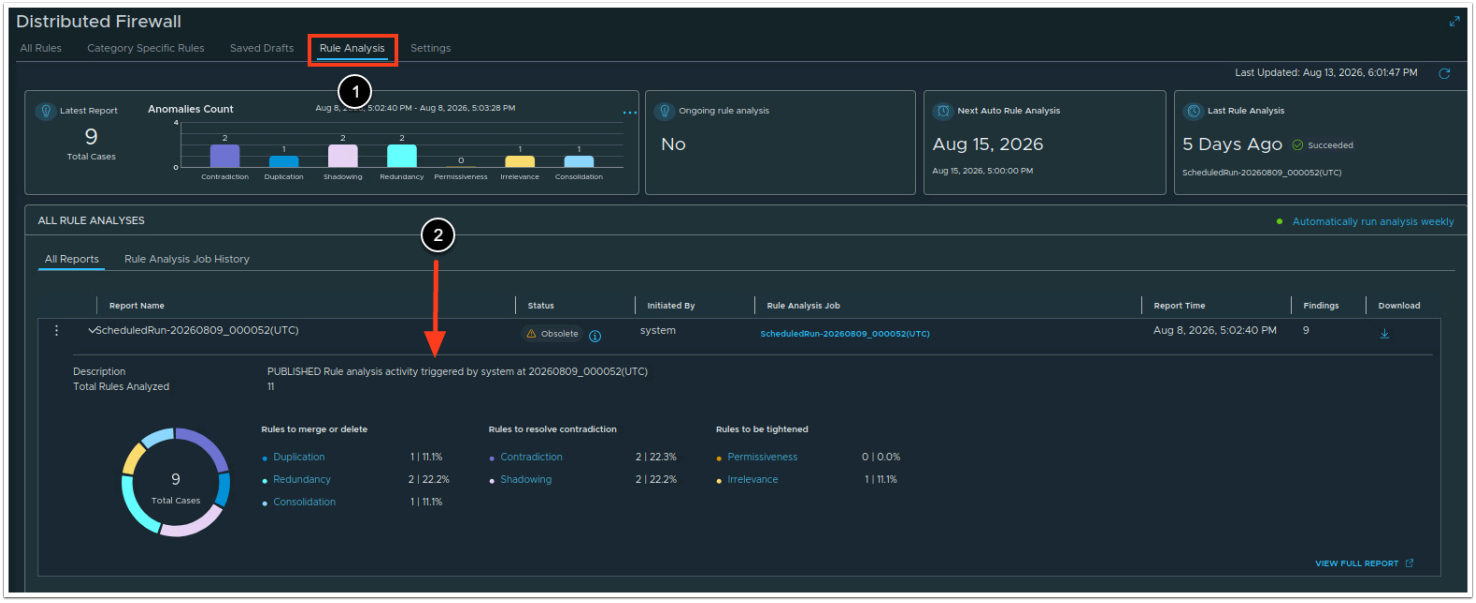


Before we start running rule analysis lets first take a look at the rules that are in place to support this analysis. These rules will be part of the RA policy.

1. Select **Security** from the menu at the top
2. Select **Distributed Firewall** from the menu on the left

3. Ensure **Category Specific Rules** is selected for the **Distributed Firewall** menu
4. Cnsure you are on the **APPLICATION** Section
5. Collapse the **3 Tier App** policy if it is expanded by clicking on the triangle
6. Expand the **RA** policy at the top by clicking on the triangle
7. Review the preconfigured policies that are designed to illustrate criteria of Rule Analysis

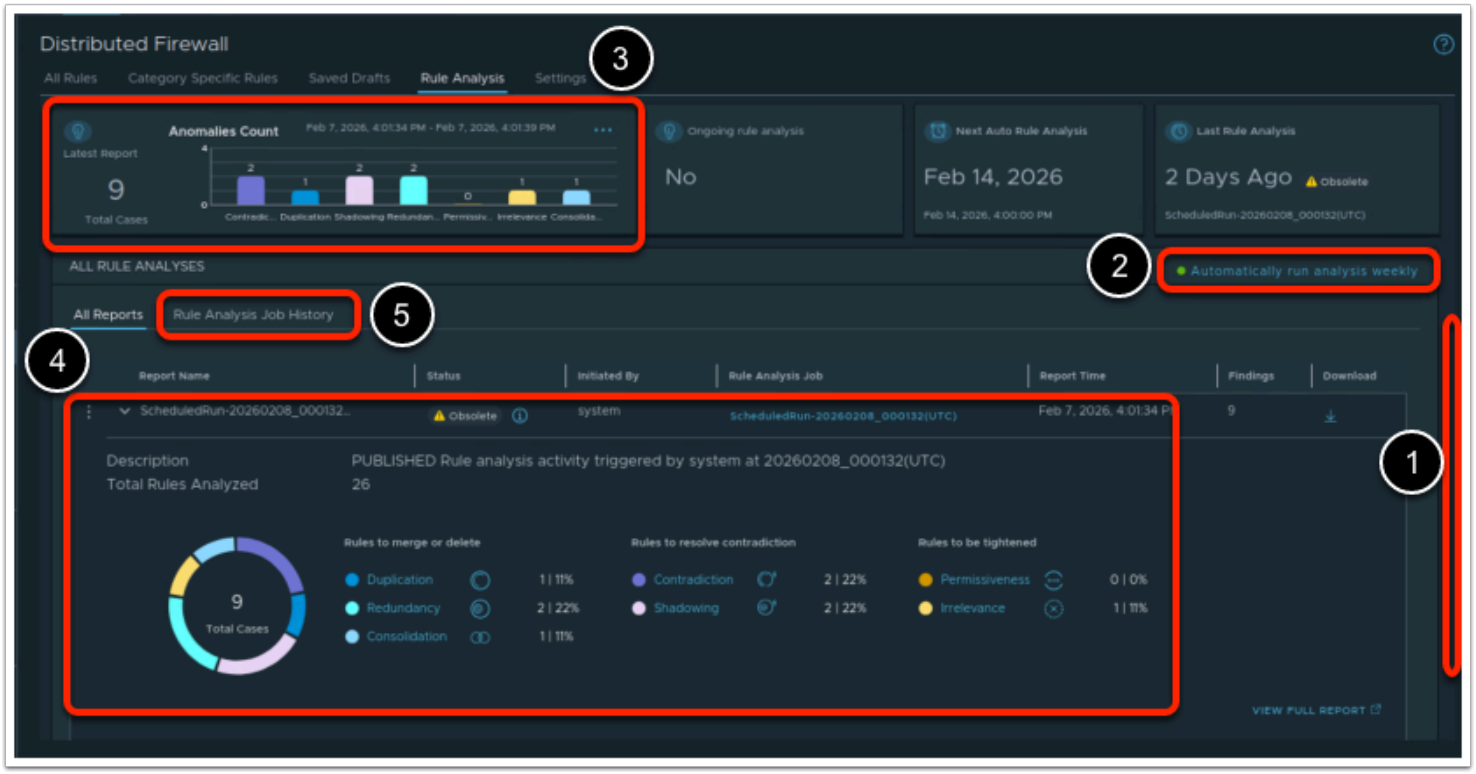
Start a Rule Analysis

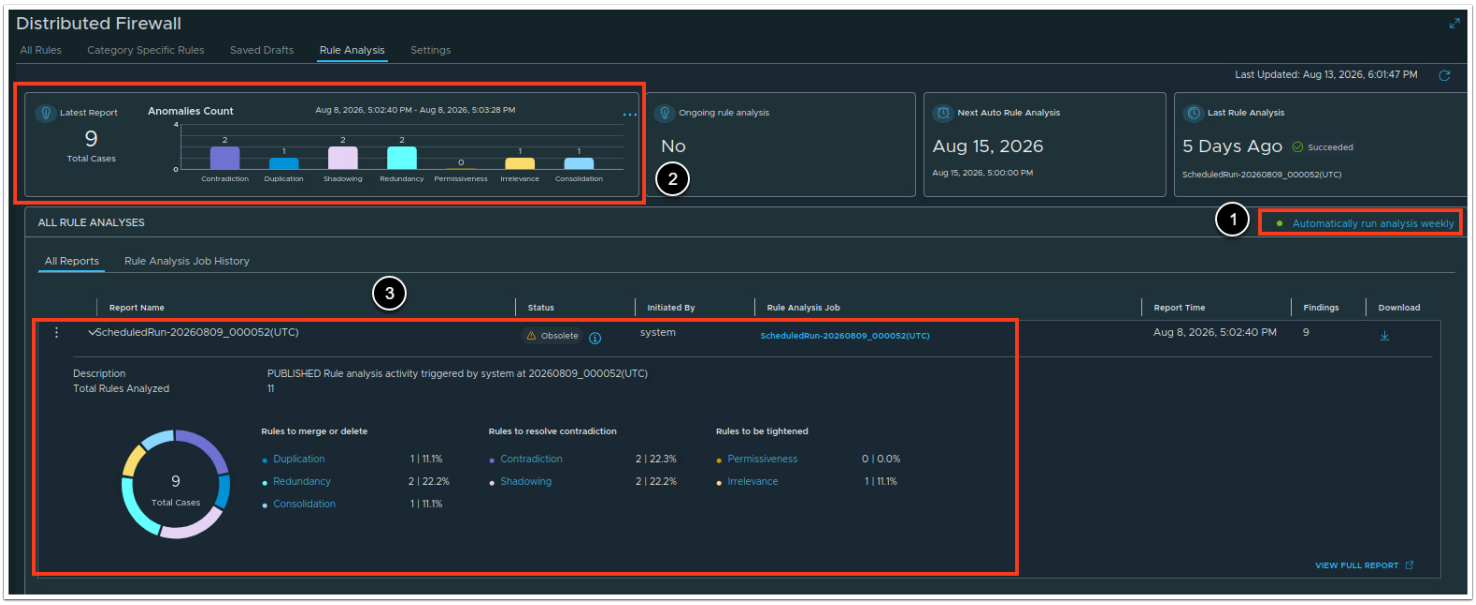


1. Select **Rule Analysis** in the Distributed Firewall menu
2. A Rule Analysis should already have been run and will be present

> The status will indicate Obsolete. This error can be ignored as the policy that is being evaluated is static and created for demonstration purposes

Review Rule Analysis Dashboard





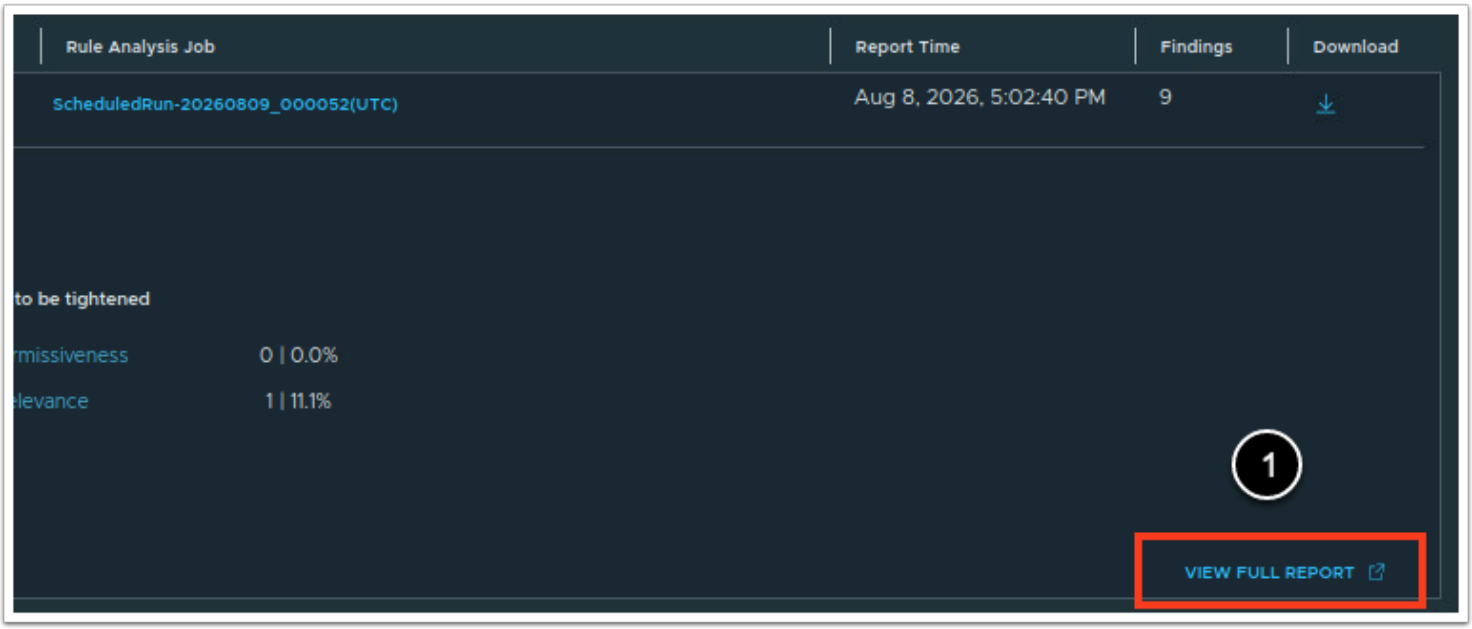
1. The default settings once enabling Rule Analysis is to run a weekly report, we see it enabled here.
2. Here we see the Anomaly Count summary of the latest report indicating 9 cases of a mix of contradicting, duplicate, shadow and/or redundant rules.

3. This is where we find the latest report and the findings.

Review Rule Analysis Results

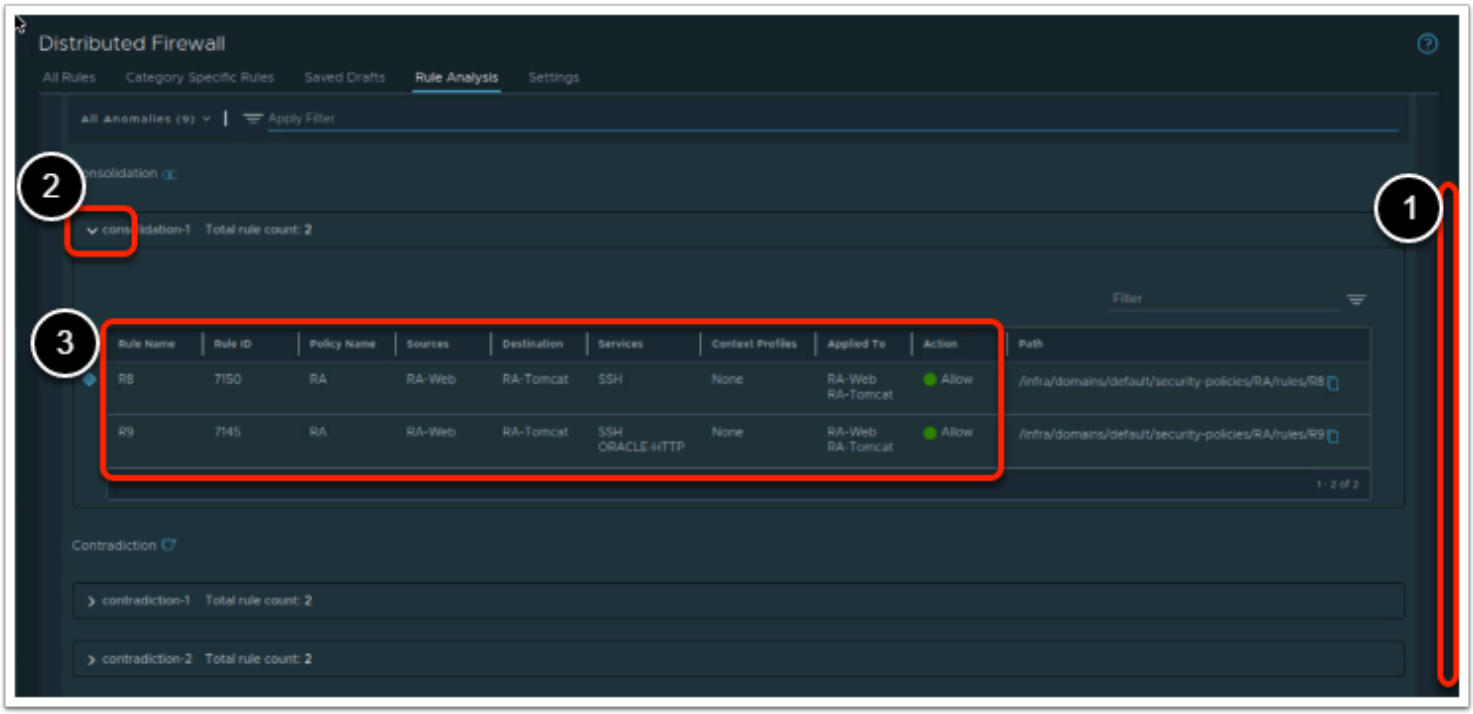
There are 7 categories that analyze rules for in this section you will have an overview of what those are.

View the Report



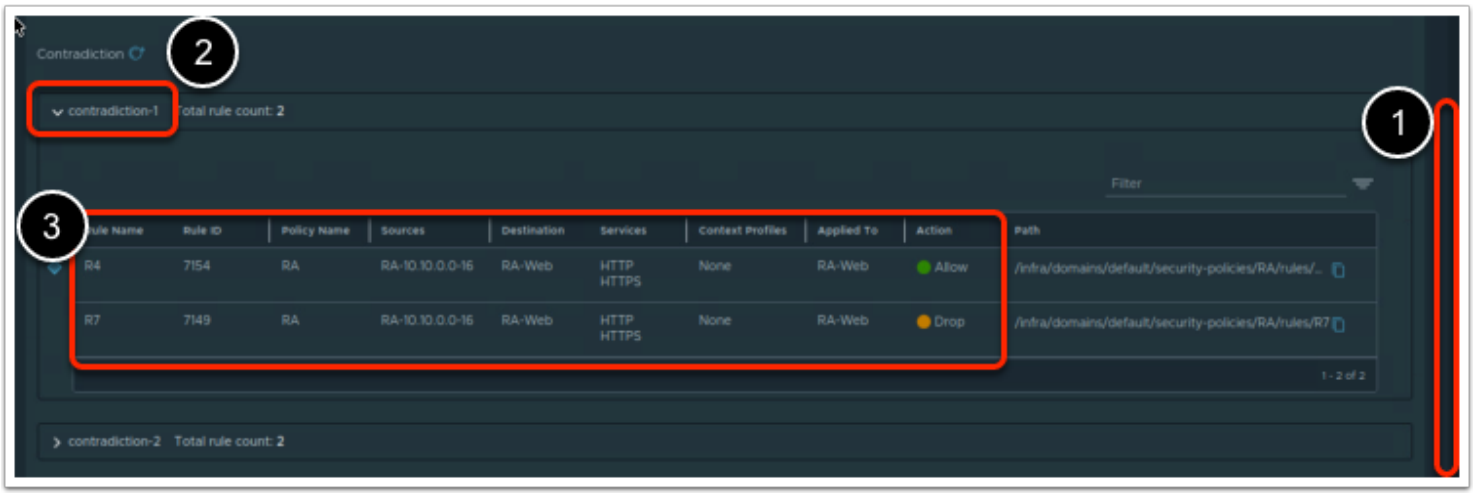
1. Click **VIEW REPORT,** this opens in a new browser TAB and you should automatically be taken to there.

Review the Report - Consolidation Recommendation



1. Click on the scroll bar on the right and page down.
2. Click on the triangle next to Consolidation and expand the findings.
3. Here we see two rules with the same Source, Destination, Applied To and Action fields but with different Services. The system has identified these two rule are candidates to consolidate.

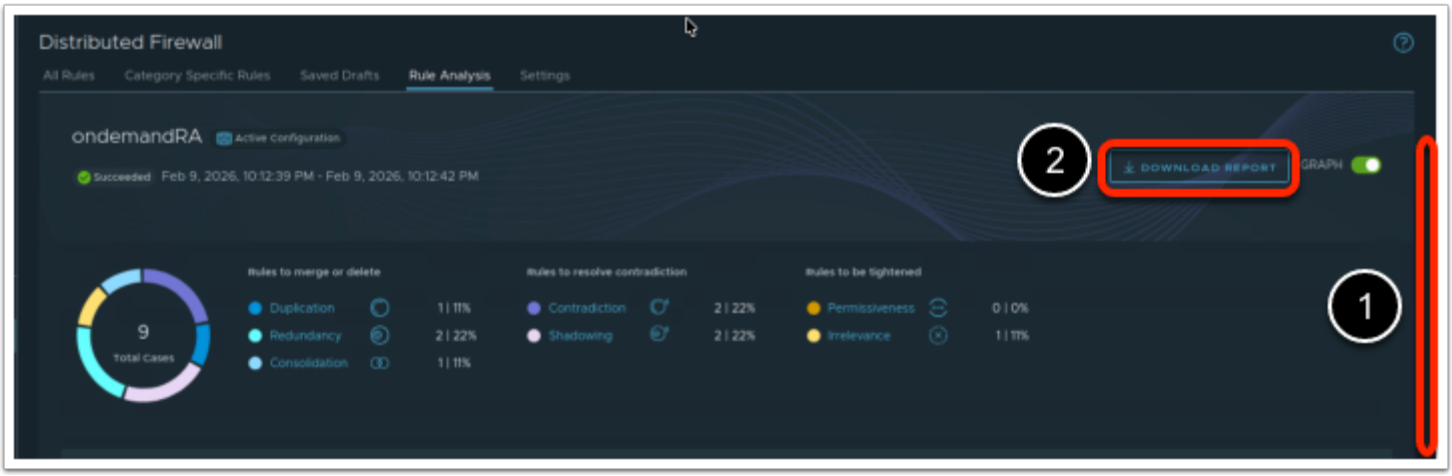
Review the Report - Contradiction Analysis



1. Using the scroll bar on the right and page further down if needed.
2. Click on the triangle to expand the Contradiction findings
3. Here we two rules which are identified as contradicting. The Source, Destination, Services and Applied To fields are all the same but the Action are different and so these rules are contradicting one another.

> Feel free to scroll through some of the others fields and see how the system performs the analysis

Download Report



The report can be downloaded

1. Use the scroll on the right and page up to the top.
2. Here we can download the report as a CSV file for offline analysis and review.

Conclusion

Conclusion

In this module you experienced how Rule Analysis feature helps identify issues such as rule contradictions, redundancies, and irrelevance, among other anomalies.

You experienced how the result of the Rule Analysis is presented on a dashboard in the NSX Manager user interface.

The Lab is Complete

You have reached the end of this module and lab.

How to End Lab

You have reached the end of this module and lab.

End of Lab Manual



vmware®

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Item No: 51227-vcf-wp-hands-on-labs-manual-2025, Jan-25

Module 6 - AI Assistant for VMware vDefend (20 Min)

AI Assistant For VMware vDefend

AI Assistant Overview

AI Assistant is a Generative AI-powered feature that is integrated into the vDefend Security Services Platform.

AI Assistant is designed to simplify security operations, accelerate troubleshooting, and streamline threat analysis using natural language interactions. You can use AI Assistant for Firewall to simplify and troubleshoot firewall operations, and AI Assistant for Threat Defense for threat explanations and remediations.

To serve different operational needs, the feature is divided into two distinct modules: AI Assistant for Firewall and AI Assistant for Threat Defense. In this lab we will be focusing on AI Assistant for Firewall.

AI Assistant for Firewall

AI Assistant for Firewall Overview

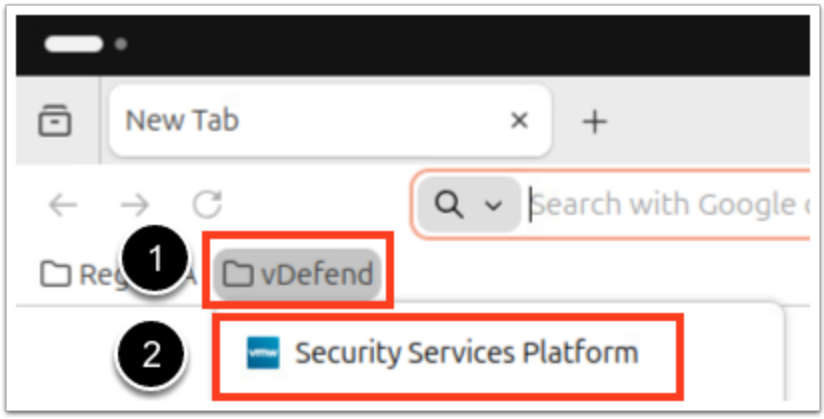
AI Assistant for Firewall simplifies firewall operations and troubleshooting by providing AI-generated insights that are based on live configurations, real-time policy violations, and alerts. AI Assistant for Firewall provides an AI-based natural language interface to help users gain operational insights for vDefend Distributed Firewall, metrics, rule analysis, and API reference retrieval. AI Assistant for Firewall does not create vDefend Distributed Firewall or IDS/IPS rules. The primary users for the AI Assistant for Firewall are Enterprise Administrators and Security Administrators.

Key Use Cases

- **Operational Insights:** Provides AI-generated insights that are based on live firewall configurations, real-time policy violations, and firewall alerts. Note that the Gateway Firewall (GFW) is not supported.
- **Broad Firewall Support:** Delivers knowledge insights for Distributed Firewall (DFW), Gateway Firewall (GFW), Identity Firewall (IDFW), IDS/IPS , App IDs, and Context Profiles.
- **Metrics & Analysis:** Assists with operations that are related to firewall metrics, capacity dashboards, and rule analysis.
- **Flow Visibility:** Offers operational intelligence for flow insights, parsing data regarding VMs, IPs, Groups, Tags, Services, and ports and protocols.
- **API Guidance:** Provides API reference lookups. It acts as a lookup tool only and does not generate code.
- **Inventory:** Explains Inventory and grouping objects to provide full context around security policies.

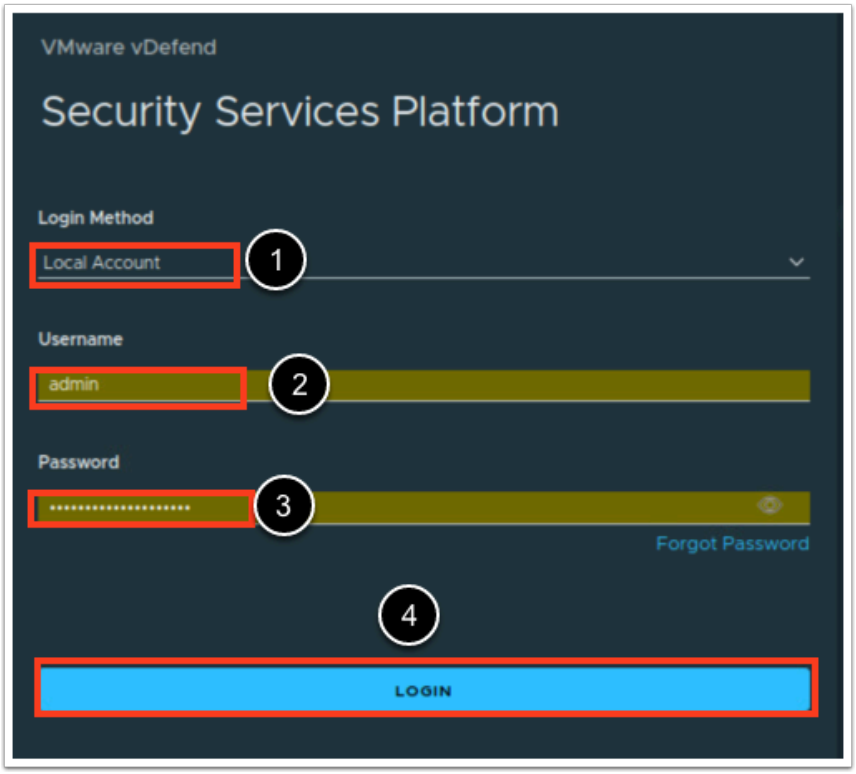
Activate AI Assistant for Firewall

Access Security Services Platform



1. From the Firefox browser, click on the **vDefend** Folder.
2. Select **Security Services Platform**.

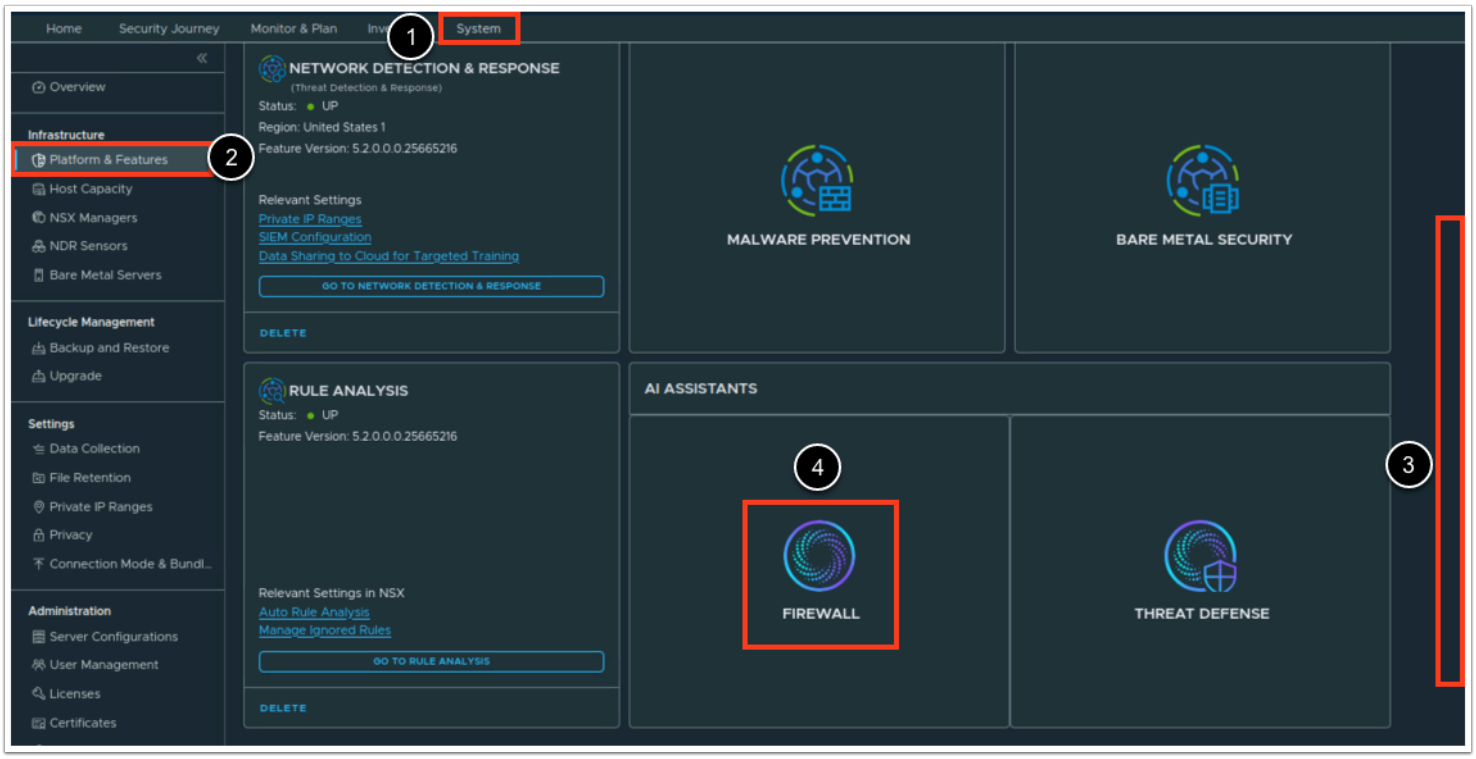
Authenticate in Security Services Platform



The Login Method, Username and Password should already be populated. If not, follow the steps below.

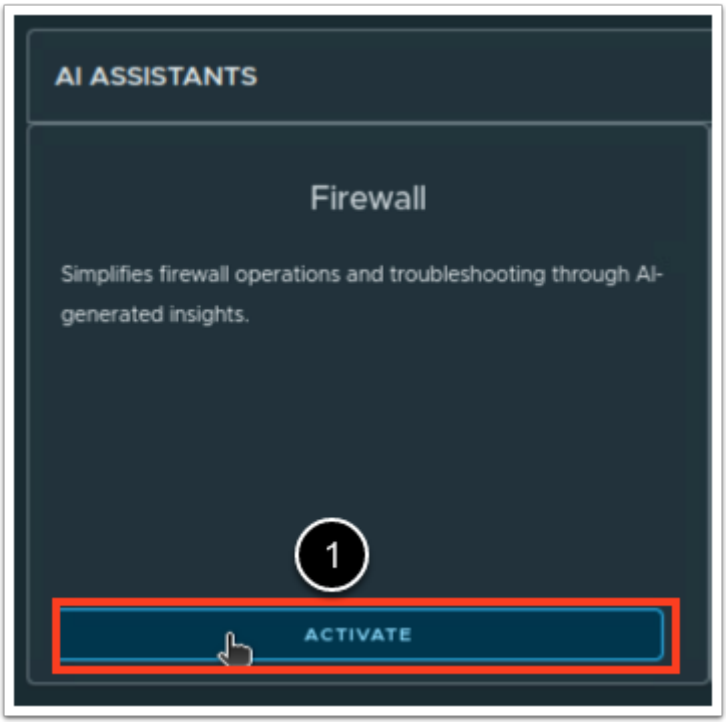
1. Confirm Login Method is **Local Account.**
2. Confirm Username is **admin.**
3. Confirm the password is **VMware123!VMware123!**
4. Click **LOGIN**

Access Platform & Features in SSP



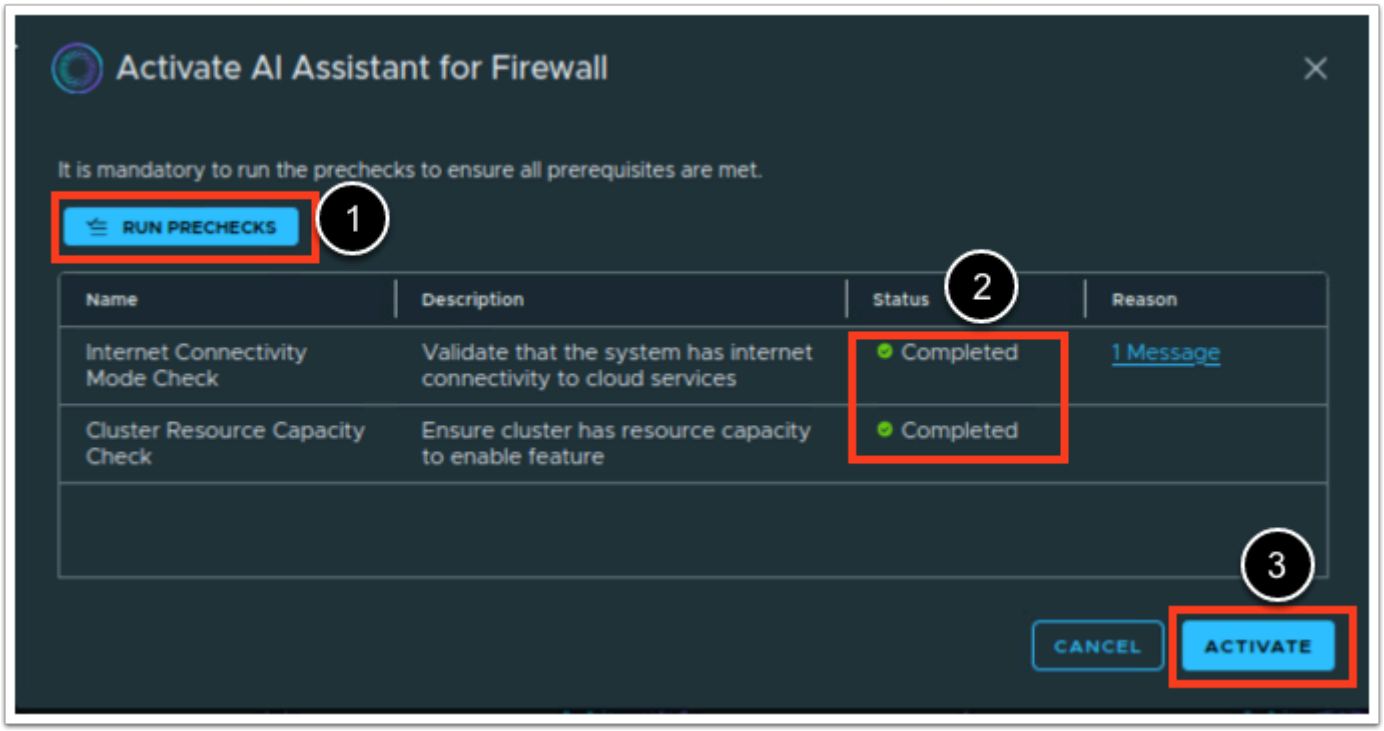
1. Click **System.**
2. Click **Platform & Features.**
3. User the **scroll bar** and pull the page to the bottom.
4. Click on **Firewall** under AI Assistans.

Activate Firewall Assistant



1. Click **ACTIVATE.**

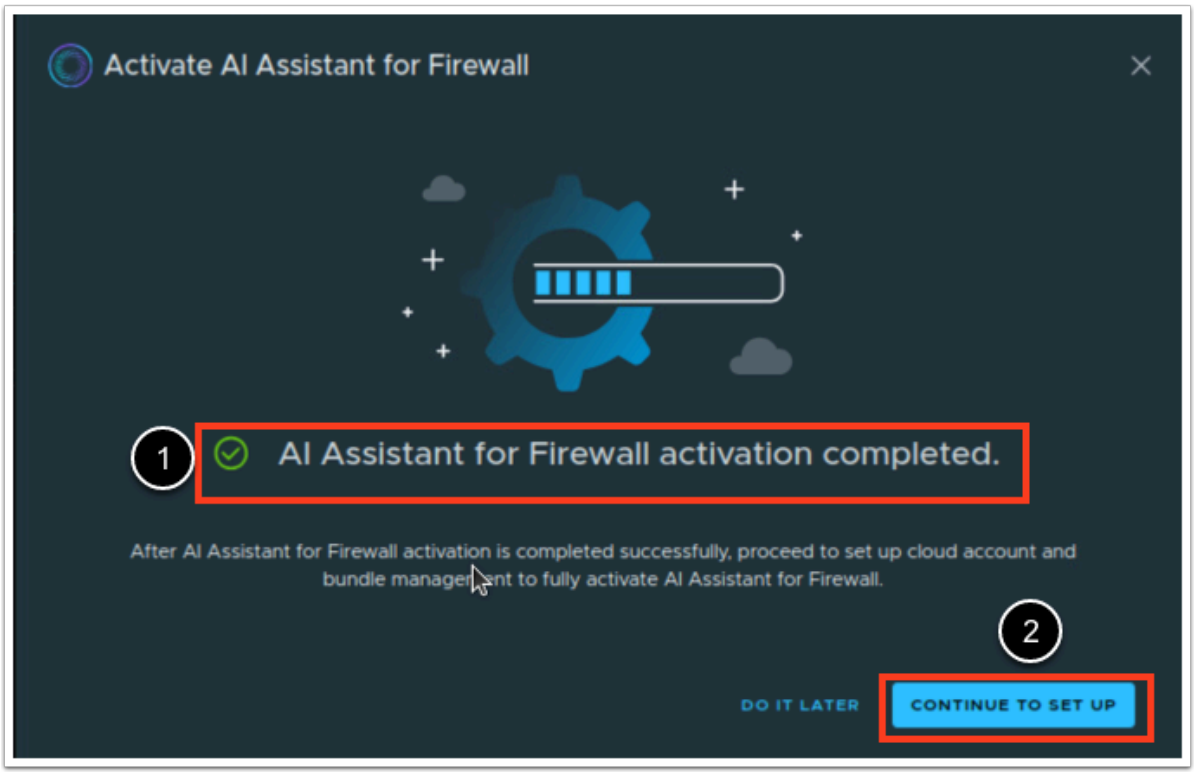
Run Prechecks



1. Click **RUN PRECHECKS.**
2. Validate Status **Completed.**
3. Click **ACTIVATE.**

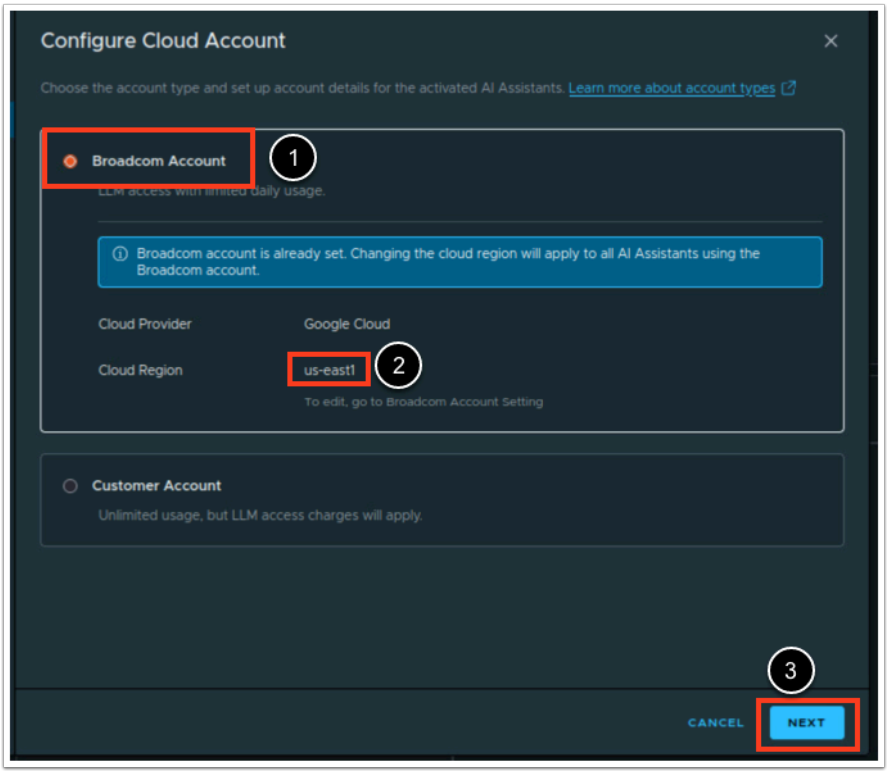
> The activation process will run for 2 to 3 minutes, you can monitor the progress on the progress bar.

Continue to Setup



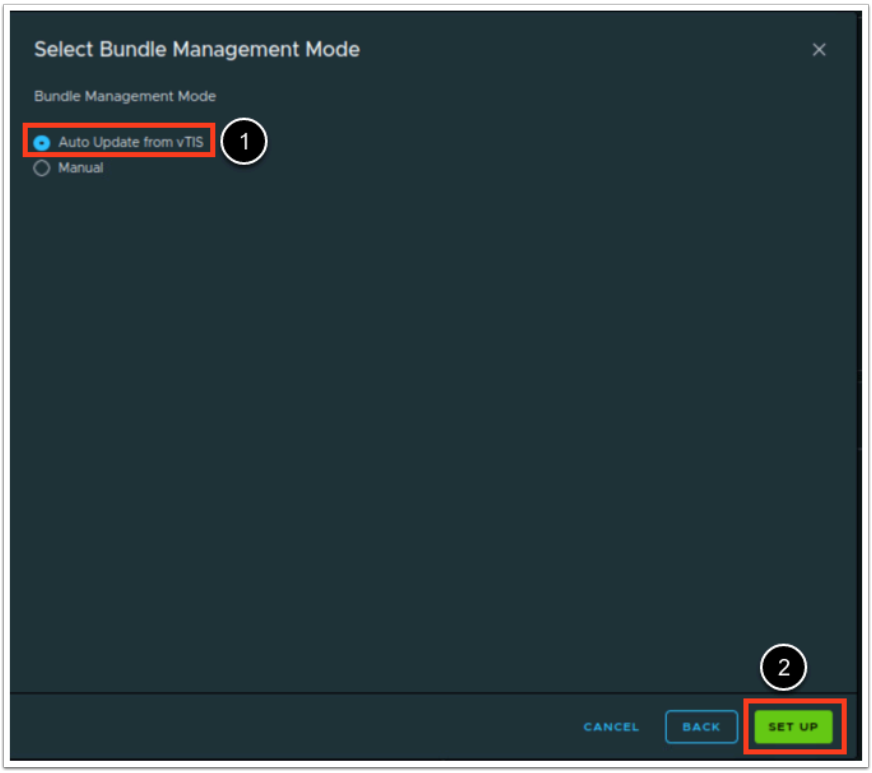
1. Validate activation completed.
2. Click **CONTINUE TO SET UP.**

Configure Cloud Account



1. Confirm **Broadcom Account** is selected.
2. Confirm Cloud Region is **us-east1.**
3. Click **NEXT.**

Bundle Management Mode



1. Confirm Bundle Manage Mode is set to **Auto Update from vTIS.**
2. Click **SETUP.**

> The AI Assistant Bundle will be downloaded from the vTIS online portal. This does take a few minutes to complete - typically 7 minutes in this lab.

Monitoring Progress of the download



1. Click on **IN PROGRESS.**
2. Bundle Import Status will show **In Progress** while downloading.

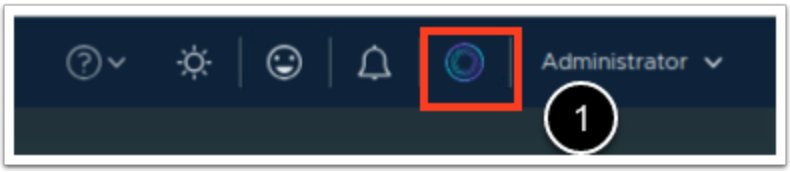
AI Assistant for Firewall Activated



1. Once successfully activated, the Status should be **UP.**
2. Once the bundle has successfully downloaded and updated you should a bundle version **5.2.0_xxxx.**

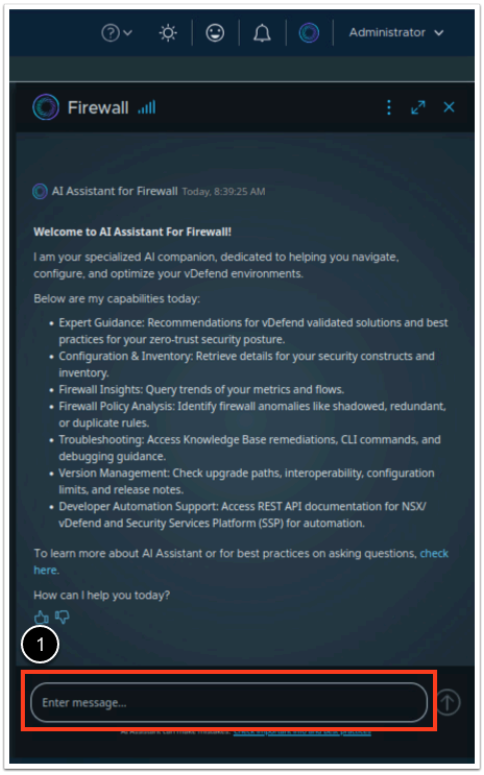
> Since this is doing an automatic update the exact numbering on the bundle versions could differ.

Using the AI Assistant for Firewall Chat With SSP UI



1. In Security Services Platform, Click the AI Assistant for Firewall icon that is located in the top-right corner of the global navigation bar.

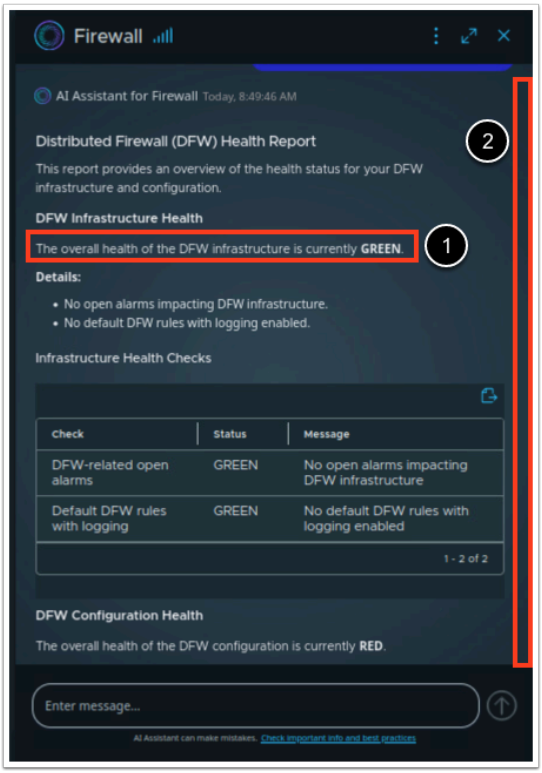
Interact with the AI Assistant for Firewall chatbot - Firewall Health Check



1. In the search bar type **Perform a dfw health check**

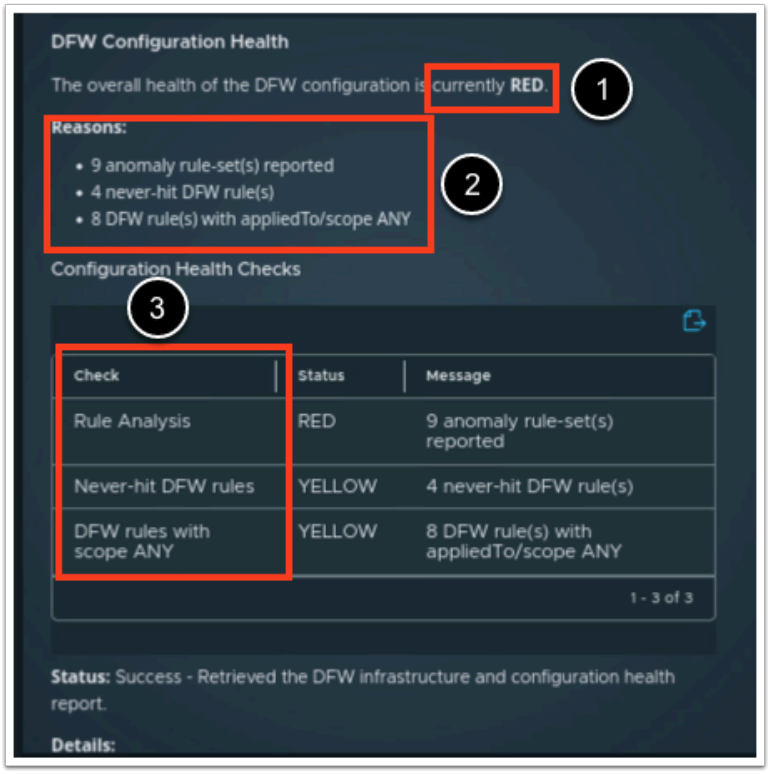
> The AI Assistant will collect statics from the Metrics collected in Security Services Platform

Monitor output from the Firewall Health Check



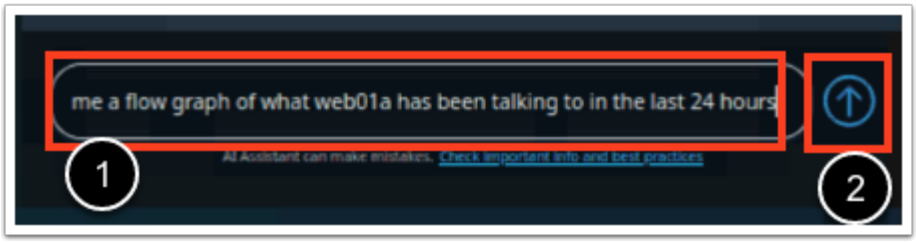
1. Here we see the overall status as **GREEN** and no major errors to report.
2. Use the scroll bar to page down and view the other findings.

Review output from the Firewall Health Check related Configurations



1. Here we see the configurations health is currently **RED**.
2. Here we see some reasons as to why the configuration score.
3. Here we see where these can be checked or the cause of the score.
    1. We notice that Rule Analysis has highlighted a number or anomaly rule-sets.
    2. We have DFW rules which have not hits.
    3. We have a number of rules configured with the apply to scope as **ANY.**

Use the AI Assistant for Firewall to monitor traffic for a selected virtual machine



1. Go and copy this prompt into the AI Assistant chat bar: ***Can you show me a flow graph of what vm 'web01a' has been talking to in the last 24 hours.***
2. Click on the **UP** arrow to send the prompt.

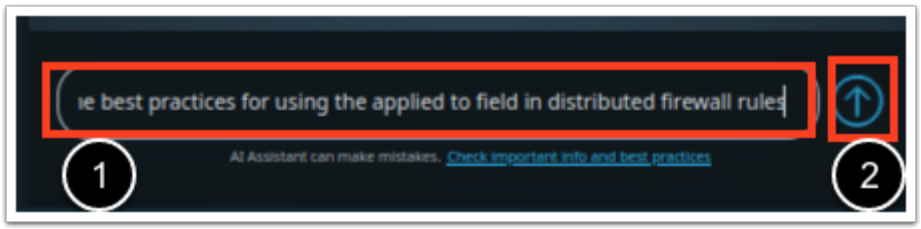
> The AI Assistant does take some time collect all the information as it processing through all the data

Review the flow data output generated



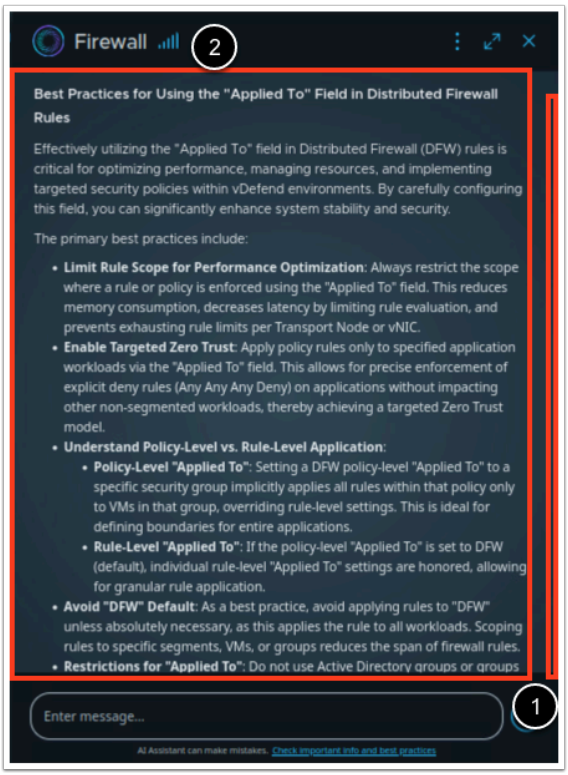
1. Use the scroll bar on the right and pull the page down as needed to view the VM Flow Graph.
2. Here we see all the entities which web01a has been community too.

Best Practices Knowledge Prompt



1. Now lets try and a best practices knowledge prompt, in the prompt bar copy and paste the following "What are the best practices for using the applied to field in distributed firewall rules"
2. Click the UP arrow to submit the prompt.

Review out the output of the best practices prompt



1. Use the scroll bar on the right and pull the page down or up as needed.
2. Here we see the Best practices to follow when using the **Applied To** Field in Distributed Firewall Rules